SAD - #901: iGnite Segment CDP Release 3.1

Solution Outline/Architecture for - VH220041 - Ignite - Segment CDP

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Document Information:

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Storage:	iGnite R3

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Restriction:	Confidential

Quick Links:

Business Requirements (BRS, Canvas etc.)	List - Ignite (CMID) - TPG Telecom Jira HLR → R 3 R equ ire me nts & T PG Wo rkin g D ocu me nts

Solution Governance:

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3.8.4.1 End-to-End Integration Health

Project Clarity Id:	VH2 200 41
Solution Governance Id:	SG# 443 7
Privacy Impact Assessment:	In Pro gres s
PCI-DSS Compliance:	N/A
Ops Acceptance (RLM ID):	
Threat Risk Assessment:	In Pro gres s

How to Complete this Document

How to Complete this Document

All typing in *blue italics* in the body of this document is for instruction and guidance and must be removed on completion of the document.

The document is structured into 7 sections

***Section 1** is the Context designed to understand the project objectives, scope, Business Context overview and critical success factors. This is required for both solution Outline and Solution architecture documents*

3.8.4.2 Synchronisation Comparison
3.8.4.3 Operational Trends
3.9 Quality of Service (QoS) / Performance
3.9.1 QoS and performance measures
3.9.2 End-to-end service indicators
3.9.3 Performance assurance
3.10 Data Backup Policy
3.11 Business Continuity and Disaster/System Recovery
3.12 Automation
3.12.1 Deployment Automation
3.12.2 GitHub Actions Automation Setup for Segment SaaS
3.12.3 Operational Activities
3.12.3.1 Operational activities will enable authorised support teams to monitor, pause, resume, rerun, recover and reconcile Segment integrations without requiring uncontrolled changes to production configuration.
3.12.3.2 Scheduled operations
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3.12.3.8 Monitoring-pipeline operations
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Section 2 *is the End to End solution description which provides solution options (if any), recommended solution rationale along with architectural overview of the recommended solution from contextual and components perspective. This section will be reviewed by the principal architects as part of architecture assessment of solutions to ascertain if the recommended solution aligns to Architecture policy and blueprints.*

Section 3 *provides the Non-functional overview section, to ensure that the recommended solution addresses the non-functional requirements of the respective areas including the design standards and standard operating environments.*

Section 4 *Security Legal and Regulatory Aspects of the Proposed Solution*

Section 5 *provides the description of the Non-production environment, the changes that are required in the environment, dependencies, assumptions and constraints*

Section 6 *provides deployment and migration strategy.*

Section 7 *provides future considerations.*

ARB presentation

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      ii. Business Context Overview
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4 Security Controls Compliance
5 Legal and Regulatory Aspects of the Proposed Solution
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6 Non-Production Environment (Optional)
6.1 Non-Production Environment Solution Overview
6.2 Changes to Non-Production Environment Architecture
6.3 Model Building blocks (including testing tools)
7 Deployment/Migration Strategy
8 Future Considerations
9 Appendix
9.1 A. Auth0 to Segment direct connector
9.2 B. Twilio Segment to Pega CDH
9.2.1 Data Flow
9.2.2 Generic Microservice forwarder
9.2.3 Non-Functional Considerations
9.3 C. RIO pipeline with Terraform option
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iv. User Groups and Domains Impacted
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c. System Register and CMDB
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e. Solution Interactions
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g. Architecture Deviations and Technical Debt
h. Architecture Policies, Standards and guidelines
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b. Operation Design Standards Exception
4. Mandatory Security Controls Compliance (Mandatory 14)
5. Legal, Regulatory and PCI-DSS Compliance
a. PCI-DSS Compliance
b. Regulatory Codes and Acts

Document Control

Change History

Change History

Version	Date	Description	Author
0.1	Apr 10, 2026	Initial version	John Mathew
0.2	Sep 4, 2026	• Appended component section for OSS platform • Updated domain impacts section • Updated solution diagram to correct integration with Splunk, added BMC helix link	John Mathew
0.3	Sep 6, 2026	• Updated data interaction details in the component interaction section • Updated the data asset components	John Mathew
0.4	Sep 7, 2026	• Updated environment details suggestions • Update mandatory 14 controls.	John Mathew
0.5	Sep 10, 2026	• Corrections to Apigee and Microservice components to use Kafka connect. • Updated Data component interactions to use redshift cluster instead of RDS Postgres.	John Mathew
0.6	Sep 11, 2026	• Update Splunk data log collection details • Under automation section added Terraform and GitHub repositories.	John Mathew

0.7	Sep 15, 2026	• Added observability monitoring for delivery metrics to compare data loaded into Redshift cluster vs data sent from segment. • Added approximate ongoing cost running redshift cluster. • Removed Terraform setup using RIO pipeline as requirement deprioritised. • Added Operational monitoring requirements especially for the warehouse destination section.	John Mathew
0.8	Sep 18, 2026	• Update the PII assessment after review. • Updated CDP Redshift cluster from RDS Postgres workload.	John Mathew
1.0	Sep 24, 2026	• Added reverse proxy and zero rating changes. • Updated mandatory 14 controls. • Added testing consideration for testing out zero rating.	John Mathew
1.1	Sep 27, 2026	• Added a security matrix to outline the different authentication and authorisation methods for each components. • Updated Monitoring, QoS, and Operational tasks. • removed reference of RDS Postgres from everywhere	John Mathew
1.2	Sep 29, 2026	• Remove reference to using fixed db user and password for redshift as Segment have enabled IAM role capability. • Added tagging and data governance guidelines for the data assets.	John Mathew
1.3	Oct 2, 2026	• Amended solution diagram to include workload for CDP redshift cluster and Splunk Enterprise Alert collector. • Amended component EDM workloads to include alerting to be done through Splunk Enterprise Alert collector.	John Mathew
1.4	Oct 6, 2026	• Update cost of Redshift MDS for additional based on revised data transfer, backup storage.	John Mathew
1.5	Oct 15, 2026	• Added Auth0 in solution diagram and components and event bridge needs to be modified to filter calls to Segment. • Added TFE API as it has some of the server-side calls.	John Mathew

		• mentioned alternate to SSM as Dynamo DB	

Contributors

Contributors

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Sankar Bhaskaran	Architect	BSS IT
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Approvers

Approvers

Name	Responsibility	Date	Signature

Distribution

Distribution

Name	Responsibility	Date

References

References

Provide a list of all other documents to consider when reading this document

The following documents are referenced:

Reference	Document	Version
SAD - #901: Ignite - Customer Data Platform Onboarding/Standalone	Onboarding SAD	1.1

Definitions

Definitions

Definitions of all terms used in this document to define the solution (glossary)

Term / Abbreviation	Description / Definition
BRS	Business Requirement Specification
ETE	End to end
SO	Solution Outline
SAD	Solution Architecture Document
HLD	High level Design
LLD	Low level design or Detailed design
BB	Building block

CDP	Customer Data Platform
MDS	Marketing Data Store
S3	Amazon Simple Storage Service
KPI	Key Performance Indicator
CLM	Customer Lifecycle Management
OSS	Operational Support Systems
CMDB	Configuration Management Database
ITSM	IT Service Management
PII	Personally Identifiable Information
SDK	Software Development Kit
CSP	Content Security Policy
SRI	Subresource Integrity
TOM	Target Operating Model
WoW	Ways of Working
MFA	Multi-Factor Authentication
SSO	Single Sign-On
UI	User Interface
UX	User Experience
DSP	Demand-Side Platform
CDH	Customer Decision Hub (Pega CDH)

ETL	Extract, Transform and Load
HTTP	Hypertext Transfer Protocol
API	Application Programming Interface
QoS	Quality of Service
SLA	Service Level Agreement
OPEX	Operating Expenditure
EDM	Enterprise Data Management
RDS	Relational Database Service
DDoS	Distributed Denial of Service
EDR	Endpoint Detection and Response
CI/CD	Continuous Integration / Continuous Deployment
SaaS	Software as a Service
TOM	Target Operating Model
VCA	Vodafone customer channel/application (confirm official expansion)
DFA	Brand acronym (confirm official expansion before publishing)
BSS	Business Support Systems

Auth0	Identity and Access Management Platform
CLM	Customer Lifecycle Management

Context

Required for both SO and SA

Purpose

To support TPG's strategic objective of becoming a leading data-driven marketing organization, the implementation of a robust and scalable Customer Data Platform (CDP) is required. The new CDP will integrate seamlessly with existing systems, enable advanced Customer Lifecycle Management (CLM) capabilities, and act as the central hub of the marketing architecture by unifying offline customer context with online channel interactions.

With the decommissioning of the existing SAS CI360 platform, Segment CDP has been selected as the strategic replacement due to its ability to deliver a comprehensive, real-time, and unified view of customer interactions across all channels. Segment will enable TPG to deliver more relevant and personalized customer experiences, driving improved commercial outcomes through increased retention, cross-sell, and upsell.

This transition is a critical component of the iGnite Program and is targeted for completion by June 2026. Successful adoption of Segment CDP will strengthen TPG's customer engagement capabilities, enhance marketing performance, and ensure alignment with TPG's long-term data and marketing strategy.

Segment CDP has been selected as the strategic replacement for SAS CI360 due to its ability to provide a **comprehensive and unified view of real-time customer interactions across all channels**. Segment's capabilities will support TPG to deliver relevant, personalized, and effective offers, enhancing the overall customer experience and driving improved commercial outcomes through higher retention, cross-sell and upsell opportunities.

Business Benefit

Required for ARB presentation Next

- Leverage rich customer engagement context across TPG systems, while adhering to privacy and consent requirements, to drive sustained digital marketing effectiveness.

- Transitioning to Twilio Segment ensures TPG's Customer Data Platform capabilities remain modern, scalable, and aligned with market trends, enabling easier adoption of new technologies and responsiveness to evolving business needs.
- Improve the identification and resolution of known and unknown individuals, clearly distinguish customers from prospects, and build accurate, trusted marketing profiles.
- Deliver consistent, clean, and reliable behavioral data matched to customer profiles and activated through PEGA, enabling automated, timely, and scalable marketing communications.
- Enable advanced Customer Lifecycle Management (CLM) use cases, including:
  - Proactive management of customer experience through enhanced churn identification and prevention.
  - Increased weighting and prioritisation of reactive customer interactions to support churn mitigation and cost optimisation.
  - Improved propensity modelling to inform churn reduction, upgrade opportunities, and targeted interventions.
  - Cross-sell and growth optimisation initiatives, including convergence and product expansion strategies.
- Store customer interaction data within the target data platform (AWS) to support KPI measurement, performance reporting, and continuous optimisation of CLM marketing campaigns.
- Enable querying of historical customer data within AWS to support customer suppression lists and creation of sophisticated lookalike and paid media segments for acquisition campaigns (future capability).
- Address current limitations in prospect segmentation and paid media activation through CDP-driven integrations with key AdTech platforms (future capability).

Scope

Required for ARB presentation Next Back

In Scope

iGnite R3 is focused on delivering the **foundational Customer Data Platform (CDP) capability using Twilio Segment for the Vodafone consumer brand only**. R3 establishes the core CDP platform required to collect, unify, store, and activate customer data by enabling **inbound and outbound channel data ingestion, identity resolution, real-time data activation to Pega CDH, and persistent storage in AWS MDS (S3 and Redshift)**.

R3 also delivers the **transition and decommissioning of SAS CI360 (CDP Lite) for Vodafone**, ensuring continuity of existing digital marketing capabilities while creating a scalable foundation for future releases.

All brands other than Vodafone (including DFA) are explicitly **out of scope for R3** and are planned for subsequent releases (R4/R5). Additionally, **enterprise OSS capabilities such as Microsoft Entra ID, BMC Helix ITSM integration, and CMDB registration are delivered through a separate enterprise solution and are excluded from R3 scope**.

In Scope Details

Brand Scope

- **Vodafone (Consumer) only**

CDP Platform Foundation

- Implementation of **Twilio Segment CDP** for Vodafone
- Segment configuration:
  - Workspaces, sources, destinations, protocols, Unify
  - Identity graph and customer profile unification
  - Data ingestion standards and schema governance
- Privacy, consent, and PII enforcement within Segment
- Platform monitoring, alerting, replay, and recovery capabilities

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Operating Model & Enablement

- Definition and rollout of:
  - Target Operating Model (TOM)
  - Ways of Working (WoW)
- Segment training and enablement for Vodafone business and IT users
- Governance artefacts and supporting tools (e.g. GitHub)

Inbound Channel Data Enablement (Vodafone)

- Digital interaction and identity data ingestion from:
  - **Auth0**
  - **OneApp** (Web & Mobile – Vodafone pre-paid and post-paid)
  - **Websites:** Shop, VCA, Salesforce (Vodafone)
- Technical enablement:

- Segment JavaScript and SDKs
- Anonymous-to-known identity stitching
- CSP, SRI, reverse proxy, and network configuration required for tagging

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Data Storage – AWS MDS

- Persistent storage of Segment data:
  - Raw event data in **Amazon S3**
  - Structured event and profile data in **Amazon Redshift**
- Schema management, retention, lifecycle policies, and performance controls
- Secure access and environment segregation within AWS MDS

Reverse Data Enablement

- Enablement of **AWS MDS → Segment** traits and identity enrichment
- Trait-ready customer profiles for activation in Pega CDH

Legacy Capability Transition

- **SAS CI360 (CDP Lite) decommissioning for Vodafone**
- Functional transition of equivalent CDP capabilities to Segment

Out Of Scope

Out of Scope Items

Brands

- **All brands other than Vodafone**

> These brands are planned for delivery in **R4 and R5**.

Enterprise OSS / Shared Services (Separate SAD)

- **Microsoft Entra ID (SSO / MFA) integration**
- **ITSM integration with BMC Helix**
- **CMDB registration and lifecycle management**

> These are dependencies provided by an enterprise platform solution, not R3 deliverables.

Future Release Scope (R4 / R5)

- Onboarding of additional brands (e.g. DFA)
- Additional inbound channels:
  - **Genesys**
  - Cludo
- Expanded outbound enablement for non-Vodafone brands

Advanced Analytics & Activation

- Machine learning and propensity modelling
- Paid media and AdTech platform integrations
- Lookalike audience creation and DSP activation
- Personalisation and experimentation platforms
- Creation of new derived data products using data from Segment for reporting

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Customer Experience Changes

- No customer-facing UX or UI redesign
- No front-end journey re-engineering

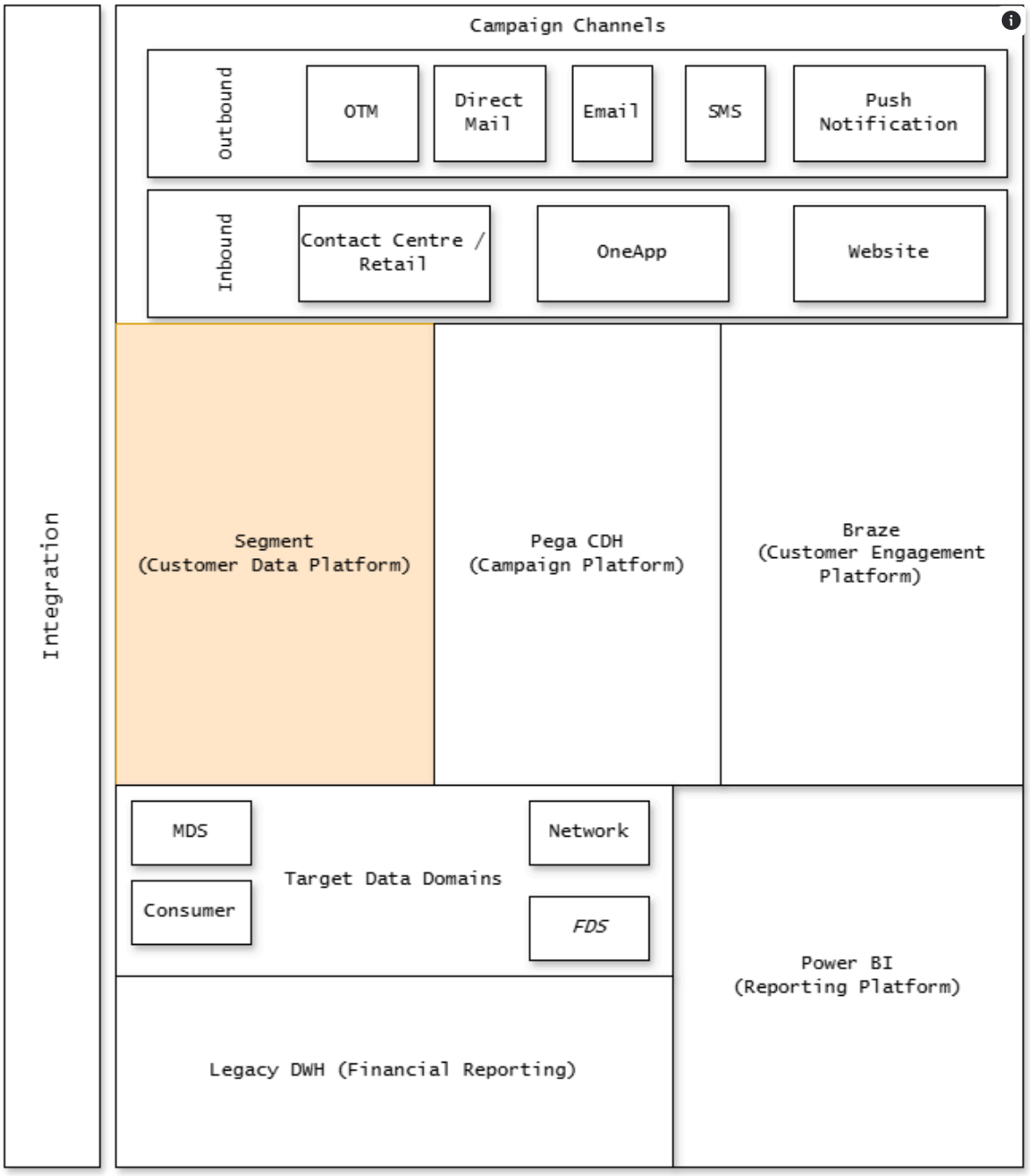
Non-CDP Transformation

- Redesign of Pega campaign logic or marketing strategies
- Broader data lake or analytics transformations beyond AWS MDS enablement

Business Context Overview

Required for ARB presentation Next Back

The below diagram describes the role of Segment in the campaign ecosystem as customer data platform. Segment along with Pega CDH and Braze as Customer Engagement platform plays key role in the outbound & digital campaigns and will collect and use data from the users of the digital platforms (websites, apps, etc.). Segment, will also collect, transform, send, and archive first-party customer data.



Target Stack

Products and Business Services

Required for ARB presentation Next Back

Contextual elements associated with the recommended solutions are as follows. Provide the brands, products and business services that are impacted by the solution.

Brands in Scope

Brand	In Scope(Y/N)	Comments
Vodafone	Yes	Vodafone customers are included in the solution.
TPG	No	DFA will be handled as part of future releases next year.
iiNet	No	DFA will be handled as part of future releases next year.
Lebara	No	This will be handled outside the current Ignite program.
Felix	No	This will be handled outside the current Ignite program.
AAPT/Internode	No	This will be handled outside the current Ignite program.
Other		

Products

Include the list of products impacted with this solution. The product catalogue is available here.

Product(s)	Comments
Mobile Postpay	N/A for this solution scope
Mobile Prepay	N/A for this solution scope
Mobile Broadband Postpay	N/A for this solution scope
Mobile Broadband Prepay	N/A for this solution scope
NumberSync	N/A for this solution scope
Upfront Postpaid	N/A for this solution scope
Fixed Wireless Access	N/A for this solution scope
National Broadband Network	N/A for this solution scope
Vision	N/A for this solution scope

Opticom	N/A for this solution scope

Customer Lifecycle Stage Impacts

Outline the customer lifecycle stage that is impacted with this solution. Provide the details of the impact.

Please refer to the customer experience framework and tools to answer this section.

No impact on customer experience blueprint.

Customer Lifecyle stage Impacted	Description of Impact
Ex: Find	Updating the recommendations logic

Business Services

The list of Business Services can be accessed here. If the Business service is not in this list, These Business Services are realized by Technical services.

Include the list of business services impacted with this solution.

Add/Modify /Delete	*Business Service Name*	*Description of the type of change*
Add	Marketing Technology	Add Twilio Segment to the martech stack and enable use cases within the Ignite and other spring programs.

User Groups Impacted

Required for ARB presentation Next Back

Examples of elements have been listed in the table below. Add, delete or modify the elements to suite the recommended solution.

Roles, functions	Description of the Impacted User Groups
Customer - Domestic Mobile	No Impact
Customer - Domestic Mobile Roaming	No Impact
Customer - International Mobile Roaming	No Impact
Customer - Fixed Consumer	No Impact
Customer - Enterprise & Government	No Impact
Customer - Wholesale Fixed	No Impact
MVNO / MVNE	No Impact
Retail / Dealer	No Impact
Customer Service Rep	No Impact
Customer Online Store	No Impact
Customer Self Care	No Impact
Credit / Collections	No Impact
Business Intelligence	Implement / support the implementation team with solution.
Revenue Assurance	No Impact
Deferred Channels	No Impact
Manage Service Partner	New system and integrations to be supported by selected MSP team until the decommissioning the SAS platform.
Workplace	No Impact
Staff / Employees	*For domain specific impacts pls use table 4*
...	*Any other area not covered in the above list*

Roles, functions, and separate table for business interfaces and associated enhancements required

Table 3: User Group Changes

Domains Impacted

Team	Team	Description of the Impact
Network & Infrastructure	Cloud Infra. Engineering (CiNS - CiE)	Implement cloud controls for logs ingestion from Segment to TPG managed AWS accounts. Create new workloads in existing marketing data store AWS account. RIO CI/CD setup support to onboard new application.
Network & Infrastructure	Ent. Gov. & Wholesale Network Services	No Impact
Network & Infrastructure	Fibre Access Networks	No Impact
Network & Infrastructure	Fibre Engineering Services	No Impact
Network & Infrastructure	Fixed Voice Engineering	No Impact
Network & Infrastructure	Mobile Data Engineering	No Impact
Network & Infrastructure	Mobile Voice Engineering	No Impact
Network & Infrastructure	Network Dev. Engineering (CiNS - NDE)	
Network & Infrastructure	Network Services Engineering (CiNS - NS)	Implement firewalls, DNS changes to support the solution
Network & Infrastructure	OSS Platforms & Design (CiNS - OSS)	Implement OSS logging, monitoring and alerts in Splunk, Status

Network & Infrastructure	Technology Centres & Major Construction	No Impact
Network & Infrastructure	Transmission Networks	No Impact
Network & Infrastructure	Wireless Networks	No Impact
Tech. Service Operations	Network Operations - Fixed	No Impact
Tech. Service Operations	Network Operations - Mobile	No Impact
Tech. Service Operations	Service Governance	TBC
Tech. Service Operations	Technology Operations IT & Digital	Support the new/additional system until decommissioning of SAS CI
Tech. Service Operations	Tech. Ops Service Design & Assurance	TBC
Tech. Service Operations	Workplace Health & Safety	No Impact
Technology Security	Technology Cyber Security	Support the solution with TRA and Security Assurance.
IT & Digital	IT -Sales & Care	Implement new integration with digital into Segment CDP.
IT & Digital	IT - ERP	No Impact
IT & Digital	IT - QA and Testing	Support testing of new platform integration as defined in the delivery scope.
IT & Digital	IT - Revenue Charging & Integration	No Impact
IT & Digital	IT - Workplace	No Impact
IT & Digital	IT- Data (Business Intelligence)	Implement new integration for Segment CDP ingestion and export. Implement RIO CI/CD data pipeline for Twilio segment

		code configuration using Terraform.
Finance	Margin Assurance& Fraud Detection	No Impact
Customer Ops	Consumer and COPS	*Consumer CLM team will need training to use new platform for data collection and segmentation to downstream systems.*
Customer Ops	Billing & Payment	No Impact
Customer Ops	Business Configuration	No Impact
Customer Ops	Care Platform Operations	No Impact
Customer Ops	Credit & Collections	No Impact
Customer Ops	Customer Care	No Impact
Customer Ops	Go To Market (GTM)	No Impact
Customer Ops	Logistics & Inventory	No Impact
Other		

Table 4 Impacted Domains

Process Impacts

Required for ARB presentation Next Back

Identify any new processes to be developed or existing processes that will need to be changed. Describe the new processes or changes required to existing processes

ETE process	New / Modified	Description of the change
Ongoing Twilio Segment NPM Package Maintenance and Upgrade Process (Vodafone Domain)	New	The Java SDK hosted within the Vodafone domain for client-side event tracking requires ongoing maintenance

		to ensure it remains current with Twilio Segment SDK releases. This includes implementing security updates, maintaining vendor supportability, and enabling access to new features and enhancements introduced by Twilio Segment.
Segment zero-rating management	New	Submit approved Vodafone CloudFront endpoints through the Service Rating Request process for Billing and Packet Core assessment, configuration and pre-production validation. Future endpoint changes must follow the agreed BAU change and regression-testing process.

Table 5: Process Impacts

Critical Success Factors

Required for ARB presentation [Next](#) [Back](#)

Describe what measures will be used to quantify this initiative was successfully implemented and are meeting associated business requirements.

1. Identity resolution with key identifiers consistently captured and stitched using **identify/alias** approaches, ensuring **computed traits are available for decisioning**.
2. Enable **data export to the Target Data Platform** to support downstream reporting and analytics requirements.
3. Govern configuration via version control and approvals with repeatable deployment and traceable change history.
4. Establish a **Target Operating Model** covering RACI, L1–L3 support, runbooks, and escalation paths to support BAU operations and ongoing governance.

End to End Solution Description

Required for both SO and SA

Solution Options

If one or more solution options have been considered, then describe each option including the associated pros and cons of each. Add or delete options as required

Solution Option 1 - <Option Name>

Provide a brief description of this option. Include an Architecture Model (View) for each architecture and describe the benefits and disadvantages of each solution.

Pros	Cons
Describe succinctly (e.g., in point form) any pros associated with this Solution Option.	Describe succinctly (e.g., in point form) any cons associated with this Solution Option.

Table 6: Solution Option 1

Solution Option 2 - <Option Name>

Provide a brief description of this option. Include an Architecture Model (View) for each architecture and describe the benefits and disadvantages of each solutions

Pros	Cons
Describe succinctly (e.g., in point form) any pros associated with this Solution Option.	Describe succinctly (e.g., in point form) any cons associated with this Solution Option.

Table 7: Solution Option 2

Solution Option 3 - <Option Name>

Provide a brief description of this option. Include an Architecture Model (View) for each architecture and describe the benefits and disadvantages of each solution

Pros	Cons
Describe succinctly (e.g., in point form) any pros associated with this Solution Option.	Describe succinctly (e.g. in point form) any cons associated with this Solution Option.

Table 8: Solution Option 3

Recommended Option and Rational

Nominate the preferred option and clearly describe the reason(s) for selecting this option using the table below. Clearly specify what considerations played a key role if the preferred architecture solutions were not selected. If the scoring matrix table was developed as part of the evaluation add a link. Please refer to the architecture policies in section 2.4 for non-functional compliance if the recommended solution.

Influence if the recommended solution is selected.

Cost of Ownership	Time to Market	Risk/Security	Architecture Alignment	Operations

Impact if the recommended solution is not selected.

Cost of Ownership	Time to Market	Risk/Security	Architecture Alignment	Operations

Table 9: Impact of Recommended Solution

Solution Overview

Required for ARB presentation

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The following diagrams provides an overview of the baseline (as is) and recommended solution

Details associated with each Building Block (SYSTEM) described in this section is available in section 3.4. Use names as per the system register. Ensure that the solution follows the Architecture Conventions Standard.

A checklist of standard integrations and patterns on public cloud are available here.

Please refer to documented Architecture Patterns: Architecture Patterns Archived

The Cloud Platform Patterns: Architecture Patterns

Segment CDP will be introduced as a Software-as-a-Service (SaaS) platform to host a unified customer profile. The platform will be deployed within the vendor's AWS environment and will integrate with both cloud-based and on-premises enterprise components.

Customer Data Unification

Segment will be onboarded as the central customer data backbone, consolidating real-time and batch customer data from all key touchpoints, including:

- Digital channels (e.g. VCA, Shop, OneApp – web and mobile)
- CRM platforms
- Enterprise Data Warehouse

All data will be governed and unified into a single customer profile, providing a trusted and consistent view of the customer across the organisation.

Decisioning and Activation

The unified customer profiles will be consumed by **Pega Customer Decision Hub (CDH)** as the primary decisioning engine. This will enable next-best-action strategies to be executed across all customer touchpoints in near real time.

Segment CDP will feed unified customer profile built based on events (e.g., clicks, purchases) to Pega CDH via Kafka (existing AWS hosted). Pega CDH processes these events to compute next-best-action (e.g., personalized offers) across various channels.

Digital Platforms Integration

Segment will ingest customer interaction data from a variety of digital customer facing components for tracking and analysis:

- **Auth0 (CIAM) integration** enables Segment to capture customer login events and identifiers, linking them to a persistent master customer ID for accurate profile matching.
- **OneApp (Mobile – iOS & Android)** uses Segment SDKs for both server-side tracking (via HTTP Tracking API) and client-side tracking (via React Native SDK), replacing the existing SAS CI360 SDK.
- **Shop integration** captures customer clickstream and interaction data using Segment's HTTP Tracking API (server-side) and Segment JavaScript Tag (client-side), replacing SAS CI360 JS tags.
- **OneApp (Web)** collects web interaction and clickstream data through Segment's HTTP Tracking API and JavaScript Tag, also replacing SAS CI360 JS tags.
- **VCA website integration** ingests prospect and customer website navigation and interaction data using a combination of server-side tracking (HTTP Tracking API) and client-side tracking

(Segment JavaScript Tag).

- **EOI Forms Submission**

OSS Integrations

- **SSO (Azure AD / Microsoft Entra ID):**
  Segment integrates with TPGT's Azure AD as the Staff Identity Provider using SAML-based SSO, enabling secure access with granular, role-based permissions across Segment workspaces, sources and destinations.
- **Splunk (Logging & Monitoring):**
  Segment integrates with TPGT's Splunk platform to capture and analyse application events, supporting real-time monitoring, performance visibility, anomaly detection and centralised logging. ITSM tree for Pega application to be updated to add Twilio segment integration.
- **SIEM (Security Monitoring):**
  User and system audit logs are captured via Splunk and forwarded to TPGT's SIEM to meet security monitoring and audit trail requirements.
- **BMC Helix (ITSM / Ticketing):**
  Segment will integrate with TPGT's BMC Helix for automated ticket creation and synchronisation for critical incidents (e.g. Kafka connectivity issues), providing centralised visibility of incident status (TBC).

The solution will:

- Minimise point-to-point integrations wherever possible
- Maximise reuse of existing integration patterns and channels
- Align with the broader enterprise integration and architecture standards

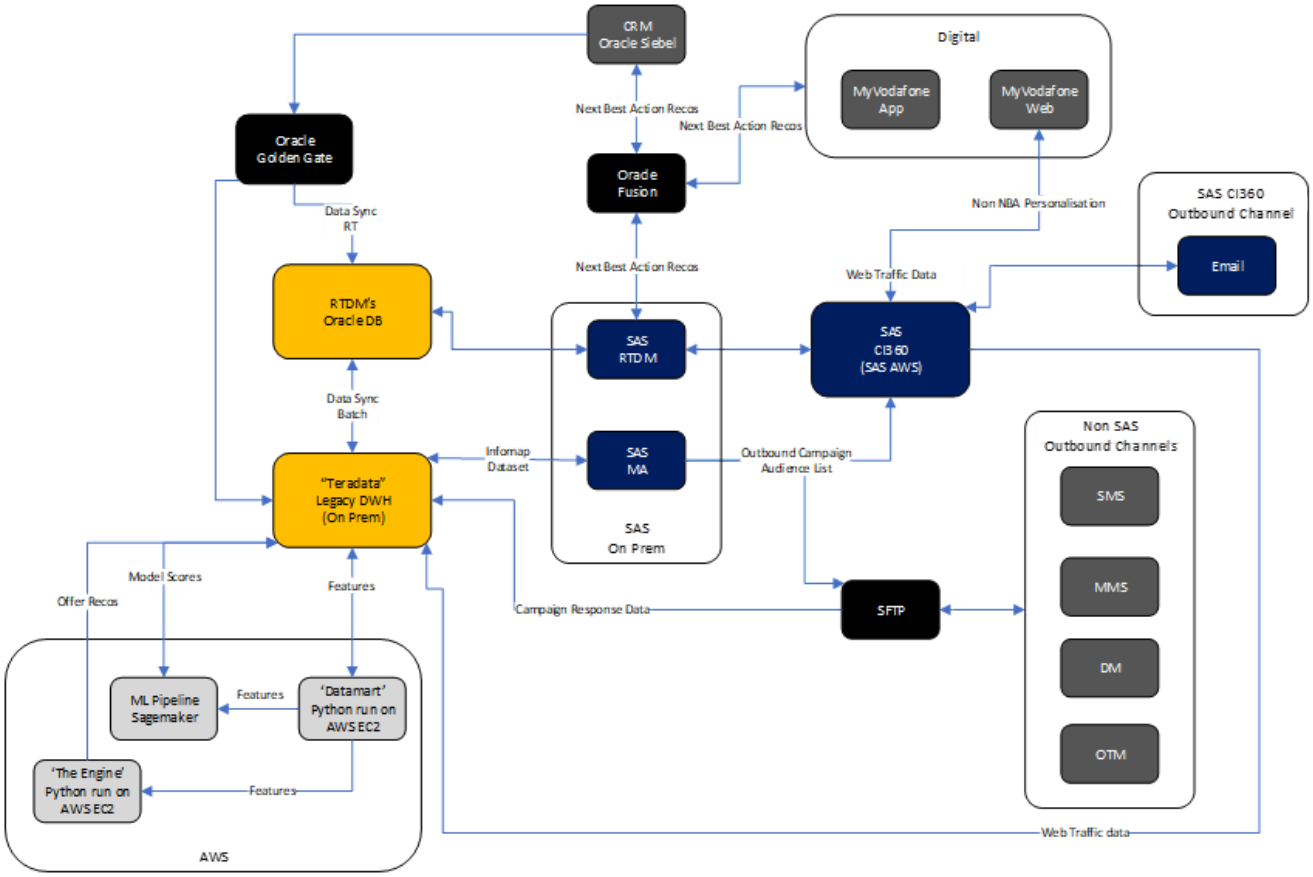


Figure 3: Baseline Solution Architecture Diagram (As is)

Please insert URL of the diagram

The diagrams can be made in Visio / Flowchart Maker & Online Diagram Software using notations that can accurately describe the model.

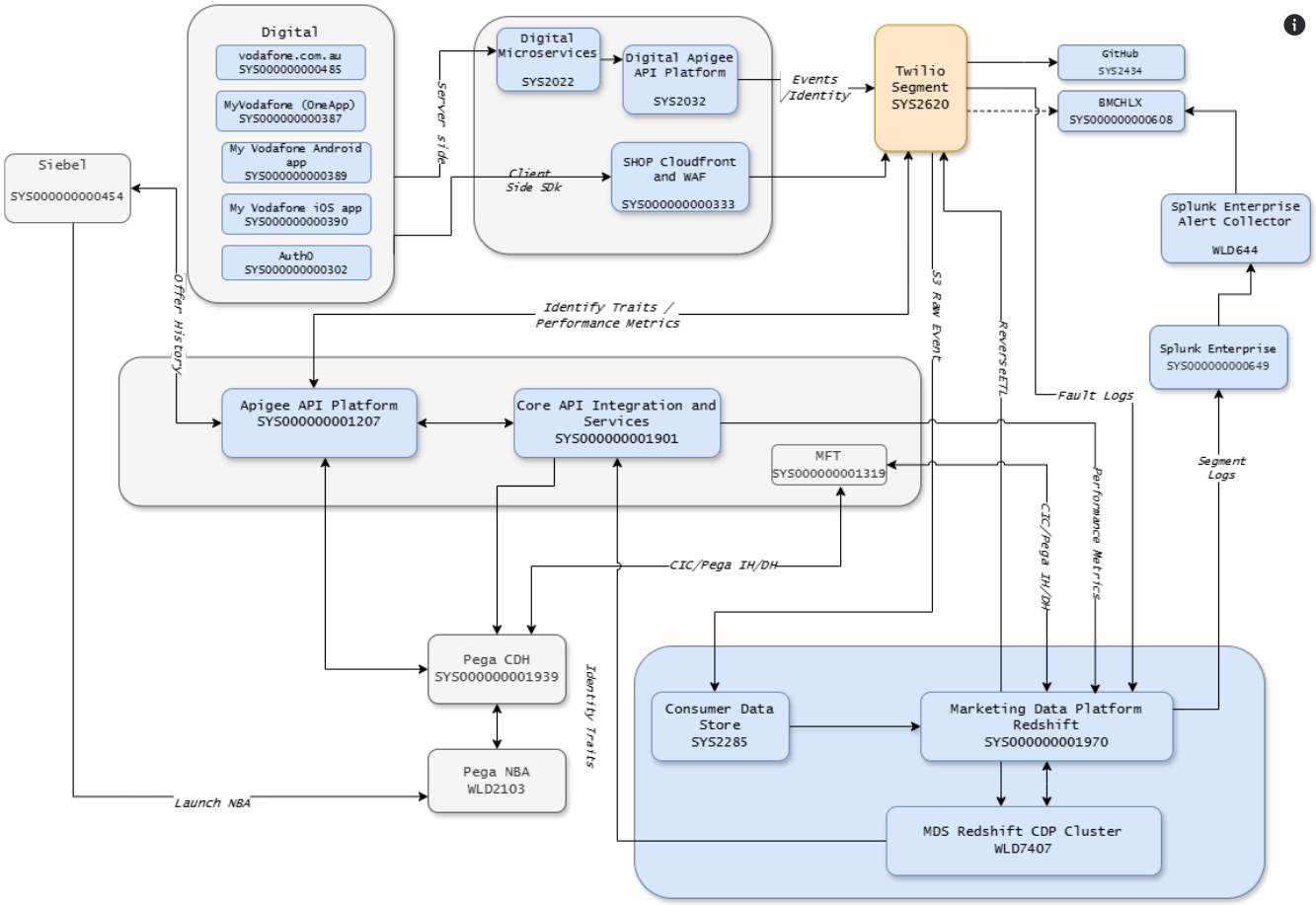


Figure 4: Proposed Solution Architecture Diagram

Alignment to Transition/Target Architecture Blueprint

Clearly show how the Proposed Architecture adds or aligns to the Transition/Target Architecture Blueprint (if available). If the proposed solution does not align with the Transition/Target Architecture Blueprint, clearly specify the rationale.

This initiative aligns with the broader Martech modernization strategy:

- SAS RTDM/MA functionality is being replaced by Pega.
- Additional RTDM/MA use cases are being progressively migrated to Pega as part of ongoing Ignite phases.
- Replace the **SAS CI 360 Discover module** which currently functions as customer data platform in SAS ecosystem.
- In parallel, the Ignite program will replace the **SAS CI 360 ESP module** with **Braze** as the new Email Service Provider (ESP). This will enable the full decommissioning of SAS CI 360.

- The combined changes support simplification, modernization, and future scalability of the Martech landscape.

Solution Assumptions

Required for ARB presentation

Next

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Where known, provide the solution assumptions for the recommended option.

Assumption	Description	Validation Approach
PrivateLink connectivity	Segment can establish the approved private connectivity path to the CDP Redshift Cluster in `ap-southeast-2`.	Complete vendor confirmation, connectivity proof of concept and end-to-end testing.

Table 10: Solution Assumptions

Solution Constraints

Where known, provide the solution constraints for the recommended option.

Constraint	Description	Validation Approach
Segment configuration must be deployed via a Framework-registered CI/CD pipeline (RIO), and is limited to what the Segment Public API exposes.	RIO is TPG's standard IaC and its adoption is a mandatory prerequisite before any implementation on TPG's AWS Cloud, and pipelines must be registered as a Registered Implementation traceable to a CI/CD Framework version. Configuration scope is bounded by the Segment Public API, on which the `segmentio/segment` Terraform provider is built; the provider does not hold full feature parity with the API.	1. Confirm the deployed pipeline is registered against the applicable Framework version, and the workload-to-pipeline relationship is traceable. [CI/CD Fram...quirements | TPG Confluence (Cloud)] 2. Evidence promotion of a representative configuration change through Dev → SIT/UAT → Pre-Prod → Prod via pipeline only, with no direct UI change in higher environments.

	Residual configuration — including Entra ID group synchronisation, which requires a step separate from Terraform — is applied via the Public API or manual configuration. All Terraform changes follow the existing change-request and approval process, with production changes permitted only under change control	3. Maintain and review a documented register of configuration items outside provider coverage, validated against the Segment scenario review 4. Confirm each production deployment is linked to an approved change record.
Segment runtime traffic is configuration-dependent	Segment CDN traffic may include source settings and runtime bundles in addition to static resources. Third-party device-mode destination calls are not automatically covered by the proxy.	Validate the full browser and application network flow using the final Segment source and destination configuration before Production deployment.

Table 11: Solution Constraints

Solution Pre-Requisites and Dependencies

Where known, describe any pre-requisites associated with the implementation of the recommended solution and the potential impact to the solution if the pre-requisite is not met.

Pre- Requisites	Impact
Segment onboarding and workspace provisioning completed (Dev, SIT/UAT, Prod)	Configuration, testing and deployment activities cannot commence until environments are available.
Entra ID SSO setup to be completed as part Ignite R3.0 Onboarding schedule.	Business and support users will be unable to access the platform securely and user provisioning cannot proceed.

Table 12: Solution pre-requisites

Where known, describe any dependencies associated with the implementation of the recommended solution and the potential impact if the solution is not implemented according to the proposed architecture.

Dependencies	Impact
Application access for L2 support teams via Azure Virtual Desktop (AVD) is dependent on the successful delivery of the Citrix Replacement Project. Due to the current embargo on publishing new applications within the existing Citrix environment, Segment access must be provisioned through AVD once the replacement capability becomes available. CLMNBA-5684: [FR] CDP Onboarding | Domain Whitelisting & Secure Remote Support Access (Drop 2 - R3.1) READY FOR CONFIRMATION	A delay in enabling Segment access through Azure Virtual Desktop (AVD) will prevent L2 support teams from accessing the application, limiting their ability to perform day-to-day operational support, incident investigation, and troubleshooting activities. This may require continued reliance on alternative support arrangements until the AVD capability is available.
Foundation **data products including identity attributes** must be validated and ready for consumption ahead of the next-phase development. CLMNBA-5145: [FR] MDS to CDP | Customer Profile Synchronisation READY FOR CONFIRMATION	Increased **delivery and cost risk** if temporary extracts or manual workarounds are required to unblock teams.
Public Cloud uplift of RIO to execute a Terraform component for a non-AWS SaaS provider. CLMNBA-5099: [FR] CDP Implementation | Infrastructure as Code (IaC) for Configuration Deployment (TBC) Deprioritised/Delayed	If not met by design close, there is no compliant deployment path for Segment configuration. Promotion reverts to manual UI configuration in each workspace — removing version control, auditability and repeatability, and placing the solution outside SC-1. The fallback is a GitHub Actions pipeline, which would require its own Framework registration and assessment, adding schedule and governance overhead
Git Hub repository setup is automated process as part of Terraform RIO pipeline.	If RIO pipeline capability cannot be met, we will need to request GIT hub team to setup repositories manually to meet requirement to

CLMNBA-5913: [FR] CDP Implementation | Git-Based Configuration Storage & Version Control READY FOR CONFIRMATION	sync configuration information into Github repositories.
Zero-rating approval and configuration. CLMNBA-7763: [FR] Digital to CDP | Vodafone.com.au - Zero-Rating Validation READY FOR CONFIRMATION	Billing Strategy and Design and Packet Core Network must approve and configure the Vodafone CloudFront endpoints. Without completion, Segment traffic cannot be treated as zero-rated and must fail gracefully without affecting Digital journeys.

Table 13: Solution Dependencies

System Register and CMDB

Required for ARB presentation

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Required for both SO and SA

Use the below information to classify the current solution SBBs in the system register updates table:

Attribute	How to populate this field?	Acceptable Values
Service Criticality	Derived from BIL questionnaire	Business Critical
SOCI Critical System	**For Existing systems**, this can be referred to from system register. **For new systems**, use the tool TPG SOCI critical assets test .xlsx to identify if your system falls under SOCI Critical asset.	Yes No
Data Sensitivity	Based on definition from system register.	

Data Classification	Based on Information Classification and Handling Policy Section 3B)	Public Confidential
Hosting Platform	Based on Solution	SaaS, TPG AWS Cloud
System Register Update Required?	Based on Solution	Yes (or) No

Service Criticality	Business Critical
SOCI Critical System (Yes/No)	No
Data Sensitivity	PII Sensitive
Data Classification	Confidential
Hosting Type	SaaS
System Register Updates required?	No

System Register Updates

System Register Update

Required for ARB presentation

*Every project is required to ensure that the **existing systems** in scope of their project need to have accurate data in the system register and CMDB. Consequently, the below two are the preconditions for ARB approval of this solution:*

1. ***Systems Register Accuracy***

All existing systems impacted by the solution must have correct and up-to-date accurate data entries in the Systems Register. This includes metadata, ownership, lifecycle status, and other mandatory attributes.

2. ***Workload and resource (subnet, hosts etc..,.) Mapping***

All existing systems within the scope of the solution must have their workloads and hosts created and related to their systems in the CMDB. This is essential for infrastructure visibility and effective risk management.

The process for managing the CMDB data along with the respective SSPs is described here. Helix CMDB Data Management Process Guide

.	*Total # Of Systems*	*Total # Of Workloads*
New		*1*
Existing	*15*	*3*
Retiring		

Confirm the followings:	***Yes***	***No***
I confirm all system mandatory attributes have been revised and updated	*Yes*	
I confirm all workloads mandatory attributes have been revised and updated	*Yes*	
I confirm all system and workload attributes are accurate and up to date.	*Yes*	

The dashboard to check compliance for the systems is available here.

NOTE:

1. *Every project needs to ensure that the above two preconditions are closed for full ARB approval.*
2. *For **New Systems,** this data will be available with the project/architect and it will be captured as part of SSP request to create system/workload. The new system should be in the planning stage. Upon ARB Approval the system shoould be moved to Design stage. This data will be available with the project/architect and it will be captured as part of SSP request to create system/workload.*

Component Overview

This section should capture all the solution building blocks impacted with this solution.

Required for ARB presentation

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The following components perform key roles in the delivery of the end to end solution. For Solution Outline it's a high-level overview and should not be taken as an exhaustive list of impacted components or the impacts on them. For Solution architecture however, a full list of all impacted components along with the impact should be included.

The proposed change to each Solution Building Block (SYSTEM) is categorized as either New, Modified, Decommissioned or Unchanged.

Where relevant the solution components must incorporate Service Security, Service Management and Legal and Regulatory requirements.

Use names as per the system register. Ensure that the solution follows the Architecture Convention Standard.

Application Objects (Application Systems, EDM Workloads)

System(s)

Add/Modify/Delete	System ID Example: SYS000000000309	System Name	System Suite (If Applicable)	Contribution / Impact description including realised capability
Add	SYS2620	Twilio Segment		• Enables Profile Sync for unified customer profiles; real-time/near real-time updates; Identity Resolution (hierarchy, rules); provides persistent Data Graph for accurate, scalable identity stitching across systems. • Provides identity stitching across channels via Query

				String API, uses `ajs_uid` for deterministic matching; retains anonymous events for attribution. • Central platform for identity resolution and profile management across omni-channel touchpoints. Maintains **persistent data graph**, executes identity stitching, computes and governs derived traits (25–40), manages profile sync to downstream platforms (incl. Pega CDH). Supports audience building for real-time activation. • Configures and manages curated behavioral audiences (approx. 25–50) derived from unified profiles, events, and computed

				traits. Provides an efficient behavioral context layer for CDH via an integrated audience/traits interface for Next-Best-Action decisioning (NBA). Supports governance, refresh, and activation pipelines. • Manages real-time **schema validation**, enforcement, and transformation of event data ingested from multiple source systems (AWS, Digital MS, Digital FE). Enables source-specific **tracking plans**, auto-validates events (required fields, data types), blocks invalid events, and applies data cleaning/standardisation before CDP ingestion.

				• Provides automated, auditable workflows for **end-user deletion** (Right to Erasure) and **data suppression** within Segment workspaces. Blocks ongoing data collection for suppressed users and maintains compliance with enterprise privacy policies. Manages identity-based deletion and suppression across all connected sources and destinations. • Enforces **workspace-level data retention policy (30-day limit)** within Twilio Segment-managed storage for event data, profile data, object data, and audit logs. Automatically deletes expired data to ensure compliance with

				privacy and regulatory requirements. • Enforces automated **PII detection and classification** leveraging Segment's **Privacy Portal** across all data ingested. Uses **configurable and out-of-box matchers** for key-based and value-based scanning. Dynamically categorises attributes/events into risk-based groups (e.g., restricted, sensitive) to maintain data visibility and compliance posture. • Implements **privacy enforcement at the ingestion layer**, blocking classified **restricted PII fields** from entering Segment-managed systems

				or propagating to downstream destinations. Integrates privacy classification policies from the **Privacy Portal** and automates governance without requiring manual rule duplication. • Provides **end-to-end visibility** into Segment data pipelines (Source → Protocols → Destination). Delivers real-time metrics, threshold-based monitoring, and actionable alerting to detect data ingestion or delivery issues impacting downstream systems like Pega CDH or MDS. • Configures **AWS (MDS)** as a governed Reverse ETL source to Segment CDP. Enables enrichment of Segment Profiles

				with curated warehouse data (identity attributes, sales, engagement history) for activation across downstream destinations. • **Near Real-Time Accessibility & Activation** ◦ Enable low-latency access via Segment Profile API for personalization and audience sync. ◦ Stream updates to downstream platforms (owned and paid) with current profile state. ◦ Segment to stream updates to Pega via Kafka • • **Behavior & Preference Enrichment:** Ensure unified profiles retain

				behavioral history during anonymous-to-known transitions and enrich them with both pre-identification actions (page views, clicks) and post-identification attributes (preferences, purchases). • **Reverse ETL Extraction and Synchronization:** Segment CDP will use Reverse ETL to extract data from the TPGT MDS (hosted within AWS Redshift) using: ◦ Parameterized SQL queries or materialized views for performance-optimized data access. ◦ Support for both Segment-managed tables (e.g., profiles, audiences) and custom user-defined tables.

				• **Segment Source API:** Segment CDP to setup source API to receive contact profile information. This will be used to create a profile within segment.
~~Modify~~	~~SYS00000000193 9~~	~~Pega Customer Decision Hub~~		• ~~Setup Kafka data set and Real Time data flow to ingest traits and audiences from Twilio Segment.~~ • ~~Define schema structure aligned with Segment profile traits and audience attributes.~~
Modify	`SYS00000000197 0`	Marketing Data Platform Redshift		• Create a new redshift cluster to sharing data to Segment and ingest data from Segment • Ingest data from Twilio segment using destination connection. • Create a share for non PII attribute for Twilio Segment consumption using ReverseETL.

				• Create a data share for PII attributes to be passed to Segment using HTTPs API function call. • Create S3 bucket to ingest raw data from segment. • Create S3 bucket under application logs for Segment logs which need to be forwarded to Splunk index as same Pega logs. • Update the existing pipeline to push logs into Splunk as per pattern outlined: Splunk & SIEM Integration • All MDS datasets within Illuminate — including campaign performance, customer engagement, and audience segmentation data — are tagged with appropriate **domain** and **sensitivity** levels.

				These tags support secure integration with **Pega** and **Braze**, ensuring only authorised marketing users access customer or campaign-related data containing PII. • Twilio Segment configuration will be managed as code and deployed via a GitHub Actions CI/CD pipeline, assessed against the TPG CI/CD Framework, to enable automated build, deployment and operational support processes. MDP changes include configuration of GitHub Actions workflows for plan and apply, management of source code repositories and release artefacts, and maintenance

				of operational documentation and support procedures. Ownership of application configuration, pipeline workflows and operational support remains with the MDP team, while GitHub platform administration and organisation-level controls are provided by the Cloud ID team, and cloud governance controls by the Public Cloud team.
Modify	`SYS2022`	Digital Microservices (DEEL MS, DEEL FE, DEEL CIAM, TFE API)		• New Apigee Proxy with DEEL to act as ReverseProxy to be used to call Segment from client side from Web and App. • Setup config using SSM in each of DEEL accounts i.e. CIAM, FE, MS to forward to Segment Track

				and Identify calls from DEEL-Apigee layer.
Add	`SYS2032`	Digital Apigee API Platform		• Digital Apigee API platform will provide the API proxy layer for Segment APIs, enabling TFE-API and Digital Microservices to securely access Segment services through the enterprise API gateway.
Modify	`SYS000000000302`	Auth0		• Provides the Event Bridge integration and event filtering required to publish selected Auth0 identity events. • Filtered events are processed by a Lambda function deployed within CIAM for onward integration with Segment.
Modify	`SYS000000000485`	vodafone.com.au		• Add Twilio Segment Javascript (`Analytics.js`) for client-side calls passed to Twilio Segment.

				• Integrate Segment Analytics.js for approved client-side interaction tracking. Host the version-pinned artefact under a Vodafone-controlled path for CSP and SRI compliance.
Modify	`SYS00000000033 3`	SHOP Cloudfront and WAF		• Route approved Segment runtime CDN and event-ingestion calls through validated Vodafone-controlled CloudFront proxy endpoints. Analytics failures must not affect Shop customer journeys.
Modify	`SYS00000000038 7`	MyVodafone (OneApp)		• Add Twilio Segment Javascript (`Analytics.js`) for client-side calls passed to Twilio Segment.
Modify	`SYS00000000038 9`	My Vodafone Android app		• Add Segment SDK for client-side event data collection and pass

				through to Twilio Segment.
Modify	`SYS000000000390`	My Vodafone iOS app		• Add Segment SDK for client-side event data collection and pass through to Twilio Segment.
Add	`SYS000000001207`	Apigee API Platform		• A proxy for Core API to call Twilio Segment HTTPs API function to pass new contact information. • A proxy for core API to call Twilio Segment HTTPs API function to delete contact details who have been inactive more than 13 months. • A proxy for core to call Twilio segment HTTPs API to download metric and other systems alert details that eventually pushed to Splunk.
Add	`SYS000000001901`	Core API Integration and Services		• New Kafka connect service to push HTTPs Api call for PII data/Non PII traits into Segment for

				new and updated contact information. • New Kafka connect service to push HTTPs Api call into Segment to purge customer contacts have been inactive for more than 13 months. • New Kafka connect Service to pull metric API call from segment into Marketing Data Platform S3 bucket for log events to be pushed to Splunk. • ~~New service to convert Segment Twilio Segment destination function call to Kafka topic for Pega CDH to consume.~~
Modify	`SYS00000000064 9`	Splunk Enterprise		• Ingest segment logs into existing Pega index. • Create new dashboards for Segment monitoring and alerting

Add	`SYS2690`	Azure Virtual desk		• Provides browser-based access to the Segment platform for L2 support teams, enabling authorised users to monitor, investigate, and troubleshoot customer data platform operations. Access is governed through enterprise authentication and role-based access controls to ensure compliance with security and data governance requirements. CLMNBA-5684: [FR] CDP Onboarding | Domain Whitelisting & Secure Remote Support Access (Drop 2 - R3.1) READY FOR CONFIRMATION
Modify	`SYS000000000842`	Entra Id		• Restrict access to the Segment web interface to authorised users accessing from TPG-managed laptops and Azure

				Virtual Desktop (AVD) environments only. Access controls are enforced through Entra ID Conditional Access policies, device compliance validation, and enterprise authentication to prevent access from unmanaged or non-corporate devices and ensure compliance with TPG security standards. CLMNBA-5684: [FR] CDP Onboarding | Domain Whitelisting & Secure Remote Support Access (Drop 2 - R3.1) READY FOR CONFIRMATION
Add	SYS2434	GitHub		• Central source code repository for managing Infrastructure as Code (Terraform/OpenTofu), Segment configuration artefacts, and deployment

scripts. Provides version control, auditability, and controlled promotion of changes through delivery environments.

CLMNBA-5913: [FR] CDP Implementation | Git-Based Configuration Storage & Version Control

READY FOR CONFIRMATION

- The GitHub component will be enhanced with GitHub Actions workflows to validate, review and deploy Segment SaaS configuration through the Segment Public API. The implementation requires repository and branch controls, approved runner capacity, environment-specific approval gates, secure API token management, controlled

				Terraform state storage, pipeline security scanning, deployment audit evidence and defined MDP operational ownership. The GitHub Actions runner will initiate outbound TLS connections to Segment SaaS; no inbound connectivity into TPG is required for this configuration-management flow. CLMNBA-5099: [FR] CDP Implementation | Infrastructure as Code (IaC) for Configuration Deployment (TBC) Deprioritised/Delayed
Add	`SYS000000000608`	BMC Helix Suite IT Service Management		1. Modify the existing BMC Helix Incident Management workload to receive correlated Segment CDP alerts from the Splunk Enterprise TSI / Alert Collector workload and automatically

				create, classify and route operational incidents using the agreed service, CI, priority and resolver-group mappings. 2. Integrates with Segment operational alerting to auto-create incidents for delivery failures and threshold breaches. [Optional] 3. Leverages Segment's Event Delivery and Delivery Overview features to trigger email-based alerts that initiate incident records and notify relevant L2 teams. [Optional] 4. Routed to the correct L2 support group based on predefined mappings for service and environment. CLMNBA-5805: [FR] CDP Onboarding | Operational Onboard

					ing READY FOR CONFIRMATION

EDM Workload(s)

Add/Modify/Delete	Workload ID Example: WLD1952	Workload Name	System ID this workload is related to	Target Hosting Platform (Ex: TPG AWS, GC, VVF etc..,.)	Contribution / Impact description including realised capability
		Please use the same object name as defined in the System Register			
Modify	`WLD5766`	Vodafone Segment	`SYS2620`	SaaS	CLMNBA-5106: [FR] CDP Implementation | Unify - Identity, Profile Sync & Data Graph - Configuration READY FOR CONFIRMATION • **Unified Customer Profile Creation & Identity Stitching:** Use Segment's Unify Identity Graph to connect identifiers

					(anonymousId, sessionId, userId, deviceId, UUID). ◦ Enable cross-device and cross-channel identity resolution. ◦ Merge anonymous and known profiles via alias and identify calls. ◦ Maintain real-time updates to profiles as data changes • **Identity Graph & Stitching:** Implement Segment's Identity Resolution engine to alias identifiers (e.g. anonymousId, userId, email, deviceId) and

					merge anonymous and known profiles across sessions and channels, supporting real-time stitching via query parameters (e.g. ?ajs_uid=). CLMNBA-6137: [FR] CDP Implementation | Unify - Profile Traits Configuration READY FOR CONFIRMATION • Handles **identity graph**, cross-channel ID stitching, and synchronisation of unified profiles in near real time to downstream platforms. Governs identity logic and profile persistence. • Executes **computed**

trait logic for enrichment: real-time audiences, event-based aggregations, lifecycle signals (tenure, recency), behaviour segmentation and propensity. Supports updates to Profile Sync for activation.

CLMNBA-6139: [FR] CDP Implementation | Protocols - Configuration

READY FOR CONFIRMATION

- Manages **tracking plan configuration** for source systems (AWS, Digital MS, Digital FE), defines event schemas, allowed properties, prohibited fields, and

					enforces naming standards with version control. • Executes **Protocols-based real-time validation** for all events, generates schema violation alerts, manages enforcement (pass/block/transform), and integrates with alerting tools (Helix). • Applies **rule-based cleanup and transformation** logic (through Protocols + Insert Functions), ensures format normalisation, default value insertion, and PII exclusion before CDP activation.

CLMNBA-5947: [FR] CDP Implementation | Privacy Enforcement - End-User Deletion & Suppression

READY FOR CONFIRMATION

- Executes user deletion requests across Segment CDP entities. Initiates deletion via **UI or API**, enforces workspace-scoped data purge, and integrates compliance audit logs.
- Ensures **real-time blocking** of new data ingestion for suppressed identities across all sources in the workspace. Maintains suppression state and audit

					logs for compliance. • Captures and stores details of all deletion and suppression requests, status, and downstream enforcement results, providing an auditable, exportable compliance record. CLMNBA-5105: [FR] CDP Implementation | Privacy Enforcement - Twilio Data Retention Period READY FOR CONFIRMATION • Applies and enforces a **workspace-configured 30-day retention policy**. Automates removal of aged-out **event and profile data**

from Twilio Segment infrastructure.

- Maintains **audit logs of retention settings**, records lifecycle changes, and provides evidence for privacy and regulatory reporting needs.

CLMNBA-5948: [FR] CDP Implementation | Privacy Enforcement - PII Detection & Classification

READY FOR CONFIRMATION

- Scans incoming profile and event data using configurable matchers for **PII detection**, applies risk classification (restricted/sensitive), and updates

					Segment's dynamic data inventory. • Maintains central repository of detected PII attributes, applies governance policies for sensitive data, and provides **auditable reports for compliance reviews**. CLMNBA-5949: [FR] CDP Implementation | Privacy Enforcement - PII Controls, Blocking & Alerts READY FOR CONFIRMATION • Enforces privacy policies from Privacy Portal to **block restricted PII fields at the source ingestion point**, ensuring restricted data

					does not enter CDP. • **Detects new/unclassified fields in real-time** and sends alerts to Privacy Officers and Segment Admins. Supports integration with collaboration tools (Slack) or routing to enterprise monitoring solutions. CLMNBA-6140: [FR] CDP Implementation | Data Delivery Observability & Alerting - Configuration READY FOR CONFIRMATION • Tracks **pipeline health metrics** across all sources/destinations: volumes, drops, retries,

					Protocols failures, delivery lag. Enables diagnosis (schema failures vs delivery issues). • Applies **configurable monitoring rules** per source/destination for failure rate, drops, latency, and retry backlogs. Supports dynamic threshold tuning by environment (Dev→Prod). • Generates alerts enriched with source/destination info, error type, impacted volumes, and remediation hints. This information should be available in the API download

that captures information for Splunk. (Monitor Your Twilio Segment Workspace Using Public API | Twilio)

CLMNBA-5100: [FR] CDP Implementation | Reverse-ETL Configuration

READY FOR CONFIRMATION

- Manages **source configuration for AWS (MDS)** as Reverse ETL source in Segment CDP. Ensures environment consistency, schema definition validation, and centralised governance (e.g., Confluence + Git).
- Executes governed

scheduled syncs (hourly) from warehouse-curated dbt/SQL models into Segment Profiles. Supports incremental sync, error handling, field mapping governance, and SLA-based retry workflows.

CLMNBA-5125: [FR] CDP to MDS | Redshift - Event Deduplication in Sync

READY FOR CONFIRMATION

- Implements **deduplication logic** for Segment CDP events before they are written into **Amazon Redshift**. Compares key attributes (timestamp +

user_id + event_name) to detect and discard duplicates without impacting valid event sequences. Reduces double-counting risks and unnecessary storage/processing costs.

CLMNBA-5130: [FR] CDP to MDS | Redshift - Segment Synchronization

READY FOR CONFIRMATION

- Manages **scheduled syncs** from Segment CDP into **AWS (MDS)** Redshift with configurable frequency (hourly minimum in Prod). Supports **Profiles Sync**

					hourly and Reverse ETL with **customisable intervals (≥15 mins)**. Enables **manual sync triggers** for operational flexibility. •
Add	`WLD7407`	Market Data Store Redshift CDP Cluster	`SYS000000001970`	TPG AWS	• Enable Twilio Segment reverseETL capability to enable non PII data transfer. Fixed redshift cluster is required by Segment to enable this feature. • Enable metrics for each sync to Redshift to be pushed application logging S3 bucket to be sent to splunk for reconciliation. • Enable observability, audit and

					monitoring logs for SIEM and Splunk as per similar setup for Redshift serverless. • Setup equivalent nonproduction environment based on number of workspaces setup for Twilio segment. • The Redshift cluster inherits LF-Tags from Lake Formation and applies **dynamic data masking** and **role-based controls** across all FDP and DDP layers. Within this workload, the **Identity** and **Segment Unify** datasets contain **PII** such as customer

					identifiers and segment attributes; these are explicitly managed under the `Sensitivity = PII` tag. This guarantees that sensitive data can only be accessed by users with approved PII privileges while maintaining analytics performance at scale.
Add	`WLD644`	Splunk Enterprise Alert Collector	`SYS000000000649`	Virtual	• Modify the existing Splunk Enterprise TSI / Alert Collector workload to ingest and correlate Segment CDP operational events, monitor agreed KPI thresholds,

					and automatically create incidents in BMC Helix using the existing Splunk-to-Helix integration.

Technology Objects (Network System)

Network System(s):

ADD/MODIFY/DELETE	Network System ID *(Ex: SYS2175)*	Network System Name *(Ex: GC60 Core Switch)*	Type of Network System *(Ex: Core Switch Nodes)*	Comments
		Please use the same object name as defined in the system register		

Data Assets

Name	Proposed Change	Sensitivity	Related System(s)	Contribution / Impact Description	Physical location where DATA is stored (data sovereignty)	Who has Access Access Controls?	Privileged/Admin Access path

Non-PII Traits	New	Non PII	Target Data Platform Marketing Data Store (MDS)	New Share and Interface	AWS Australia (TPG VPC)	TCS, TPG internal support staff Authentication via AD	There are 2 privileged accounts setup as part of illuminate program 1. AWS power user - part of ssp with short duration (8 hours) 2. redshift admin user - part of ssp with short duration (24 hours)
Identity PII	New	PII	Target Data Platform Marketing Data Store (MDS)	New Share and Interface	AWS Australia (TPG VPC)	TCS, TPG internal support staff Authentication via AD	There are 2 privileged accounts setup as part of illuminate program

							1. AWS power user - part of ssp with short duration (8 hours) 2. redshift admin user - part of ssp with short duration (24 hours)
Purge Known identity older than 13 months	New	Non PII	Target Data Platform Marketing Data Store (MDS)	New Share and Interface	AWS Australia (TPG VPC)	TCS, TPG internal support staff Authentication via AD	There are 2 privileged accounts setup as part of illuminate program 1. AWS power user - part of ssp with short duratio

							n (8 hours) 2. redshift admin user - part of ssp with short duration (24 hours)
Segment Event Data	New	PII	Target Data Platform Marketing Data Store (MDS)	New Data feed from Twilio Segment	AWS Australia (TPG VPC)	TCS, TPG internal support staff Authentication via AD	There are 2 privileged accounts setup as part of illuminate program 1. AWS power user - part of ssp with short duration (8 hours) 2. redshift admin user - part of ssp with

							short duration (24 hours)
Segment Raw events to S3	New	PII	Consumer Data Store (CDS)	New Data feed from Twilio Segment	AWS Australia (TPG VPC)	TCS, TPG internal support staff Authentication via AD	There are 2 privileged accounts setup as part of illuminate program 1. AWS power user - part of ssp with short duration (8 hours) 2. redshift admin user - part of ssp with short duration (24 hours)

Table 15: Data Component Contribution / Impact

Contact Identifiers - HTTP API Source

Contact Identifiers relates to profile traits from a redshift cluster schema. This will include keys and other traits that will be useful for segment to stitch up identity.

The primary data that is being shared would look like below but not limited too

- `Contact_Id: 1-75308ZF`
- `Cust_Row_Id: 1-75308W1`
- `Dgtl_UUID_Nm: 8d125cfd-b72d-4c20-9252-5a420097d3b4`
- `Email_Addr_Hash_Val: 15bdc24cd659e8fd8b17a3d24bdb36dbe231bdb81631cc232ca80ff06c10e730`
- `MSISDN(61 format): 61400000000`

Start with a refresh period for this is every 60 mins and include only changed records to pull into Segment. Period should be configurable. The final listed is being addressed as part of illuminate delivery.

CLMNBA-5145: [FR] MDS to CDP | Customer Profile Synchronisation READY FOR CONFIRMATION

Purge Identity - HTTP API

The inactive customer past 13 months needs to be purged from segment platform. The data share from MDS will derive the `Contact_Id` or `Cust_Row_Id` that are inactive post 13 months. The purge should be just for identity traits and keep it as anonymous.

Use the micro service to do this call once a week to DELETE_INTERNAL command.

Details in documentation link: User Deletion and Suppression

CLMNBA-7315: [FR] CDP to MDS | De-Identify Inactive Customer profile Data

READY FOR CONFIRMATION

Offline Traits

CLMNBA-6137: [FR] CDP Implementation | Unify - Profile Traits Configuration

READY FOR CONFIRMATION

Offline traits to be passed to segment would be attributes like `Acct_Type_Cd, Contract_Type, Pega Interaction History latest, Latest Order interaction (this is near real-time) etc.`

Warehouse Destination copy from Segment

Events are collected in Segment and copied over to Redshift cluster in a modelled format for easy consumption.

CLMNBA-5119: [FR] CDP to MDS | Redshift - Structured Event Data Delivery

READY FOR CONFIRMATION

For Profiles Sync, navigate to Unify > Profiles Sync and select Redshift as your warehouse. Twilio Segment This is a separate sync path from the standard Connections warehouse sync, and exports unified customer profile data built from identity resolution across sources.

CLMNBA-5120: [FR] CDP to MDS | Redshift - Profile Sync Data Storage READY FOR CONFIRMATION

These map to tables created in your Redshift schema:

- **Track calls →** `tracks` **table + individual event tables** Each Track call is stored as a distinct row in a single Redshift table called `tracks`. Additionally, each unique `.track()` event creates its own dedicated table Twilio Segment, with each property becoming a column.
- **Identify calls →** `identifies` **table** The `identifies` table stores Identify method calls — you can query it to find out user-level information. Twilio Segment A materialized `users` table is also built from these, representing the latest state per user.
- **Page calls →** `pages` **table** The `pages` table stores Page method calls. Query it to find out information about page views. Twilio Segment
- **Screen calls →** `screens` **table** The `screens` table stores Screen method calls — used for mobile app screen views. Twilio Segment
- **Group calls →** `groups` **table +** `accounts` **table** Group calls generate the `accounts` table in Warehouses — used to query group-level information like company or organisation traits. Twilio Segment
- **Alias calls** Used to associate one user identity with another (e.g. anonymous → identified user on sign-up).

Engage Data (Audiences, Computed Traits, Journeys)

With Engage, you can send Computed Traits and Audiences to Redshift. Segment offers two ways: Storage Actions Destinations (recommended) for Linked Audiences and Event-Triggered Journeys, and Classic Storage Destinations for Computed Traits and Profile Audiences. Twilio Segment - engage warehouses

Specifically:

- **Audiences** — synced as `identify` or `track` calls; audience membership is written as a boolean trait (true/false as users enter/exit)

- **Computed Traits** — customer-level attributes computed from event history (e.g. "total orders last 30 days")
- **Linked Audiences** — entity-level data tied to the Data Graph
- **Event-Triggered Journeys** — journey state data written after each run

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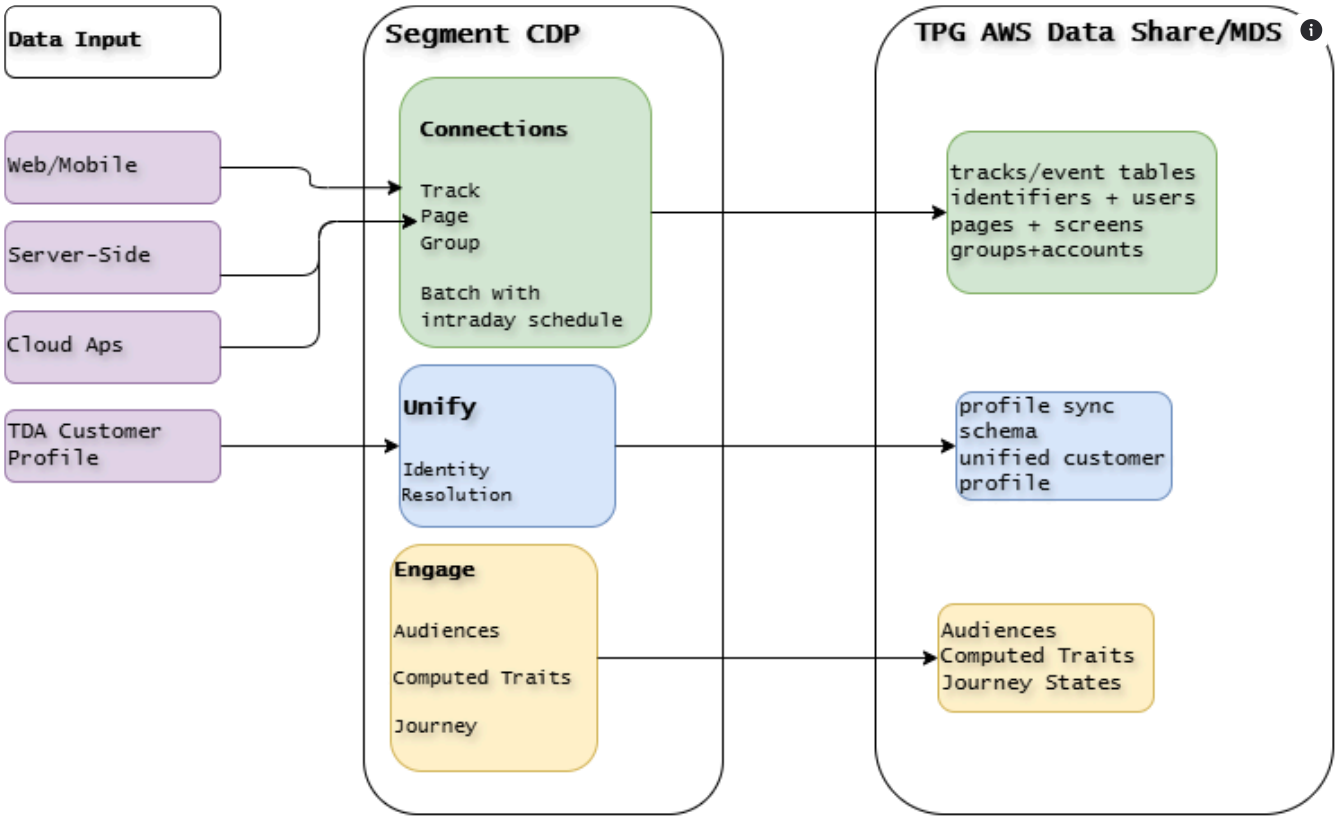


Figure 5: Data flow with segment

Data layer — CDP Redshift Cluster (the "data share")

Multi-AZ, private subnets only, KMS CMK encryption at rest, TLS in transit, dedicated Redshift database IAM role per environment and integration, with schema-scoped privileges., audit logging on.

Schema	Contents	Consumer
`pii_identify_share`	email, contact number, MSISDN, SHA-256	Poller Lambda

	hashed identifiers (email), plus a change log / `updated_at` watermark column. No clear-text PII.	
`nonpii_traits_share`	Non-PII trait model views — tenure, plan family, propensity scores and other similar data features.	Segment Reverse ETL
`contact_delete_share`	Source system identifies like contact Siebel row id (or an equivalent) that is used to delete records from Segment if the customer has been inactive for more than 13 months.	Poller Lambda
`segment_export`	Landing schema written by the Segment Warehouse destination — `identifies`, `tracks`, `pages`, `screens`, `users`, `accounts`, Unify profile traits and Profiles Sync tables.	MDS load-back
`__segment_reverse_etl`	Segment-managed sync-state schema. Reverse ETL's DB user needs write access to this schema only.	Segment (internal)

Redshift MDS — system of record and analytics

MDS is both the **upstream source** of customer traits and the **downstream sink** for Segment output.

Data flow into the CDP Redshift Cluster:
`Redshift MDS → UNLOAD to S3 using SSE-KMS → AWS Glue/DMS transformation, including hashing, masking and share-schema preparation → COPY/load` into the `pii_identify_share` and `nonpii_traits_share` schemas within the CDP Redshift Cluster.

The process is orchestrated through AWS Step Functions. Purge gates ensure that staged files are deleted after successful validation and loading.

Data flow back from Segment:
Segment writes Unify profiles, traits and event data into the `segment_export` schema within the CDP Redshift Cluster. The data is then staged in S3 and loaded into Redshift MDS using COPY, where it is integrated with the MDS data model for analytics, reporting and downstream consumption.

Supporting components: **S3 Gateway VPC Endpoint** (private S3 access), **KMS CMKs** per data store, and lifecycle rules on the staging bucket.

Data Ownership

Ownership matrix within the Data Assets.

Data Class	System of Record
Customer master data	MDS
Consent data	MDS
Identity stitching	Segment
Behavioral profiles	Segment
Audiences	Segment
Decisioning outcomes	Pega
Reporting data	MDS

Data Refresh Batch vs Real-Time

Illuminate initial phase MVP attributes to be provided in batch. Depending on the frequency change from illuminate program for DDP's, the transfer method to Segment can be modified to

run more frequently.

Data Class	Frequency
`pii_identify_share`	Uses Kafka connect. Frequency can be set based on data refresh frequency.
Consent data	MDS
Identity stitching	Segment
Behavioural profiles	Segment
Audiences	Segment
Decisioning outcomes	Pega
Reporting data	MDS

Data Governance and Tagging

All Segment-related data products, schemas, tables and attributes will be classified and tagged in accordance with the iLLuminate Data Classification and Tagging process and Metadata Management Framework. The approved `Source`, `Classification` and `AccessLevel` tags will be applied to supported platform objects, with PII classification and related governance metadata maintained at product, table and attribute level.

The CDP Redshift Cluster will be treated as a Highly Confidential, Highly Restricted environment because it contains PII-bearing identity, profile and event data. Lower-sensitivity schemas such as `nonpii_traits_share` will retain their specific classification and least-privilege access controls, but will not reduce the security classification applied to the overall cluster.

Where classification is unresolved, the highest classification will be applied until Data Governance confirms the correct treatment. Customer-linkable system identifiers, hashed identifiers, Segment Unify profiles, audience membership and behavioral event data will be treated as PII-bearing.

Possible design time data asset/schema classification; this is to be validated at design time.

Data Asset / Schema	Source	Domain	Domain Type	Classification	Access Level	PII	Required Controls

CDP Redshift Cluster	MDS / Segment	Marketing Data Store	Producer and Consumer	Highly Confidential	Highly Restricted	Yes	Treat as a mixed-classification environment containing PII. Apply private-subnet deployment, KMS encryption, TLS, IAM-based access, schema-level RBAC, audit logging and environment segregation.
`pii_identify_share`	MDS	Marketing Data Store	Producer	Highly Confidential	Highly Restricted	Yes	Restrict access to the dedicated Kafka Connect IAM role. Apply attribute-level PII tagging to all direct and linkable customer identifiers.
`nonpii_traits_share`	MDS	Marketing Data Store	Producer	Confidential	Restricted	No, subject to field validation	Provide read-only access to the Segment Reverse ETL IAM role. Prevent unapproved PII or identity attributes from being added without Data Governance review.
`__segment_reverse_etl`	Segment	Marketing Data Store	Consumer support schema	Confidential	Restricted	No expected PII	Restrict access to synchronisation-state processing only. Validate that customer payloads and direct identifiers

							are not stored in this schema.
`segment_export`	Segment	Marketing Data Store	Consumer	Highly Confidential	Highly Restricted	Yes	Restrict write access to the Segment Warehouse Destination IAM role and read access to approved MDS ingestion and validation roles. Apply attribute-level classification and masking.
Segment Identity Data Product	MDS	Marketing Data Store	Producer	Highly Confidential	Highly Restricted	Yes	Include only approved identity and linkage attributes. Apply data minimisation, API allow-listing, retention, deletion and field-level PII classification.
Segment Trait Data Product	MDS	Marketing Data Store	Producer	Confidential	Restricted	Conditional	Apply field-level classification. Customer-linkable identifiers or sensitive attributes must be escalated to PII and Highly Restricted treatment.

Segment Unify Profiles	Segment	Marketing Data Store	Consumer	Highly Confidential	Highly Restricted	Yes	Treat resolved profiles, identity links and associated traits as PII. Apply RBAC, masking, retention, deletion and lineage controls.
Segment Computed Traits	Segment	Marketing Data Store	Consumer	Highly Confidential where customer-linked	Highly Restricted	Yes where linkable	Treat customer-linked scores, propensities and inferred traits as PII-bearing. Classification may only be reduced where the data is demonstrably non-identifiable.
Segment Audience Membership	Segment	Marketing Data Store	Consumer	Highly Confidential	Highly Restricted	Yes	Restrict access because audience membership is linked to customer profiles and may contain inferred customer characteristics.
Segment Identity and Behavioural Events	Segment	Marketing Data Store	Consumer	Highly Confidential	Highly Restricted	Yes	Classify customer IDs, anonymous IDs, IP addresses, device context and customer-linked event data as PII-bearing. Apply retention, masking and deletion controls.

Segment Raw Events in S3	Segment	Marketing Data Store	Consumer	Highly Confidential	Highly Restricted	Yes	Apply SSE-KMS, restricted IAM access, environment-specific prefixes, lifecycle controls and deletion coverage. Do not classify raw events as Non-PII by default.
Deletion and Suppression Data	MDS / Segment	Marketing Data Store	Producer and Consumer	Highly Confidential	Highly Restricted	Yes	Limit content to required identifiers, request status and audit evidence. Restrict access and avoid retention of unnecessary customer attributes.
Aggregated Operational Metrics	Segment / TPG Integration	Technology Operations	Consumer	Confidential	Restricted	No	Include aggregated counts and technical metadata only. Exclude customer payloads, customer identifiers, tokens, credentials and Write Keys.

Failure Records and DLQ Data	MDS / Segment / Kafka Connect	Integration	Consumer	Highly Confidential	Highly Restricted	Potentially Yes	Segregate from standard telemetry. Restrict access, encrypt at rest and apply retention and purge controls because failed records may contain full customer payloads.

Proposed Tag Structure

Where supported by the platform, apply the following tags:

Tag Key	Permitted Values	Purpose
`Source`	`MDS`, `Segment`, or `approved originating system`	Identifies the authoritative source of the data
`Classification`	`Public`, `Confidential`, `Highly Confidential`	Identifies the information-classification level
`AccessLevel`	`General`, `Restricted`, `Highly Restricted`	Determines the applicable access-control tier

The iLLuminate tagging process uses these tags to classify data and support IAM-based access enforcement.

Twilio Segment - Raw Events into S3

CLMNBA-5129: [FR] CDP to MDS | S3 - Segment to S3 Synchronization READY FOR CONFIRMATION

CLMNBA-5114: [FR] CDP to MDS | S3 - Raw Event Data Export to in NDJSON Format READY FOR CONFIRMATION

CLMNBA-5115: [FR] CDP to MDS | S3 - File Partitioning by Timestamp, Source, and Environment READY FOR CONFIRMATION

The Segment Tracking API processes data from your sources and collects events in batches. Segment then uploads the batches to a secure Segment S3 bucket, from which they are

securely copied to your own S3 bucket in small bursts. Individual files will not exceed 100MB in size.

PII Security Controls

Is this initiative modifying the access to, creation of, introduction of, handling, transfer, storage, masking, or purging, of any type of PII?

If not, you can skip this section.

Reference Table:

This table is taken from the data security standard that is published on the central policy hub. Please refer to the standard for detailed information.

PII Type	Data Type	Treatment & Exceptions	Retention
Type 1 PII	Examples: Name, E-Mail address, contact number, physical address. Customer service interaction data Government ID, and other sensitive identifying data Finance Sensitive information	Summary: Access Control (prod and non-prod), Mask in non-prod, Encrypt (at rest and in transit), Purging.	Refer to Document Retention and Destruction Manual
Type 2 PII	Example: Employee login credentials	Summary: Access Control (Prod and non-prod), Purged as part of Identity & Access Management.	Refer to Document Retention and Destruction Manual
Type 3 PII	Including, but not limited to, MSISDN, Fixed Number, IMEI (Device Identifier), IMSI (SIM Identity),	Type 3 Only (e.g Mobile Core): Access Control (prod and non-prod), Purging.	Refer to Document Retention and Destruction Manual

	Network CDRs, IP Address (Including time & date of use).	Type 3 & Type 1: Access Control (prod and non-prod), Mask in non-prod, Encrypt (at rest and in transit), Purging.	

System Name	Proposed Change	Type of PII (see Data Security Standard)	Change / Impact Description	Proposed Security Controls	Describe Access & Access Restriction (Design Details)	Describe Encryption (Design Details)	Describe Masking (Design Details)	Describe Purging (Design Details)
Twilio Segment	*PII data transfer*	*Type 1 - Email Address, Customer Name (First and Last), Contact Number,* *Type 2 - Employee Email address from Azure AD.* *Type 3 - Service*	*Marketing Data Store will create new workload will provide contact (or party) data.* *New workload for Redshift cluster created*	*API is over HTTPS/ TLS. TLS1.2 is supported.* *The workload authenticates to Apigee using OAuth 2.0; Apigee retrieves Segment*	*TCS, TPG internal support staff* *Remote access via AVD* *Authentication via AD*	*Encryption in transit & at rest*	*Production PII is irreversibly masked before transfer to non-production using metadata-driven AWS Glue rules. Masking methods include SHA-2*	*Contact information that has remained inactive for more than 13 months is identified through scheduled polling. A deletion request*

		Number	*to provide PII data and Non PII data.* *PII data will be pulled from Redshift cluster and pushed to segment using Kafka connect/ Apigee and HTTPs API.* *Non PII data will be pulled into Segment using direct connection to CDP Redshift cluster.*	*writekey from encrypted KVM, Apigee passes the write key to the segment tracking api.* *Viewing of PII data is limited to approved Azure AD groups. TPG AWS MDS access is controlled through RBAC, ensuring permissions are granted based on assigne*			*hashing, fixed-value substitution and format-preserving email pseudonymisation. Amazon Macie validates the output, quarantines files containing residual PII, and permits only cleansed data to proceed to non-production.* Data Masking Detail Design	*is then initiated through the Segment API to permanently purge the information from Segment.*

			Each share within the Redshift cluster will have separate user access setup.	*d roles and responsibilities.* *Data is encrypted at rest using AES-256* *Backup is encrypted in transit and at rest using volume level AES-256 stored in EU availability zones* *Segment runs on AWS and security program is governed under* ***ISO/IEC 27001****, with ISO*				

				27017/27018, SOC 2 Type II and PCI DSS attestations available via the Twilio Trust Centre. [twilio.com], [security.twilio.com] *Data ingested into Segment is from non production data layer.* *Inactive customer contact information will be purged from Segment after 13-*			

				month period.				

Controls on SBBs handling PII

Solution Interactions

Required for ARB presentation

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Table 14: Active Component Contribution / Impact

	Interaction Method (Ex: API, HTTPS, MFT etc..,.)	**From Workload**	**To Workload**	**Relationship type (Ex: Data Sharing, Orchestration etc..,.)**	**Provider (which workload)**	**Contribution / Impact Description including impacted realised capability**	**Version**
New	API	MDS	Twilio Segment (Profile updates or source connections)	Data Sharing	API Microservice/Apigee	Orchestration of new capability	1
New	API	MDS	Twilio Segment (Identity deletes)	Data Sharing	API Microservice/Apigee	Orchestration of new capability	1
New	API	MDS	Twilio Segment (Customer Traits non PII)	Data Sharing	API Microservice/Apigee	Orchestration of new capability	1

New	AWS privatelink	MDS	Twilio Segment - Offline Traits	Data Sharing	AWS Workload	Access to Redshift cluster	1
New	AWS Privatelink	Twilio Segment	MDS (Or illuminate)	Data Sharing	AWS Workload	Access to Redshift cluster	1
New	AWS IAM Role	Twilio Segment	S3 Consumer Domain	Data Sharing	AWS Workload	Access to S3 bucket is through IAM role.	1
New	Segment SDK	MyVF iOS Journeys	Twilio Segment	Data Sharing	Reverse Proxy AWS Workload	New client-side integration with Segment	1
New	Segment SDK	MyVF Android Journeys	Twilio Segment	Data Sharing	Reverse Proxy AWS Workload	New client-side integration with Segment	1
New	Javascript `<Analytics.js>`	Vodafone Web, MyVF Web	Twilio Segment	Data Sharing	Reverse Proxy AWS Workload	New client-side integration with Segment	1
New	API	AuthO	Twilio Segment	Data Sharing	DEEL Microservice Apigee	New Server-Side integratio	

						n with segment	
New	API	Vodafone.com.au Shop Journeys, VCA Journeys, Salesforce form	Twilio Segment	Data Sharing	DEEL Microservice Apigee	New Server-Side integration with segment	
New	API	MyVF Web/iOS/ Android Journeys	Twilio Segment	Data Sharing	DEEL Microservice Apigee	New Server-Side integration with segment	
~~New~~	~~API/Kafka~~	~~Twilio Segment~~	~~Pega CDH~~	~~Data Sharing~~	~~API Microservice~~	~~Microservice to forward Segment Function API call to Kafka topic for Pega CDH to consume.~~	

Component Interactions

Twilio Segment - Target Data Platform

This interaction follows two patterns

- PII data transferred using API to call Segment function

- Privatelink connection to transfer non PII traits to Segment and Segment destination to data platform

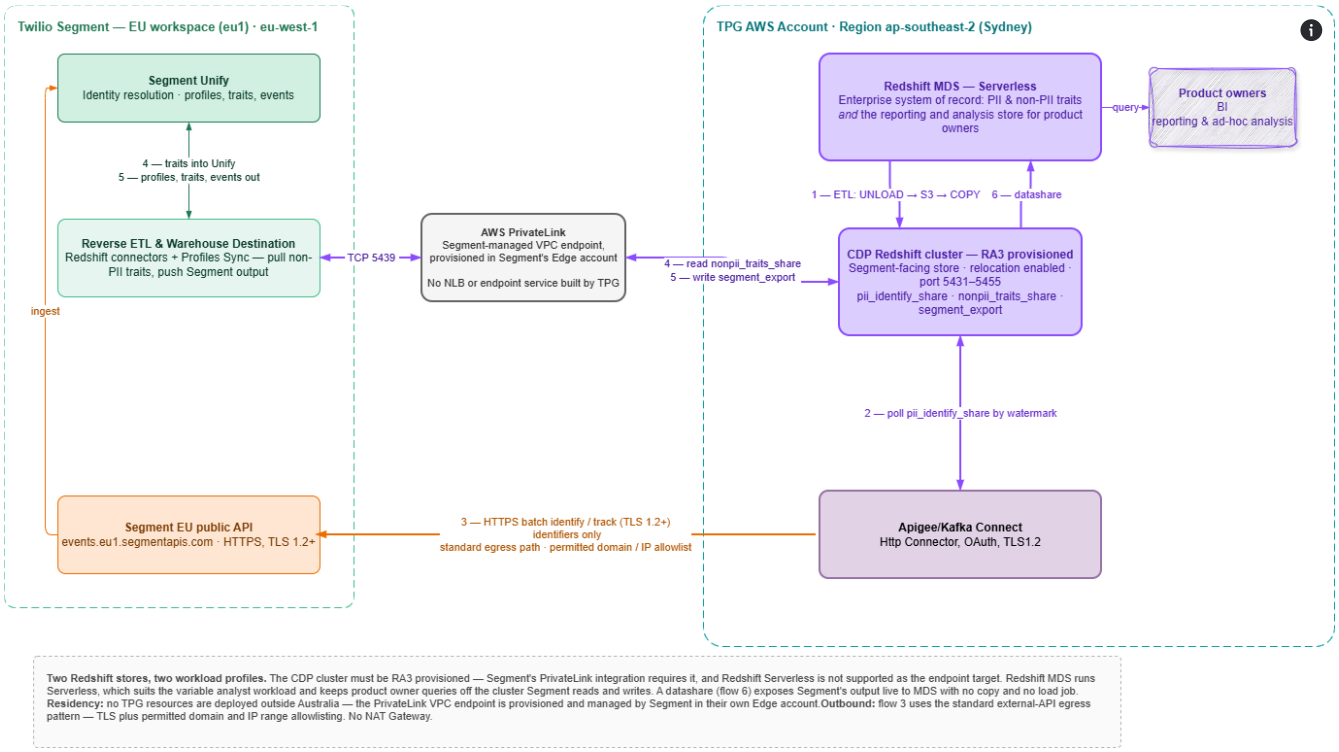


Figure 5.2: Twilio Segment to TPG AWS

Redshift MDS to Redshift fixed cluster data share

CLMNBA-5145: [FR] MDS to CDP | Customer Profile Synchronisation READY FOR CONFIRMATION

CLMNBA-6137: [FR] CDP Implementation | Unify - Profile Traits Configuration

READY FOR CONFIRMATION

Redshift MDS is the system of record for both PII and non-PII traits, and it is the upstream source for everything Segment eventually receives. The interaction is initiated inside the AWS account by a scheduled Step Functions execution, and it is entirely within ap-southeast-2 — no external party participates. Any hashed PII data will follow the same SHA 256 pattern used in TPG applications.

Instead of using exporting and importing data from MDS to the new CDP Redshift cluster, publish a data share from MDS that the CDP Redshift cluster consumes. Then build the share schema within inside the CDP Redshift cluster. If data share between Redshift components is not a preferred path, glue/MWAA/dbt will have to be used to move data from MDS to the CDP Redshift cluster.

An additional watermark (load date time as an example) needs to be included in the published data to CDP Redshift cluster that would be used by the kafka connect service to identify records to be processed for the identify calls to Twilio Segment.

For CDP Redshift Cluster, **Redshift Graviton** should be considered as it provides cost benefits compared to other types of Redshift cluster.

Kafka Connect reading the PII identify share (Source poll → HTTP Sink push)

This solution implements a scheduled polling integration where a Kafka Connect JDBC Source Connector queries the PII Identify data-share layer (CDP Redshift Cluster, populated from Redshift MDS via `UNLOAD` / `COPY`) for new or changed records, identified using the `updated_at` watermark / change-log column on the `pii_identify_share` schema. Each polling cycle retrieves only incremental deltas — identifiers such as `Email`, `Email_Addr_Hash_Val` and `MSISDN` plus associated non-clear-text attributes — and publishes them as JSON messages to an internal Kafka topic. A companion HTTP Sink Connector task then consumes each message and performs an authenticated **HTTPS POST** call to the Segment HTTP API `identify` endpoint, submitting the payload as the record body. Failed deliveries are retried with exponential backoff and, once retries are exhausted, routed to a Dead Letter Queue topic for investigation.

Segment Reverse ETL reading non-PII traits

CLMNBA-5100: [FR] CDP Implementation | Reverse-ETL Configuration READY FOR CONFIRMATION

Twilio Segment initiates the Reverse ETL connection to the CDP Redshift Cluster through the approved AWS PrivateLink path. The connection uses the Redshift-compatible database protocol with TLS encryption and authenticates using a dedicated IAM role for each environment.

Reverse ETL reads incremental records from the `nonpii_traits_share` schema using an `updated_at`, sequence or equivalent watermark. Segment maintains synchronisation state in the `__segment_reverse_etl` schema. The database role has read-only access to `nonpii_traits_share`, minimum required write access to `__segment_reverse_etl`, and no access to `pii_identify_share`, `segment_export` or administrative functions.

Synchronisation is scheduled to avoid overlap with the MDS data-load process. Monitoring captures execution status, record counts, watermark progression, authentication failures and abnormal access frequency or source-path deviations. Final PrivateLink connectivity, Redshift port, IAM role setup and credential-rotation arrangements must be confirmed during detailed design.

> *If there is a delay in reverseETL setup, use kafka connect to pass information to Segment API function.*

Segment Warehouse destination and profile sync to CDP Redshift Cluster

This interaction is also initiated by Segment and travels the same PrivateLink path in the opposite data direction. This is Segment-initiated, running between one and twelve times a day depending on the workspace plan.

The content delivered is the full picture of what Segment holds: Unify profile traits and the resolved identity graph via Profiles Sync, together with the `identifies`, `tracks`, `pages`, `screens`, per-event tables and the `users` and `accounts` collections. Sync cadence is governed by the workspace plan, typically between one and twelve times per day.

On the Redshift cluster set a specific schema like `segment_export` to store this information and ensure that database role used her holds write permission to only this schema.

CDP Redshift cluster → Redshift MDS

The final interaction closes the loop. There are couple of ways to copy the data to MDS listed in order of preference.

Option 1: UNLOAD → S3 → auto-copy job. The producer runs `UNLOAD ... FORMAT PARQUET PARTITION BY (dt)` into an S3 prefix; MDS has a `COPY JOB` configured against that prefix and ingests new files automatically as they land. No orchestrator between the two ends at all. This is the least moving parts of anything available to you.

Option 2: UNLOAD → S3 → COPY driven by Step Functions and the Redshift Data API. Same mechanism, explicit control. Worth the extra piece if you need the load sequenced against Segment's sync windows or gated on a validation step — which, given you're already sequencing around Reverse ETL, you probably do.

Option 3. External table, then `INSERT INTO ... SELECT`. UNLOAD to S3, register it in the Glue Data Catalog, and have MDS read it as an external table — either materialising into a local table or just querying it in place. This is the same plumbing as your archive tier, so it's less a separate option than a variation.

The data needs to be retained for 5 years and purged after that.

A proposed tier of storage

1. `CDP Redshift Cluster: 14 days landing only`
2. `MDS Redshift Serverless: Upto 25 Months.`

3. `S3 Parquet, Glue Data Catalog, Queries via spectrum: The balance upto 5 years.`

Twilio Segment - Target Data Platform - Raw Events into S3.

CLMNBA-5129: [FR] CDP to MDS | S3 - Segment to S3 Synchronization READY FOR CONFIRMATION

CLMNBA-5114: [FR] CDP to MDS | S3 - Raw Event Data Export to in NDJSON Format READY FOR CONFIRMATION

CLMNBA-5115: [FR] CDP to MDS | S3 - File Partitioning by Timestamp, Source, and Environment READY FOR CONFIRMATION

The Segment Tracking API processes data from your sources and collects events in batches. Segment then uploads the batches to a secure Segment S3 bucket, from which they are securely copied to your own S3 bucket in small bursts. Individual files will not exceed 100MB in size.

Segment only needs **write access** to your bucket — it never reads from it after writing. Setup it up as AWS S3 Actions.

To enable this the below steps, need to be done.

- Provision new S3 in Consumer workload. All external access to be blocked for the bucket, versioning to be enabled to detect for deletes, enable data encryption at rest.
- Create the IAM Role. Click the Require External ID checkbox and enter your Segment Workspace ID in the External ID field. If using SSE-KMS add the policy to the role.
- Update AWS IAM trust relationship to include the segment workspace id.
- Connect in Segment App and do the steps: Connections > Destinations > Storage > + Add storage destination > AWS S3 with IAM Role Support. Enter AWS Region, bucket name (not URI), and the IAM Role ARN. Enable and verify — data will appear approximately 1 hour after enabling.

One IAM role per brand workspace — each with its own trust relationship scoped to that brand's Workspace ID as External ID, and its permissions policy scoped to that brand's S3 prefix:

```
Resource: "arn:aws:s3:::mds-cdp-raw-events/brand-a/segment-logs/*"
Resource: "arn:aws:s3:::mds-cdp-raw-events/brand-b/segment-logs/*"
Resource: "arn:aws:s3:::mds-cdp-raw-events/brand-c/segment-logs/*"
```

The use of this information is for detailed investigation and future use case of large ML model input. To enable this we would look at a strategy to store data for maximum period with less

storage cost and storage is in optimist format.

For Go Live

S3 Zone Structure Right from Day One

Create the full three-zone prefix structure in the bucket **even if Gold is empty**. This costs nothing and avoids a bucket restructure later.

```
mds-cdp-raw-events/
├── brand-{a,b,c}/segment-logs/        ← Segment writes here (Bronze)
├── brand-{a,b,c}/segment-parquet/     ← Glue ETL writes here (Silver)
└── brand-{a,b,c}/ml-features/         ← Empty now, Feature Store writes here later (Gold)
```

An empty prefix in S3 costs $0. Restructuring a year of Parquet data costs significant Glue compute and engineering time.

Get the Parquet Partitioning Right from Day One

This is the single most important decision to get right now. The Glue ETL job that converts raw JSON to Parquet must partition by the dimensions ML training jobs will need. If you partition incorrectly now, you cannot efficiently query historical data later without rewriting it.

Partition structure to implement now:

```
segment-parquet/
  event_type=track/
    brand=brand-a/
      year=2026/month=04/day=29/
        part-00001.parquet
```

The critical dimensions are `event_type`, `brand`, and date hierarchy (`year/month/day`). ML training queries will filter on all three — e.g. "give me all `track` events for Brand A between Jan 2025 and Dec 2025". Without this partitioning, Athena scans the entire dataset on every query, which is both slow and expensive at ML training scale.

Also include these columns in every Parquet file even if not used in reporting today:

Column	Why It Matters for ML Later
`user_id`	Identity anchor for joining features across events
`anonymous_id`	Pre-login behaviour linkage
`received_at`	Server-side timestamp — canonical time for point-in-time correct training
`timestamp`	Client-side event time

`event`	Track event name — used for label generation (e.g. `Order Completed` = positive label)
`context_ip`	Geolocation signal
`context_user_agent`	Device/platform signal
All custom properties	Do not drop any — ML models often use properties analytics ignores

The Glue ETL job today should be configured to **preserve all fields from the raw JSON**, not just the ones needed for current Redshift reporting. Fields dropped now cannot be recovered from Silver later — only from Bronze re-processing, which is expensive.

Register the Glue Data Catalog from Day One

Run a Glue Crawler on the Silver Parquet prefix and register all tables in the Glue Data Catalog now. This means:

- Schema is tracked and versioned from the start
- Athena can query Silver data immediately without setup
- SageMaker Feature Store can reference the same catalog when enabled later
- Schema evolution (new Segment event properties) is automatically detected by the crawler on a scheduled run

Enforce File Compaction in the Glue ETL Job Now

Segment writes many small files — a known ML performance killer. The Glue ETL job should compact output Parquet files to 128–512MB per partition from go-live. This cannot be easily fixed retroactively on a year of data without rewriting all files.

Enable S3 Versioning and Object Lock Decisions Now

Versioning must be decided at bucket creation — it cannot be retroactively applied to existing objects, only to new objects written after it is enabled. Since the bucket is being created this year, enable versioning now.

Object Lock (WORM mode) similarly must be decided at bucket creation. If there is any chance regulatory requirements will mandate immutability for audit data, enable it now at bucket level even if no retention rules are configured yet. You cannot enable Object Lock on an existing bucket.

Tag All Objects with Brand and Source Metadata

Configure the Glue ETL job to write S3 object tags on output Parquet files:

```
brand = brand-a
source_type = cdp_segment
data_classification = pii
environment = production
```

These tags cost nothing and enable future IAM attribute-based access control (ABAC), S3 Storage Lens cost attribution per brand, and SageMaker data channel filtering. Retrofitting tags onto a year of existing objects requires a bulk tagging job that touches every object individually.

Full Storage Tier Picture Across All Layers

CLMNBA-5135: [FR] CDP to MDS | S3 - Data Retention and Lifecycle Policies

READY FOR CONFIRMATION

Apart from regulatory and deep investigation use cases, at this stage we are looking at a storage policy that has least cost impact.

What This Means Day by Day

```
DAY 0          Glue ETL writes Parquet → S3 Standard
DAY 1 - 30     Lifecycle moves to Standard-IA (brief hop, low cost)
               Glue Crawler runs → Data Catalog updated
DAY 31-60     Glacier Instant Retrieval (millisecond access, ~80% cheaper than Standard)
DAY 60-180     Still Glacier Instant (same tier, 90-day minimum duration satisfied at day 90)
DAY 180+       Deep Archive (12-48hr retrieval, ~95% cheaper than Standard)
```

With this policy in place, storage architecture now looks like:

Layer	Prefix	Day 0–30	Day 31–180	Day 180+
Bronze	`segment-logs/`	Standard	Glacier Instant (from Day 30)	Deep Archive (from Day 730)
Silver	`segment-parquet/`	Standard	Glacier Instant (from Day 1)	Deep Archive (from Day 180)

Access to the S3 files would be using existing AWS tools allowed in the account like Athena, glue.

CLMNBA-5117: [FR] CDP to MDS | S3 - File Compatibility with AWS Analytics and Batch Tool

S READY FOR CONFIRMATION

Twilio Segment - Digital Components

Segment has two methods of data collection.

S. No	Client-Side Collection	Server-Side Collection
1	**Analytics.js (JavaScript SDK)** The primary web library. Deployed on your website, it captures browser-based interactions automatically — page views, clicks, form submissions, scroll depth, mouse movements, UTM parameters, cookies, device type, operating system, and returning visitor status. **Mobile SDKs** Client-side SDKs include easy-to-implement iOS, Android, and React Native SDKs that capture browser/device context automatically. Twilio Segment	**Backend Libraries** Server-side libraries are available in every major language — Node.js, Python, Ruby, Java, PHP, and Go — for reliable backend tracking. Twilio Segment **HTTP API / REST API** Segment exposes a direct HTTP API, so any server or system capable of making HTTP requests can send events — useful for custom integrations, microservices, or systems without a native SDK.
2	**What it's best for:** Good things to send from client-side are things you wouldn't usually store in your database — page views, button clicks, scroll length, mouse movements, social shares, UTM tags, OS/device type, and cookie-based data like returning visitors. Twilio Segment **Limitations:** Client-side data is fine for watching general trends, but it's never going to be perfect — especially if customers use ad blockers or non-standard browsers.	**What it's best for:** For business events such as purchases, subscriptions, and support tickets, server-side tracking ensures accuracy even if the user closes their browser or loses internet connection. It also protects sensitive data from client-side vulnerabilities. Server-side tracking tends to be more accurate than user devices since it's a more controlled environment. Good data to send server-side includes things that need to be calculated from a database query, and mission-critical events like payments where accuracy is essential. Twilio Segment

Hybrid approach: Requirements for the program need both client-side and server-side tracking. Segment makes hybrid implementations straightforward — capture UTM parameters and initial user behaviour client-side, send mission-critical events like purchases and subscriptions server-side, and automatically combine both into unified customer profiles.

The key design principle: use **client-side for behavioral/UX signals**, and **server-side for authoritative business events** where data integrity, security, and completeness matter most.

Client-Side Collection Pattern

CLMNBA-5162: [FR] Digital to CDP | Vodafone.com.au (Shop, VCA) - Data - Customer Navigation READY FOR CONFIRMATION

CLMNBA-6035: [FR] Digital to CDP | Vodafone.com.au (Shop, VCA) - Data - User Interactions READY FOR CONFIRMATION

CLMNBA-6034: [FR] Digital to CDP | Vodafone.com.au (VCA) - Data - EOI Lead Capture READY FOR CONFIRMATION

CLMNBA-6112: [FR] Digital to CDP | MyVodafone (Apps) - Data - Map Device IDs (from SDK) READY FOR CONFIRMATION

CLMNBA-6036: [FR] Digital to CDP | Vodafone.com.au (Shop) - Data - SQ Address Checker READY FOR CONFIRMATION

CLMNBA-6049: [FR] Digital to CDP | MyVodafone (Web) - Data - Customer Navigation READY FOR CONFIRMATION

CLMNBA-6119: [FR] Digital to CDP | MyVodafone (Apps) - Data - App (Lifecycle) Interactions READY FOR CONFIRMATION

CLMNBA-6050: [FR] Digital to CDP | MyVodafone (Apps) - Data - Customer Navigation READY FOR CONFIRMATION

CLMNBA-6111: [FR] Digital to CDP | MyVodafone (Web, Apps) - Data - Customer Interactions READY FOR CONFIRMATION

1. Client-Side Web (Browser – analytics.js) Flow.

CLMNBA-5159: [FR] Digital to CDP | MyVodafone (Web) - Data - Identity Graph READY FOR CONFIRMATION

CLMNBA-5160: [FR] Digital to CDP | MyVodafone (Apps) - Data - Identity Graph READY FOR CONFIRMATION

CLMNBA-5161: [FR] Digital to CDP | Vodafone.com.au (Shop) - Data - Identity Graph READY FOR CONFIRMATION

High-Level Steps:

1. **User Access:** Customer navigates Vodafone website (`vodafone.com.au` , `my.vodafone.com.au`).
2. **Script Delivery:** Vodafone Web/CDN delivers `analytics.js` (integrity-protected via **CSP + SRI**).

3. **Client Execution:** Browser executes `analytics.js` to initialise tracking.
4. **Data Capture:** Browser SDK collects user interactions (e.g., page views, product views, add-to-cart).
5. **Context Enrichment:** Adds metadata (anonymousId, device type, browser, URL, referrer).
6. **Event Transmission:** Events sent to **Segment Tracking API over HTTPS/TLS 1.2+**, authenticating via **Segment Source Write Key** (OAuth not used).
7. **API Response:** Segment validates, processes event, and returns **HTTP 200 OK**.

Key Controls:

- Secure delivery (CSP, SRI)
- HTTPS/TLS for all API calls
- Static **Segment Write Key** authentication

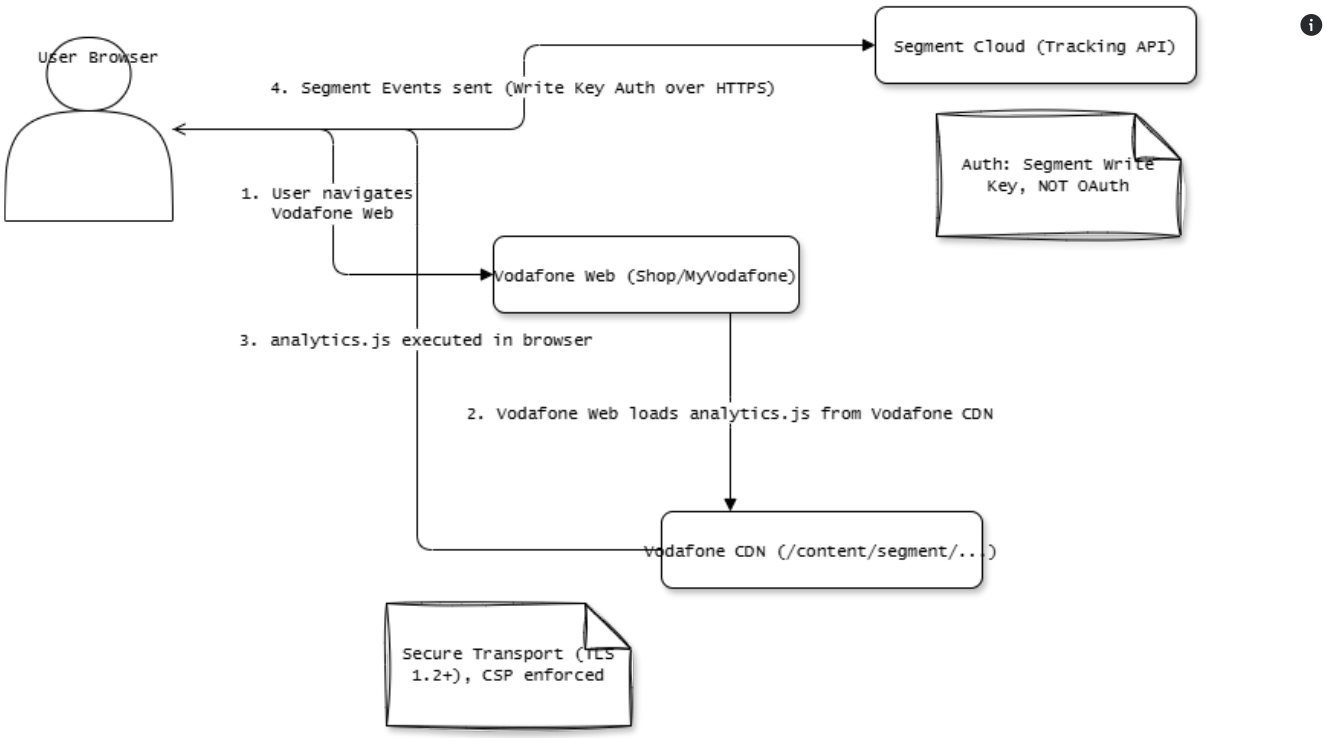


Figure 5.3a: Client-side analytics.js data collection for browser

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2. Client-Side Mobile SDK (iOS and Android) Flow

CLMNBA-5156: [FR] Digital to CDP | MyVodafone (Apps) - Configuration - Segment SDK

READY FOR CONFIRMATION

High-Level Steps:

1. **App Launch:** User opens Vodafone Mobile application (iOS/Android).
2. **SDK Initialisation:** Segment Mobile SDK is loaded during app startup.
3. **Data Capture:** SDK tracks in-app activity (screen views, transactions).
4. **Context Enrichment:** Adds device info (OS type, app version, user context).
5. **Event Transmission:** Events sent to Segment Tracking API over **HTTPS/TLS 1.2+** using **Segment Write Key**.
6. **API Response:** Segment validates and responds with **HTTP 200 OK** confirming ingestion.

Key Controls:

- HTTPS/TLS 1.2+ enforced
- Lightweight authentication using Write Key (no OAuth)
- Security compliance aligned with Vodafone policy

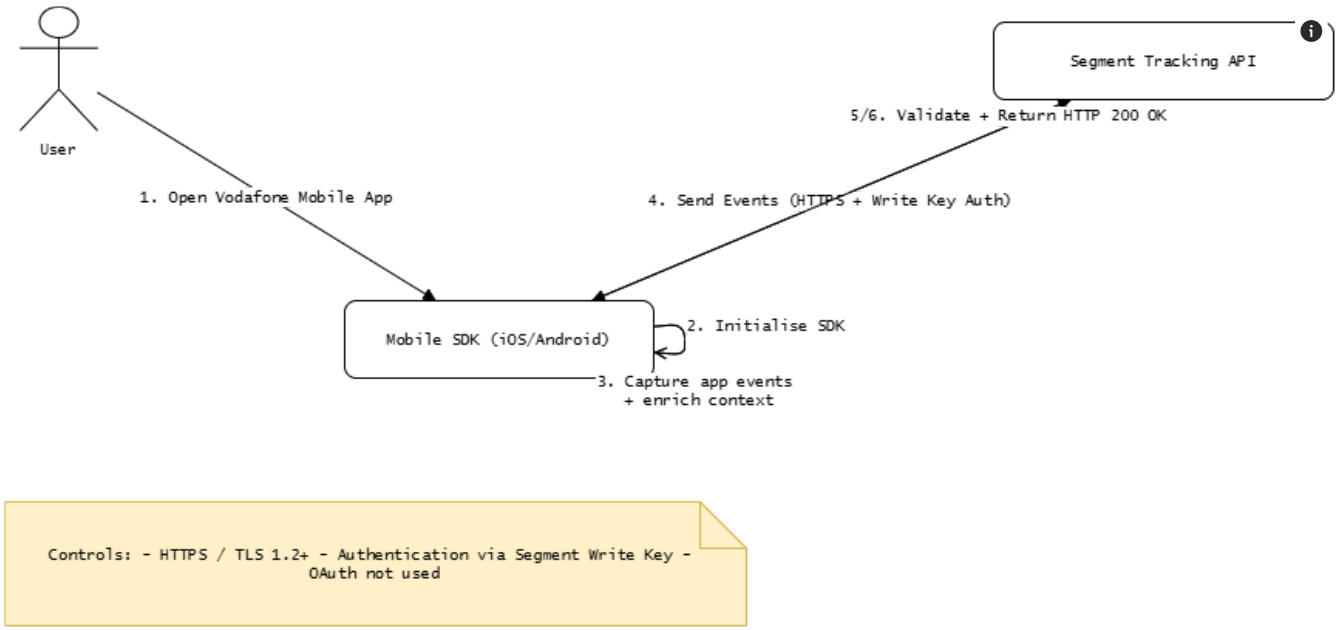


Figure 5.3b: Client-side Mobile App analytics.js data collection

Content Security Policy (CSP) And SRI

CLMNBA-6031: [FR] Digital to CDP | MyVodafone (Web) - Configuration - Content Security Policy READY FOR CONFIRMATION

CLMNBA-6032: [FR] Digital to CDP | MyVodafone (Web) - Configuration - Tag & SRI READY FOR CONFIRMATION

CLMNBA-5155: [FR] Digital to CDP | Vodafone.com.au - Configuration - Content Security Policy READY FOR CONFIRMATION

CLMNBA-5154: [FR] Digital to CDP | Vodafone.com.au - Configuration - Tag & SRI READY FOR CONFIRMATION

Twilio Segment Analytics.js is designed to operate under a **strict Content Security Policy (CSP)** and supports modern browser-side security controls required by PCI DSS. Segment explicitly supports **nonce-based CSP authorisation**, allowing customers to load Analytics.js without using `unsafe-inline` and enabling per-request script approval through the CSP `script-src` directive.

Segment Analytics.js is delivered as a **dynamic script**, including dynamically loaded destination bundles. As a result, **static Sub resource Integrity (SRI) hashes cannot be reliably applied** when the library is loaded from the Segment CDN or a delegated first-party domain (for example, self-hosted in `vodafone.com.au`). In these scenarios, script integrity must be addressed using **approved compensating controls**, rather than SRI.

For customers that require full cryptographic integrity, Segment provides **Analytics.js / Analytics Next as a npm package**, allowing the library to be bundled and self-hosted as a first-party, immutable artefact. This enables **build-time SRI hashing** and represents the **preferred integrity control** for high-risk pages such as payment journeys.

To satisfy **PCI DSS 11.6.1**, Segment is compatible with **CSP violation reporting** (`report-to` / `report-uri`). When combined with a hardened CSP and external monitoring, this enables detection and alerting of unauthorised script execution or tampering and forms part of the compensating control set where SRI is not technically feasible.

CloudFront Reverse Proxy and Zero Rating

CLMNBA-7763: [FR] Digital to CDP | Vodafone.com.au - Zero-Rating Validation READY FOR CONFIRMATION

CLMNBA-6033: [FR] Digital to CDP | Vodafone.com.au - Configuration - Reverse Proxy READY FOR CONFIRMATION

Client-side Segment traffic will be routed through Vodafone-controlled HTTPS endpoints hosted on AWS CloudFront rather than directly from the customer device to shared Segment domains. Separate CloudFront behaviours will forward approved Segment CDN runtime requests and event-ingestion calls to the applicable Segment origins. The proxy will allow only validated paths and methods, use AWS WAF and TLS controls, and must not operate as an unrestricted forward proxy.

The Vodafone-facing proxy hostname provides a uniquely identifiable destination for zero-rating assessment and avoids zero-rating the shared Segment domain. The approved hostname will be submitted through the Vodafone service-rating process and configured by the relevant Network team following approval. Zero-rating effectiveness, including barred and zero-balance customer scenarios, will be validated before production deployment.

Segment collection remains non-critical and must fail promptly and gracefully if the proxy, Segment service or zero-rating configuration is unavailable, without blocking or materially delaying the underlying Shop, web or mobile customer journey.

The primary Segment endpoints currently under consideration are:

- `cdn.segment.com` – delivery of Segment JavaScript libraries and SDK resources.
- `api.segment.io` – ingestion of customer interaction and analytics events.
- `events.eu1.segmentapis.com` – EU regional event ingestion endpoint used by web and mobile applications.

These endpoints would be called directly from Vodafone Shop, My Vodafone Web and My Vodafone App as part of the customer event collection process.

Server-Side Collection Pattern

CLMNBA-5973: [FR] Digital to CDP | Server-Side - Source Configuration READY FOR CONFIRMATION

CLMNBA-5148: [FR] Digital to CDP | IAM - Data - Login Interactions (Server-Side) READY FOR CONFIRMATION

CLMNBA-6128: [FR] Digital to CDP | Vodaofone.com.au (Shop) - Data - Customer Interactions (Server-side) READY FOR CONFIRMATION

CLMNBA-6127: [FR] Digital to CDP | MyVodafone - Data - Customer Interactions (Server-side) READY FOR CONFIRMATION

CLMNBA-5932: [FR] Digital to CDP | IAM - Configuration READY FOR CONFIRMATION

This section describes how server-side microservices within the Digital Enhancement and Engagement Layer (DEEL) AWS account sets forward **track** and **identify** events to the Twilio Segment CDP via the centralised **DEEL-APIGEE** proxy layer. The design is config-driven — each microservice reads its source identifier from AWS SSM Parameter Store, constructs a Segment-spec payload, and POSTs it to a single Apigee proxy endpoint. Apigee injects the Segment Write Key (stored in an encrypted Key Value Map) and forwards the request to `api.segment.io/v1`. Digital - AWS Architecture

Based on the current Digital AWS Architecture, the following account sets host microservices that will emit server-side events: Digital - AWS Architecture

Account Set	Components / Workloads	Compute Model	Example Events
DEEL-MS	Microservices, TFE-API, RPM	ECS Fargate (80 tasks), API Gateway + Lambda (30M req/mo), EC2-ASG (RPM)	Order Completed, Plan Changed, Service Activated
DEEL-FE	TFE, Self-Service (Web)	API Gateway + Lambda (20M req/mo)	Page Rendered (SSR), Feature Flag Evaluated
DEEL-CIAM	Auth0, IAM-MS	Lambda	Login Completed, Login Failed, Signup Completed, Session Refreshed

DEEL-APIGEE hosts the Apigee proxy (`segment-tracking-api`) that acts as the single egress point to Segment. [Digital+-+...chitecture | Word]

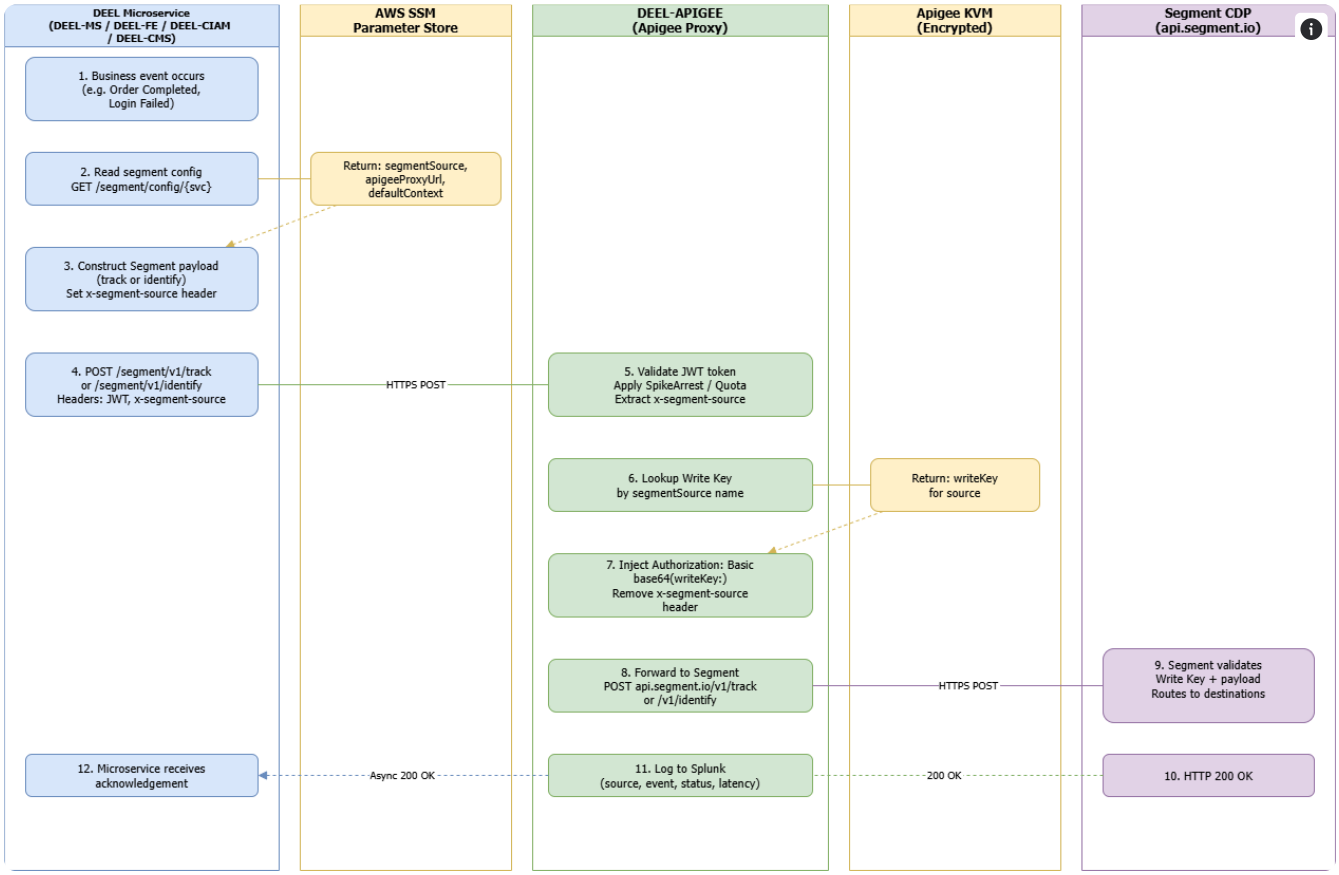


Figure 5.4: Digital DEEL server-side calls to Segment

Config Resolution (Startup)

Each microservice, regardless of account set, reads its Segment configuration from **AWS SSM Parameter Store** at startup (ECS Fargate / EC2) or cold-start (Lambda).

- `segmentSource` — a logical name (e.g. `shop-server-side`, `ciam-server-side`) that maps to a Segment Source
- `apigeeProxyUrl` — the Apigee proxy base URL
- `defaultContext` — app name, version, and `channel: "server"`

OAuth 2.0 is used to authenticate and authorise server-side calls to the HTTPS API integration path for submitting customer trait information to Twilio Segment. The applicable Segment Source Write Key is then used to authenticate the downstream request to the Segment HTTP Tracking API.

DynamoDB may be considered as an alternative to AWS Systems Manager (SSM) for storing the required configuration. The final implementation approach will be determined during detailed design and build.

Event Construction & Forwarding

When a business event occurs (e.g. Order Completed in DEEL-MS, Login Failed in DEEL-CIAM), the microservice:

6. Constructs a Segment-spec JSON payload (see §5 pseudo code)
7. Sets the `x-segment-source` HTTP header to its configured source name
8. POSTs to the Apigee proxy endpoint: `https://api.tpgt-apigee.aws.vodafone.com.au/segment/v1/track` or `/identify`

Apigee Proxy Processing

The `segment-tracking-api` proxy in DEEL-APIGEE performs:

Step	Policy	Action
1	VerifyJWT	Validates the IAM-issued JWT bearer token
2	SpikeArrest	Burst protection (20 per second)
3	Quota	Per-source rate limit (configurable)
4	ExtractVariables	Reads x-segment-source header value
5	KVM Lookup	Retrieve the Source Write Key and the OAuth app ID, key ID and private key from encrypted KVM, using the source name as the key
6	GenerateJWT / ServiceCallout (token)	Sign a client-assertion JWT and request an access token from oauth2.eu1.segmentapis.com; cache the token until it expires
7	AssignMessage	Injects Authorization: Bearer <access_token>; adds the applicable Segment Source Write Key to the request payload; removes the internal x-segment-source header before forwarding the request to Segment.
8	Target Forward	Forwards to https://api.segment.io/v1/{track|identify|batch}

9	MessageLogging	Logs source, event name, HTTP status, and latency to Splunk via Kafka

Segment Processing

Segment authenticates the request via Basic Auth (Write Key), validates the payload against the tracking plan, and routes the event to configured destinations (Redshift, Pega CDH, etc.). [twilio.com], [twilio.com]

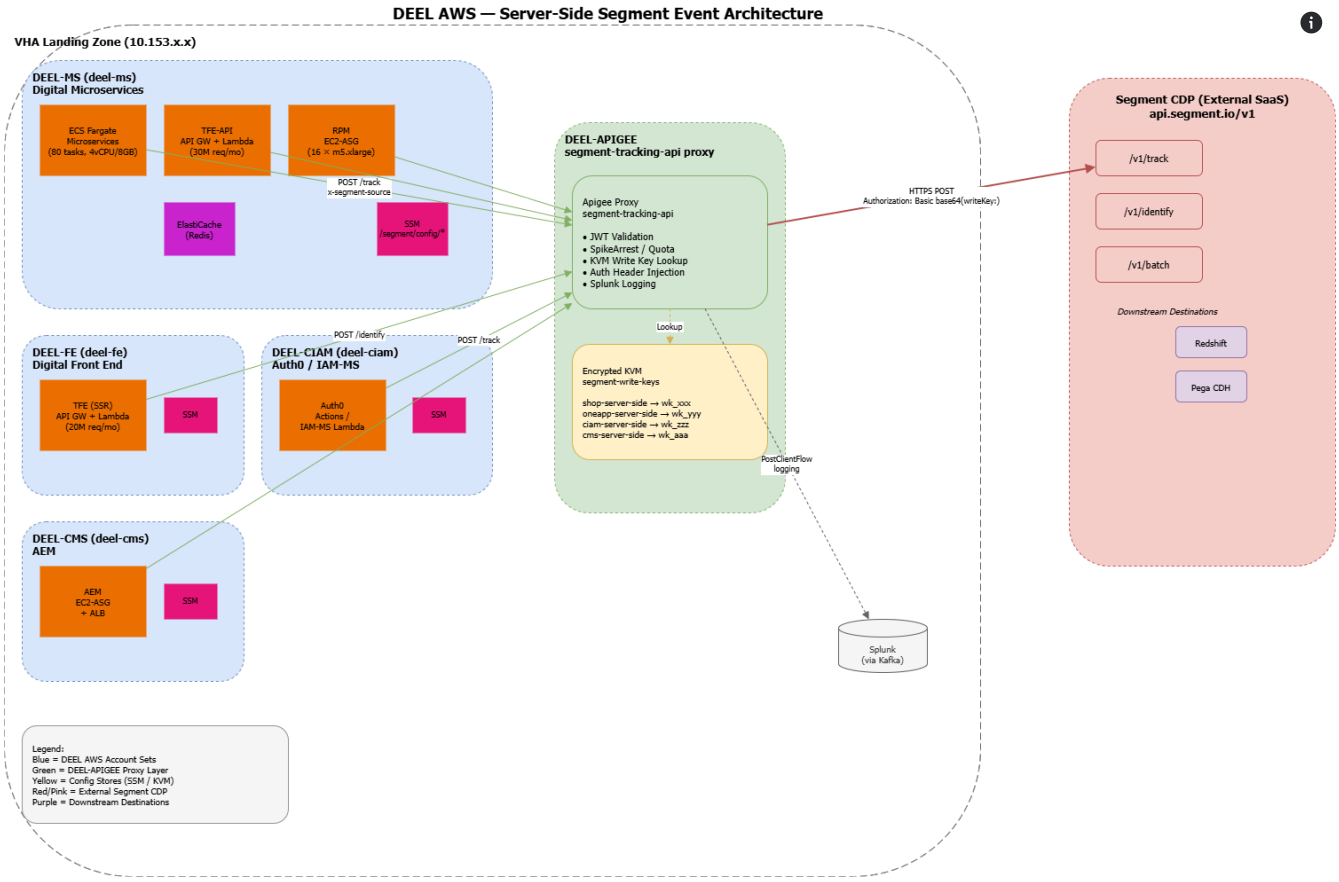


Figure 5.5: Ntwk Digital DEEL server-side calls to Segment

Twilio Segment - Splunk

CLMNBA-5672: [FR] CDP Onboarding | Application Logging & Monitoring (Drop 2 - R3.1)

READY FOR CONFIRMATION

CLMNBA-5805: [FR] CDP Onboarding | Operational Onboarding READY FOR CONFIRMATION

Splunk provides centralised operational monitoring, comparison dashboards, alert correlation and investigation support for data synchronisation between Redshift, Twilio Segment and

associated integration components. Splunk dashboard primary monitoring mechanism by correlating source extraction, Segment processing and target validation metrics for each scheduled synchronisation run.

The Splunk integration will ingest **operational telemetry and aggregated reconciliation metrics only**. Customer event payloads, customer attributes and personal information will not be sent to Splunk. Monitoring data will be classified as operational telemetry, although identifiers such as workspace names, destination names and data-product names may remain commercially sensitive and must be protected through standard Splunk access controls.

The existing observability design proposes that Segment Public API metrics are polled by the Core Integration Layer, written into an encrypted AU-region S3 landing bucket. From here the pattern used for pushing logs to Splunk index for Pega logs would be updated to include the segment related logs.

Kafka Connect HTTP Source Connector with scheduled polling will initiate the HTTP call. Segment Public APIs are protected through Bearer Token authentication. API tokens are generated within the Segment Workspace and provide authenticated and authorised access to Segment services, ensuring that only approved users and automation processes can perform API operations.

- **Base URL:** `https://api.segmentapis.com` (production; recipe examples use `api.segmentapis.build`, the sandbox). Same host for EU workspaces.
- **Auth:** `Authorization: Bearer <token>`, server-side only. Token in Secrets Manager, rotated.

Details to setup public API from Segment side is outlined: Monitor Your Twilio Segment Workspace Using Public API

Certain logs from Segment would be landed onto S3

- **Blocked Events Forwarding** — Enforce Documentation
  Source → DWH/S3
- F**uture (once EU Region Supports the feature):**
  - Granular Failure Logs — Documentation
- **Feed Details pertaining to Forward Violations Feed:**
  - Validation & Forwarding Details
- Any other feature for log forwarded which will be available at the time of delivery.

Proposed S3 layout & Splunk sourcetypes

```
s3://<bucket>/segment-obs/source_volume/dt=YYYY-MM-DD/hh=HH/…      → sourcetype segment:source_volume
```

```
s3://<bucket>/segment-obs/delivery_metrics/dt=…/hh=…/…              → sourcetype segment:delivery_metrics
s3://<bucket>/segment-obs/retl_sync/dt=…/hh=…/…                     → sourcetype segment:retl_sync
s3://<bucket>/segment-obs/audit/dt=…/hh=…/…                         → sourcetype segment:audit
s3://<bucket>/segment-obs-restricted/failure_logs/dt=…/hh=…/…       → governed; masked; NOT in the Splunk-index prefix
```

Security Interaction Matrix

Below table consolidates the component interactions to and from Segment Twilio outlining Authentication, Authorisation, Encryption, etc.

#	From Component	To Component	Interaction / Data Flow	Authentication	Authorisation / Least Privilege	Encryption	Credential Management	Logging and Auditability	Compensating Controls / Clarifications
1	Vodafone Web, My Vodafone Web	Vodafone CloudFront proxy → Twilio Segment	Client-side `track`, `identify` and `page` events using Analytics.js	Segment Source Write Key. The Write Key identifies the approved Segment Source.	Write-only event ingestion into the nominated Segment Source. No profile retrieval, configuration or administration permission. Separate Source and	HTTPS using TLS 1.2 . CloudFront-to-Segment origin traffic will use HTTPS.	Client-side Write Key is embedded in distributed JavaScript. Rotate keys every year. Rotate following confirmed misuse, abnormal ingestion, key	Log Source, event type, response status, rejection reason, latency and ingestion volume. Do not log Write Keys, authorisation headers or complete	AWS WAF, approved paths and methods, request-size controls, rate limiting, CSP, Tracking Plans, Protocols validation and anomalous-volume monitoring. CloudFro

					Write Key per application and environment.		exposure requiring response, or Source decommissioning.	customer payloads. Monitor anomalous traffic in Splunk.	nt must not operate as an unrestricted forward proxy.
2	My Vodafone iOS and Android applications	Vodafone CloudFront proxy → Twilio Segment	Mobile SDK `track`, `identify` and `screen` events	Segment Source Write Key embedded in the mobile application.	Write-only event ingestion into the nominated mobile Source. Separate Source and Write Key per platform and environment. Mobile identity assertions require server-side corroboration before authoritative profile	HTTPS using TLS 1.2. Certificate validation must not be disabled or bypassed.	The Write Key is embedded in the application. Rotate keys every year. Emergency replacement must account for older application versions through a controlled overlap and retirement process.	Monitor application version, Source, event type, response status, schema violations and unusual ingestion volume. Credentials and complete PII payloads must not be logged.	Tracking Plans, Protocols validation, edge rate controls, environment segregation, application release governance and server-side identity corroboration.

					processing.				
3	DEEL Apigee	Twilio Segment HTTP Tracking API	Apigee forwards approved `track`, `identify` and batch requests to Segment	The server-side interaction uses OAuth 2.0 for the authorised upstream call into Apigee and a Segment Source Write Key for the downstream call to the Segment Tracking API. Apigee retrieves and injects the	The OAuth-authenticated workload must be mapped to an approved Segment Source. The corresponding Write Key permits write-only event ingestion into that Source and provides no read, configuration or administrative permission.	HTTPS using TLS 1.2 , including server-certificate validation.	Segment Write Keys are stored in an encrypted Apigee KVM and are accessible only to the approved proxy runtime and authorised administrators. Use separate keys per Source and environment. Define owner, rotation, emergency revocation and recovery	Log the authenticated calling workload, approved Source, operation, status, latency, throttling and error category. Do not log OAuth tokens, Write Keys, `Authorization` headers or complete customer payloads.	Dual control boundary comprising OAuth 2.0 at Apigee and Source-scoped Write Key authentication at Segment. Additional controls include source-entitlement validation, encrypted KVM, quotas, schema enforcement and anomaly monitoring.

				Source Write Key when forwarding the request.			procedures.		
4	CDP Redshift identity share	Kafka Connect JDBC Source → internal Kafka topic	Incremental polling of identity records using an `updated_at`, sequence or change-log watermark	The Kafka Connect runtime assumes a dedicated IAM role and uses temporary IAM credentials through the Amazon Redshift JDBC driver to	The mapped Redshift identity has read-only access to the approved `pii_identify_share` views and required watermark fields only. It has no DDL, write, superuser, cluster-administration or unrelated	JDBC/PostgreSQL database connection over TLS. Kafka topics and supporting storage encrypted at rest using approved KMS controls.	No static Redshift password or long-lived AWS access key is stored in the connector configuration. The Kafka Connect workload obtains temporary credentials through its assigned IAM role. Use a separate role per environment and connector	CloudTrail records role-assumption and relevant Redshift API activity. Redshift audit telemetry records connections, mapped database identity, queries, schema access and privilege failures. Connect	Restrict the IAM trust policy to the approved Kafka Connect workload identity. Alert on unexpected role assumption, access outside the approved polling schedule, abnormal polling frequency, failed authentication, unauthori

				authenticate to the CDP Redshift Cluster. The assumed role maps to an approved Redshift database identity or role.	-schema access.		purpose, with trust and permission policies managed through approved infrastructure code and change control.	or monitoring records task status, polling runs, watermark movement, records selected and produced, query duration, retries and failures.	sed schema access, unusual query behaviour and stalled watermark progression. Confirm the connector's Redshift JDBC driver version and IAM-authentication configuration during build.

5	Kafka Connect HTTP Sink / approved server-side API integration	Twilio Segment Identify API / HTTP Tracking API	PII identity or profile update using server-side HTTPS API calls	OAuth 2.0 is used for the authorised server-side call path, together with the applicable Segment Source Write Key for authentication to the Segment ingestion endpoint. If the call is routed through Apigee,	Restrict the calling workload to the approved identity Source, environment and required API operation. The integration identity and Write Key must not provide Segment Public API, profile-read or workspace-administration permission.	HTTPS using TLS 1.2.	OAuth credentials must be held in an approved secret manager. The Segment Write Key must be held in encrypted Apigee KVM or an approved server-side secret store. Both credentials require defined ownership, rotation, emergency revocation and non-logging controls.	Log workload identity, message ID, correlation ID, approved Source, response status, latency, retry count and final DLQ outcome. Identity payloads and credentials must be masked or excluded.	OAuth workload authentication, Source entitlement validation, Source-scoped Write Key, schema validation, exponential back-off, DLQ, deduplication, rate limits and reconciliation of selected, accepted, failed and pending records.

				OAuth is validated by Apigee and the Write Key is injected at egress.					
6	Twilio Segment Reverse ETL	CDP Redshift cluster	Segment reads approved non-PII traits from `nonpii_traits_share`	Segment assumes a dedicated cross-account IAM role authorised by the TPG AWS account. IAM-based authentication provides tempor	Read-only access to `nonpii_traits_share`. Write access is limited to the Segment-required `__segment_reverse_etl` synchronisation-state schema. No access to the PII identity	Redshift database protocol over TLS. Data at rest encrypted using an approved KMS CMK. PrivateLink and the network	No static Redshift username/password is stored in Segment. A dedicated IAM role must be created per environment and integration. The trust policy must restrict role assumption to the approved	AWS CloudTrail will record IAM role-assumption and relevant Redshift API activity. Redshift audit telemetry will record database connections, query execution, schema	PrivateLink, private subnets, TLS and schema-level least privilege remain mandatory. Alert on role assumption by an unexpected principal, access outside the approved schedule, unusual session frequency, failed

				ary credentials for access to the CDP Redshift Cluster. The assumed role must map to the approved Redshift database identity or role used by Reverse ETL.	schema, Segment export schema, system administration or unrelated schemas.	path must not terminate or downgrade database TLS.	Segment AWS principal and, where supported, include an External ID or equivalent confused-deputy protection. IAM policy changes, trust-policy changes and role revocation follow approved change control.	access and privilege failures. Monitoring must correlate the IAM role session, Segment workspace/source, Reverse ETL sync ID, environment, database role, query duration, watermark and record counts.	role assumptions, unauthorised schema access and abnormal queries. Confirm the exact Segment AWS principal, supported trust-policy conditions, IAM-to-Redshift mapping and temporary credential lifetime.
7	Twilio Segment Warehouse Destination and	CDP Redshift cluster	Segment writes events, identity and profile-sync tables	Segment assumes a separate dedicated	Dedicated database role with only the DDL and DML permissi	Redshift database protocol over TLS.	Use a separate role from Reverse ETL and separate roles per environm	CloudTrail records role assumption and relevant Redshift	CloudTrail records role assumption and relevant Redshift API

	Profiles Sync		into the export schema	cross-account IAM role authorised by TPG. IAM-based authentication provides temporary credentials for Warehouse Destination and Profiles Sync access. The role must map to a Redshift identity restrict	ons required inside `segment_export`. No permission to access MDS source schemas, PII input schemas, Reverse ETL trait schemas or cluster-administration functions.	Data at rest encrypted using an approved KMS CMK.	ent. No static Redshift password is stored in Segment. Restrict the role policy and trust relationship to the required cluster/workgroup and approved Segment principal. Role creation, policy updates and revocation must be managed through approved infrastructure configuration and change control.	API activity. Redshift audit telemetry records connections, object creation, DDL/DML activity, target schema access, privilege failures and session duration. Correlate the IAM role session with the Segment workspace, warehouse/destination ID, sync ID, target table,	activity. Redshift audit telemetry records connections, object creation, DDL/DML activity, target schema access, privilege failures and session duration. Correlate the IAM role session with the Segment workspace, warehouse/destination ID, sync ID, target table, records reported as synchronised and

				ed to the export schema.				records reported as synchronised and Redshift-validated row counts.	Redshift-validated row counts.
8	Twilio Segment	MDS raw-event S3 bucket	Raw-event export using S3 Actions and IAM role	AWS role assumption using an IAM trust relationship and Segment Workspace ID as External ID.	Dedicated IAM role per workspace, brand and environment. Permit write access only to the nominated bucket or prefix and required KMS operations. No read, delete, broad bucket-list or cross-	HTTPS/TLS in transit and SSE-KMS at rest.	No static AWS access key. Manage the IAM role, trust relationship and External ID through approved infrastructure configuration. Restrict KMS use to the required encryption operations.	CloudTrail records role assumption and S3/KMS operations. Monitor denied requests, unexpected workspace identifiers, writes to unexpected prefixes, missing files and unusual	Block Public Access, restricted bucket prefix, External ID, bucket-owner controls, versioning, lifecycle controls and appropriate PII classification for raw events.

					environment permissions unless separately justified.			event volume.	
9	Core Integration monitoring poller / Kafka Connect	Twilio Segment Public API	Download delivery metrics, Source volumes, destination results and synchronisation telemetry	Dedicated Segment Public API bearer token.	Token restricted to the minimum read operations required for observability and monitoring. It must not create, modify or delete Sources, Destinations, Tracking Plans, users, tokens or workspace settings.	HTTPS using TLS 1.2 or higher. The S3 telemetry landing bucket is encrypted using KMS.	Token stored in AWS Secrets Manager or another approved enterprise secret store. It must be separated by environment and from the configuration-deployment token. Define owner, rotation and emergency revocation.	Record request endpoint, workspace, polling run ID, response status, rate-limit response, S3 write result and processing failures. Do not log the bearer token or customer event payloads.	Read-only token, outbound endpoint allow-listing, scheduled polling, response validation, rate-limit handling and monitoring-feed health alerts.

10	GitHub Actions / Terraform	Twilio Segment Public API	Validate, plan and deploy Segment workspace configuration	Dedicated Segment Public API bearer token supplied to the workflow at runtime.	Token restricted to the Segment configuration resources managed by the pipeline and separated by environment. The production token must only be available to the protected production deployment job.	HTTPS using TLS 1.2 or higher. Remote Terraform state encrypted in transit and at rest.	Store the token in protected GitHub Environment secrets or retrieve it from an approved enterprise secret manager. Do not store it in code, workflow YAML, variable files, plans, artefacts, logs or Terraform state. Define scheduled rotation and emergency revocation.	Retain commit, pull request, reviewers, Terraform plan, environment approval, workflow execution, change record and apply result. Monitor failed access, direct UI configuration and drift.	Private repository, branch protection, mandatory review, environment approvals, read-only workflow permissions by default, pinned approved actions, secret scanning, remote-state protection and scheduled drift detection.
11	Authorised TPG	Twilio Segment	Administrative and	Entra ID SAML	Segment RBAC using	HTTPS/TLS for	User access governed	Record successful and	MFA, device complian

	and support users	administration interface	operational access to Segment	SSO with MFA, Conditional Access and managed-device or AVD restrictions.	approved groups and minimum required roles. Separate administrator, developer, operator, marketer, read-only and audit responsibilities. Local accounts and break-glass access must be separately controlled.	administrative access.	through enterprise joiner, mover and leaver processes. Define privileged-access controls, access-review frequency, break-glass custody and emergency removal.	failed logins, role changes, user/group changes, token creation, Source and Destination changes, Tracking Plan changes, deletion requests and direct production UI configuration. Forward relevant events to SIEM where technically available.	ce, AVD, periodic access recertification, segregation of duties, restricted direct production changes and configuration drift detection.

12	Apigee / Core API	Twilio Segment deletion and suppression API	Inactive-customer deletion and suppression requests	OAuth 2.0 is used for the authorised server-side call path, together with the Segment credential required by the applicable deletion or suppression endpoint. The exact downstream credential must be confirmed	Restrict the calling workload to the approved deletion or suppression operation, workspace and environment. It must not receive broader configuration, profile-read or administrative permissions.	HTTPS using TLS 1.2 or higher.	Dedicated credentials held in encrypted KVM or an approved secret manager. Define ownership, rotation, emergency revocation and any required dual control.	Record request ID, system identity, request time, Segment acceptance, downstream result, completion status and failure reason. Avoid retaining deleted customer attributes in operational logs.	Controlled deletion source dataset, eligibility validation, idempotency, retry controls and reconciliation of requested, accepted, completed, rejected and pending deletion records.

				against the final Segment API selected.					

Technology Service KPIs

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Describe the changes (if any) to Technology KPIs due to proposed solution and its components

The list of Technology Service KPIs (Network and IT) reported to execs can be accessed here. If a new line of business/function is being introduced and no existing Service KPIs are available, it will be added through Operations Service Management Process.

***NOTE*:** The project has the responsibility for the following:*

1. *Ensure that the Service KPIs reported to execs are available with the proposed solution and clearly articulate any impacts to those KPIs.*
2. *Any new Service KPIs being introduced by the solution need to be captured as part of Non-Functional Requirements and adhere to the ODS (Operational Design Standards).*

S. No	Impacted KPI	Variations (different Scenarios)	Impact (Functional change or Regression test or NA)

Table 17: Impacts of KPIs

Architectural Decisions

List all the architectural decisions taken to address architecturally significant requirements

1. **CI/CD pipeline and configuration management mechanism for Segment CDP**

   Decision: Segment workspace configuration will be managed as code using the `segmentio/segment` Terraform provider, executed by a GitHub Actions pipeline.

Configuration not supported by the provider will be applied via Segment Public API scripts or manual configuration, under change control.

Context: The Segment Terraform provider is built on the Segment Public API and provisions no AWS resources, authenticating with a Public API bearer token over HTTPS . RIO's Terraform pattern assumes CloudFormation-provisioned central pipeline resources and an AWS-resident control plane , and the RIO FAQ permits GitHub Actions for any activity that does not require interaction with AWS accounts . RIO adoption is mandated as a prerequisite for implementation on TPG's AWS Cloud, which is not engaged by a SaaS-only component.

Rationale: GitHub Actions is the closer technical fit and avoids a platform uplift for controls with no object to act on. Terraform is retained over pure scripting because the plan provides a reviewable change set prior to approval, together with state-based drift detection and repeatable workspace rebuild — none of which the scripted or Git Sync options provide.

Consequences: The pipeline is not a Registered Implementation and requires its own registration and assessment against the applicable CI/CD Framework version, with CMDB integration, SAST, RBAC, metrics and Helix change-management capabilities independently evidenced. A residual set of configuration will remain outside pipeline coverage . Where no AWS state backend is adopted, the pipeline holds no short-lived credential — refer SR-1.

2. **Segment Data Ingestion and Synchronisation Model**

   Decision

   The solution will adopt a dual integration model for data exchange with Twilio Segment, aligned to data sensitivity, processing requirements and platform-native capabilities.

   - **Warehouse Connectivity Pattern**: Twilio Segment will act as the consuming client and connect to TPG-hosted Redshift environments for Reverse ETL, Profiles Sync and warehouse-based integrations. This pattern will be used for scheduled synchronisation of approved customer attributes, profile enrichment data, audiences and analytical datasets.
   - **API Ingestion Pattern**: TPG will act as the client and publish customer identity, profile update and data deletion events to Twilio Segment APIs over HTTPS via approved enterprise integration services. This pattern will be used for identity resolution, profile synchronisation, privacy controls and other event-driven data ingestion requirements.

   Rationale

   This approach leverages Twilio Segment’s native integration capabilities, aligns integration mechanisms to the characteristics of the underlying data flows, reduces custom integration

development, and maintains the Marketing Data Store (MDS) as the authoritative system of record for customer and profile data.

Consequences

The solution will utilise both warehouse-based and API-based synchronisation patterns. Appropriate monitoring, reconciliation, lineage and data governance controls must be implemented to ensure consistency, traceability and operational integrity across MDS, Segment and downstream consumer platforms.

Solution Scalability

Clearly specify how the solution meets the scalability requirements based on the expected business forecast. Clearly specify any limitations that may impact the solutions from scaling.

Twilio Segment SaaS has been sized up to cater for traffic for all brands.

Shared Infrastructure

Segment SaaS compute & storage has been sized & licensed with forecasted end users, subscriptions volume for Vodafone, TPG and iiNet as of end of Nov 2025, the primary factor for license is Monthly Active Users. If & When the MAU is exceeded, additional resources are allocated and overage charges is applied.

Redshift Serverless will be enhanced to provide and store data to and from Segment SaaS. Additional Redshift cluster would be setup to enable privatelink connectivity between Segment SaaS and TPG AWS. The reason for not using Redshift Serverless, is that privatelink connection does not support Redshift Server less and requires a Redshift cluster to be setup.

Re-use of existing Architecture Capabilities

Data products setup in illuminate will be reused for data feed to Segment. Additional space and storage will be allocated in marketing data store system to store data from Segment.

Cost Efficiency

Licensing

No new licensing required. Segment SaaS license has been procured.

Operating Expense (OPEX)

Does the solution require any ongoing opex or will change current Opex? If yes, describe how the Opex has been optimised.

Ongoing license renewal will be required for Twilio Segment SaaS. Segment function API usage will need to tracked.

Cost efficient Solution

Describe the cost efficiencies embedded in the solution architecture and design. Have you considered cost optimisation in the design? E.g.:

- *the use of serverless services? E.g., only consume resources for the compute that is required,*
- *the mechanical sympathy of the design, is it the most efficient database type for the service? e.g. Using NoSQL for maintaining state*
- *Reviewed the logging metrics parameters to balance the cost vs data collection parameters*
- *Use of managed service APIs to focus on the value of the service vs maintenance of the underpinning service components*
- *End to end service cost attributed to the business/technology service driving the demand*
- *Schedule the startup/shutdown of resources to align to demand, e.g. schedule development environments to be stood up during development hours and shutdown overnight*

All new TPG components will be stood up AWS cloud or the on target aligned platforms to avoid rework.

Other Decisions

List any other architectural decisions that played a major role in the way the solutions were designed.

Open Solution Issues / Items

List any open issues still to be resolved at the time of document sign off.

Issue	Priority	Description	Resolution
Twilio Segment Cross-Region PrivateLink Connectivity for Reverse ETL and Warehouse Destinations	Medium	Twilio Segment's cross-region PrivateLink capability is currently not available and may not be delivered in time for the Release 3.1 implementation. As a result, connectivity requirements for Reverse ETL and Warehouse Destination integrations requiring	The cross-region PrivateLink requirement will need to be deferred to a future release once the capability becomes available or an approved alternative solution is adopted. Potential implementation options and associated recommendations

		private connectivity between AWS regions cannot be met within the current release scope.	have been captured in the Ignite decision log under: TRAID-65: Segment connectivity to TPG Data components Under Review
GitHub Actions deployment approach	Low	Confirm the approved pipeline pattern, CI/CD Framework requirements, Terraform state backend, API token management, security controls, approvals and operational ownership.	Agree and document the production-ready GitHub Actions design with the relevant platform, security and governance stakeholders before implementation.
CloudFront proxy and zero-rating design validation	Medium	Confirm the Vodafone proxy hostname, Segment CDN runtime paths, EU ingestion origin, cache behaviour, WAF controls, operational ownership and residual third-party endpoints. Zero rating must not be assumed until formally approved and validated.	Confirm the Vodafone proxy hostname, Segment CDN runtime paths, EU ingestion origin, cache behaviour, WAF controls, operational ownership and residual third-party endpoints. Zero rating must not be assumed until formally approved and validated.

Table 18: Open Issues / Items

Architecture Deviations

(Optional for SO required for SA)

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Description of Architecture Deviation:

No deviation from target architecture.

The solution aligns to target architecture as it replaces current SAS CI360 discover capability and integrates to Target data platform (through illuminate program). It uses the core integration services to enable data transfer between TPG and Segment SaaS.

Description of the Technical Debt: (*please include architectural debt as well as debt related to implementation approach.)*

Technical Debt Name	Debt Owner	Impact	Architectural Risk

Remediation Plan

Technical Debt Name	Debt Owner	Remediation Plan	Scope	Cost Estimate (optional)	Time Estimate (optional)

Table 19: Technical Debt

Architecture Policies Standards and Guidelines

All the policies, standards and guidelines published are available in Central Policy Hub on intranet.

https://tpgtelecom.sharepoint.com/sites/Policies/

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Policies

Name	Policy Name	Compliant (Y/N)	Comments

TO_POL_01	Technology Resilience Policy	Y	Resilience and graceful failure are defined, but Segment connectivity, retry, PrivateLink availability and recovery testing remain subject to validation.
TO_POL_03	Service Assurance Policy	Y	Operational telemetry, API failure monitoring, reconciliation and alerting are described in section 3.8. Security-event integration and retention require confirmation in section 3.17.
TS_POL_33	Cyber Security Policy	Y	Design compliant: Authentication, authorisation, secrets management, logging and monitoring are defined by interaction in section 2.5.6.
TS_POL_36	Secure Software Development Policy	Y	Design Compliant: TPG-managed proxies, connector configuration, functions and deployment workflows follow approved CI/CD, peer review, scanning, testing and change-control requirements.

Table 20: Architecture Policies

Standards

Name	Policy Name	Compliant (Y/N)	Comments
CINS_STA_03	Inventory Management Standard	Y	Twilio Segment has been selected as the SaaS product for integration into the MarTech ecosystem, enabling: • Customer trait provisioning for Pega Decisioning Engine

			• Data enablement for AdTech platforms Procurement Status: The procurement process has been successfully completed in line with TPG Telecom's sourcing and governance requirements.
CINS_STA_01	IP Addressing Standard	Y	AWS components using PrivateLink and Redshift (VPC endpoints, subnets) will follow TPG IPAM hierarchy and be registered in IPAM. No deviations.
CINS_STA_02	Quality of Service Standard	Y	Integration uses AWS networking and Apigee, adhering to DiffServ principles; no latency-sensitive degradation.
CINS_STA_03	Workload Placement standard	Y	SaaS (Twilio Segment) integrates via Apigee-managed proxy for secure API calls. Hosting follows standard decision matrix (SaaS use case).
CINS_STA_04	Standard Operating Environment (SOE)	Y	AWS workloads and Apigee/Kafka Connect use approved imaged and includes mandated security agents.
CINS_STA_05	Cryptographic Standards - Technical Implementation	Y	TLS 1.2 is required for API and database connections; TPG data stores use KMS-backed encryption.
IT_STA_01	Information Management Standard	Y	Data classification, event PII treatment, location, retention and deletion must be reconciled across Segment, S3, Redshift, Kafka, DLQs, Splunk and

			backups. Refer to sections 2.5.3 and 2.5.4.
NPE_STA_01	NPE IT Environment Standard	Y	Separate Sources, credentials, roles, buckets, schemas and endpoints are required per environment. Production PII must not be copied to non-production without approved irreversible masking.
TSO_STA_01	Operations Design Standards	Y	Monitoring and reconciliation are defined; security audit-event coverage, retention, review ownership and automated incident creation require confirmation.
CINS_STA_06	Cost Optimisation Standard for Public Cloud Infrastructure	Y	AWS resource provisioning adheres to TPG tagging policy, cost allocation, and show-back models for PrivateLink and Apigee proxy setup.

Table 21: Architecture Standards

Guidelines

Name	Policy Name	Compliant (Y/N)	Comments
CINS_GDL_01	Automation Guideline	Y	Segment configuration should be managed through an approved pipeline. Manual higher-environment changes are restricted and monitored as exceptions.

CINS_GDL_02	Configuration Management Guideline	Y	Git is the authoritative configuration source; Terraform plans, approvals and execution records provide traceability. Items outside provider coverage require an exception register.
IT_GDL_01	New Capability Guideline	Y	Security assurance, support onboarding, operational acceptance, data governance, CMDB and service ownership must be complete before production.

Table 22: Architecture Guidelines

Proposed Solution Non-Functional Overview (Relevant areas required)

Required for SA

This document describes the Solution Architecture. Hence all relevant non-Functional aspects must be described

Availability

Clearly specify how the solution meets the following availability requirements based on the business criticality of the solutions (please refer to policies APP-ENT-10 and APP-ENT-11):

- *99.99% availability for any application or infrastructure supporting Mission and Business critical services*
- *Solution to be architected with sufficient redundancy such that no single component, group of components or single site outage will impact the provision of services to TPG Telecom customers.*
- *No Single Points of Failure in the Production Environment*
- *Solution to be architect such that any requirements to upgrade or maintain the application can be performed in a modular process with no impact to customer-facing and critical business services support or delivered by the solution*

If the solution hosted in AWS or other Public Cloud provider, the Public Cloud provider SLA's should be clearly specified. If solution leverages a SaaS deployment, the SaaS vendor

committed availability should be specified. Clearly specify if the solution does not comply with associated architecture policy and standards.

Twilio is contractually obligated to comply with its Service Level Agreement (SLA). The platform is designed with multiple geographically diverse regions and fault-independent availability zones to maintain service availability.

the Data Ingestion API being available and operational to Customer with a threshold of 99.99%, calculated on a calendar month, per-minute basis.

Archive and Purging

Clearly specify how the solution complies with the TPG TELECOM Data Retention policy, what scheduled tasks should be in place to perform archiving and purging in automated and scheduled manner without impacting performance or availability or end customer responsiveness. Clearly specify if the solution does not comply with associated architecture policies.

Error Handling

For Business Process or Service added or customised, Distinct Error Code and Error Messages should be generated for every failure scenario, across Application stack and meaningful message displayed on front end for end customer consumption.

Each Error should have unique code and message to be stored in respective originating application for debugging and reprocessing. In Event of high volume of errors, Application should not crash, jeopardising healthy transactions or background processes.

System should have capability to clear Errors, Failures, Retry automatically or with minimal manual intervention once error causing cause is alleviated (e.g. failed downstream component is up or failed file transfer)

Error Conditions, its message and way to reprocess should be documented in SOP and demonstrated to Operations prior to go live.

Database

Ensure, all database optimisation and performance/tuning parameters specified in the TPG TELECOM Oracle Database SOE have been implemented and:

(a) Tested for effect (i.e.) with the parameter set to the Oracle default and with the parameter set to the TPG TELECOM SOE value – if different.

(b) CPU, IOPS and elapsed time recorded using the default and TPG TELECOM-recommended parameters against a known test suite of database transactions.

(c) Document assumptions and the methodology to determine how estimates are calculated for the database concurrent session limit parameter.

(D) Optimization methodology of databases

(E) All database housekeeping activities required to maintain the integrity and performance of the database must be able to run at any time without impact to customer-facing services

Deviation from Standard Solution Support Model

Describe any non-standard support task under each level

Level	Key Activities	Description
Level 1		
Level 2		
Level 3		
Level 4		

Table 23: Solution Support

Operational Support Functions

Describe the different tools that will be utilised to provide the operation support functions.

		Twilio Segment
Service Monitoring	Target System	Splunk Enterprise
	Alerting	Splunk Enterprise Alert Collector & Segment Alerts
	KPI Reporting	Splunk Enterprise - Dashboard
Performance Management	Performance Monitoring	Splunk Enterprise - Dashboard
	Alerting	Splunk Enterprise Alert Collector & Segment Alerts
Capacity Management	App Capacity Reporting	Twilio Segment is SaaS and infra is managed by vendor

	Infra Capacity Reporting	Twilio Segment is SaaS and infra is managed by vendor
Fault Management	Target System	Splunk Core & Segment Alerts
	Auto Ticketing	Splunk Enterprise Alert Collector & Segment Alerts Alert emails that can be used to raise ticket to be assigned to relevant team. At the integration to Helix, the use of API would be accessed and implemented instead of email
Restoration	Backup solution	As this is SaaS provider, the backup solution is provided by vendor
Security	SIEM Integration	Splunk ES
	Active Threat Management	Splunk ES
	Forensics	Splunk ES
Access Management	User Identity access via SailPoint	Azure AD, Entra Id

Table 24: Operational Support Functions

Operations Asset Inventory Management

(Mandatory for SO/SAD)

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Based on the inventory management standard published on the intranet, please specify the target inventory system that is impacted with this solution:

The list of inventory management systems is based on the Inventory Management Standard that is published in the Central Policy Hub.

Asset Management Type	Definition/Examples	Target	Impacts on Target system
CMDB	Physical or virtual configurations items (CIs) required to support the operational ITIL processes.	BMC Helix	Adding new workloads
Data Centre Infrastructure/Cabling	Database of equipment, racks, suites, cabling and ancillary equipment in supported Data Centres	Cormant-CS or LSD	No Impact
Active RAN and Tx Network Inventory	Inventory Database which records the telco active network element inventory information including rack, subrack, Card location, manufacturer, type, serial number and connectivity	Orion-T / Orion- R / LSD / InfraDB	No Impact
Passive RAN and Tx Network Inventory	Inventory database which records passive equipment Eg Antennas, wave guides, batteries, huts, pipes, pits	Orion-T / Orion- R / PipeDB / ArcGIS / LSD / InfraDB	No Impact
Telco Spares Management	Database of spares location, status, numbers	SPMS (VHA)	No Impact

Telco Build Inventory	Database containing material on hand (not yet installed) to build sites eg racks, huts, antennas, cable	CEVA / Orion-R	No Impact
Software Licenses	Database that records software license entitlements, usage, license expiry dates, etc	SNOW	SNOW tool no longer used
Employee Asset Management (Ex: PCs)	The database of what equipment is held by employees such as PCs	SNOW, BMC Helix, JAMF, InTune	No Impact
Leased Lines	Leased lines used by TPGT, cost, lease dates, lessor details	Orion-T (VHA)	No Impact
IPAM			No Impact
ERP Cloud (Oracle Cloud)			No Impact

Asset Inventory Management

Service Dashboards & Alerting

The following to be considered in the proposed solution in case business services or technical services are modified or newly created:

1. *Any newly on-boarded system or change in existing system impacting listed Service transactions needs to be integrated into Splunk providing transactional activity logs & metrics.*
2. ***Monitoring dashboards** should be designed for the monitoring of critical business services. These dashboard should provide a view of the business services status so that service interruption/degradation that are not detect through alerting can be identified by L0 monitoring teams / NOC teams.*

3. ***Troubleshooting dashboards*** *should be designed for the purpose of providing a view of technical services that support business services and assist with identifying the root cause of an incident.*
4. ***KPI reporting dashboards*** *should be created or modified to support the reporting of critical business services*
5. ***Alerts*** *should be created to automate the detection of service availability or service performance degradation related to critical business services. Alerts should be integrated with the Event Management platform and auto-ticketing in the ITSM platform enabled.*

The monitoring design will provide end-to-end visibility across data ingestion, synchronisation, storage, configuration deployment and monitoring collection. Monitoring will distinguish platform availability, process completion, data completeness, data quality, data freshness, security-relevant access behaviour and the health of the monitoring pipeline itself.

A successful application, API, connector or Segment synchronisation status does not independently demonstrate successful end-to-end processing. Operational assurance therefore requires correlation and reconciliation across the source system, integration layer, Segment processing and target data store.

Splunk will provide the central TPG operational dashboarding, alert correlation and investigation capability. Segment-native dashboards and notifications, AWS monitoring, Apigee analytics, Kafka Connect metrics and Redshift system views will act as source telemetry. Component-native notifications may be retained as secondary diagnostic or fallback mechanisms but will not be the authoritative indication of end-to-end service health.

Splunk dashboards will provide operational visibility of Segment CDP service health, interface performance, event-processing failures and applicable data-transfer conditions. Continuous alerting will be limited to agreed critical KPIs, while supporting metrics will remain available for reactive troubleshooting. When an actionable threshold breach or service degradation is detected, the Splunk TSI workload will correlate the event and initiate an incident in BMC Helix with the applicable severity, affected service and resolver-group information.

Monitoring Data Sources

Splunk will receive operational telemetry from the following logical sources:

Source	Information supplied

Segment Public API	Event volume, successfully received events, ingestion failures, source and destination filtering, successful delivery, failed delivery, retry metrics and discard reasons
Segment Reverse ETL	Sync status, source model, timing, records selected, records processed, rejected records, retry result, failure reason and watermark
Segment Warehouse Sync	Warehouse and source identifiers, sync status, Ping and Load status, duration, synced rows, notices and failures
Segment Profiles Sync	Profiles selected, exported, rejected or failed, execution status, duration, mapping errors and schema errors
Redshift source validation	Eligible records, selected records, invalid records, duplicate records, source-query status, refresh timestamps and watermark
Redshift target validation	Actual records in Segment-managed target tables, distinct keys, duplicate keys, maximum load timestamp, table freshness and query status
Kafka Connect	Connector and task status, records polled, records produced, failures, restarts, lag and failed-record backlog
Apigee and Segment APIs	Request volume, response status, latency, authentication failures, timeouts, rate limits and retries
AWS infrastructure	Redshift availability, PrivateLink and NLB health, connection failures, query errors, load errors and capacity indicators
Configuration and deployment processes	GitHub Actions workflow status, Terraform execution result, Segment configuration change and release reference

Deletion and purge processes	Records submitted, accepted, completed, rejected and pending
Monitoring ingestion pipeline	Poller execution, API response, S3 write status, SNS/SQS status, queue backlog and Splunk ingestion latency

Splunk Event and Correlation Attributes

Every monitoring record should include sufficient context to associate it with the correct environment, integration and processing window.

The logical event attributes could include:

- monitoring timestamp;
- environment and region;
- data-flow name and direction;
- integration type;
- source and target systems;
- Segment workspace ID;
- Segment source and destination IDs;
- warehouse ID;
- sync ID;
- enterprise run ID;
- batch ID, where applicable;
- common correlation ID;
- data-product or source-model name;
- Redshift database, schema and table;
- event, identity, trait, profile or deletion category;
- schedule name;
- scheduled, start and completion times;
- comparison-window start and end;
- execution status;
- retry attempt;
- error category and error code;

- change or release reference;
- service owner and support group;
- data classification.

A common `sync_id`, `run_id`, `batch_id` or `correlation_id` must be available across the source extraction, Segment processing and Redshift validation telemetry. This enables Splunk to compare records belonging to the same execution rather than unrelated totals from different processing windows.

The final physical field names, formats and mandatory-field rules should be defined in the Splunk Integration and Dashboard Specification.

Actual Redshift Target-Count Capture

Splunk will not directly calculate the actual data count from Redshift.

A TPG-controlled, scheduled, read-only validation process will query the actual Segment-managed tables in the provisioned Redshift cluster after the relevant Segment Warehouse or Profiles Sync window. The process will generate aggregated count and freshness telemetry and send the results to Splunk through the approved ingestion path.

The target validation process should capture:

- Redshift cluster, database and schema;
- target table;
- Segment sync ID or associated reconciliation run ID;
- comparison-window start and end;
- actual records within the comparison window;
- distinct business-key or event-key count;
- duplicate-key count;
- records with missing mandatory keys;
- minimum and maximum source event timestamp;
- maximum Redshift load or received timestamp;
- inserted, updated and deleted counts, where reliably measurable;
- validation-query status;
- validation-query duration;
- table-freshness age;
- query failure or timeout category.

The validation query should use the appropriate incremental field for each target table, such as a load timestamp, received timestamp, event timestamp or other approved sync attribute. The specific column and business key may differ between event, identify and profile tables and must be defined during detailed design.

A full-table count should not be the default operational method because it does not isolate the affected sync window and may introduce unnecessary Redshift workload.

AWS documents that Redshift system and monitoring views can provide query, workload and load information. These views should be used as supporting evidence for load execution, errors and performance. `SYS_LOAD_DETAIL`, for example, provides file and load-processing information, although AWS notes that scanned lines may not equal rows actually loaded. Consequently, system-view metrics alone should not replace validation against the actual target tables.

Splunk Dashboard Views

End-to-End Integration Health

The dashboard will show:

- overall health by environment and data flow;
- current and last successful execution;
- last telemetry received;
- data-freshness status;
- watermark lag;
- source, Segment and target counts;
- failed and pending-retry records;
- reconciliation variance;
- active alert or incident;
- monitoring-feed health.

Synchronisation Comparison

The comparison view will show one logical record per synchronisation execution:

Field	Dashboard purpose
Run or sync ID	Correlates the execution across components
Source records	Shows records eligible or selected at the source

Segment processed records	Shows records successfully handled by Segment
Filtered records	Explains intentional exclusions
Failed records	Shows records requiring investigation
Pending records	Shows records awaiting retry
Segment synced rows	Provides the Segment warehouse-side measurement
Redshift validated records	Provides the actual target-side measurement
Variance	Shows unexplained difference
Watermark	Confirms incremental processing progression
Freshness	Confirms data availability within SLA
Status	Provides Green, Amber, Red or Unknown classification

Operational Trends

The trend view should include:

- processing volumes over time;
- source, Segment and Redshift count trends;
- variance trends;
- success and failure rates;
- synchronisation duration;
- data-freshness and watermark lag;
- retry and failed-record backlog;
- API response and latency trends;
- Redshift query and load performance;
- Splunk ingestion latency.

Quality of Service (QoS) / Performance

The integrity of the solution is dependent on Quality of Service (QoS) / Performance contributions from all elements spanning the breath of this solution. Hence the description should address all the elements listed in section 2.4.

The solution's Quality of Service is dependent on the performance of Segment SaaS, Digital collection components, CloudFront, Apigee, Kafka Connect, Redshift, PrivateLink, S3 and the operational monitoring pipeline. Performance must therefore be measured end to end rather than solely from the availability or response time of an individual component.

The solution must demonstrate that peak campaign activity, product launches and other forecast business-volume events do not materially degrade customer-facing journeys, event collection, identity processing, synchronisation, data freshness or existing MDS workloads. Client-side analytics collection remains non-critical and must fail promptly and gracefully without blocking or materially delaying Vodafone web or mobile customer journeys.

System	Describe the impacts, if any, to the SYSTEM because of the QoS requirements
Twilio Segment	Segment-native and TPG-controlled telemetry will be used to measure availability, throughput, latency, error rates, synchronisation duration, data freshness, reconciliation variance and capacity against agreed operational thresholds. Monitoring provides evidence of QoS performance but does not itself guarantee availability, performance or security.

Table 25: QoS / Performance

QoS and performance measures

Component / Interaction	Measures to monitor	Design expectation
Client-side Analytics.js and mobile SDK	Collection request latency, HTTP error rate, timeout rate, WAF blocks and Source-volume change	Analytics failure must not block or materially delay the customer journey
CloudFront reverse proxy	Origin latency, error responses, cache behaviour, WAF action and request volume	Event-ingestion requests must not be cached; only approved paths and methods are permitted
DEEL → Apigee	Request rate, latency, OAuth failure, quota use, throttling and target failure	Support forecast peak volume without degrading the calling service

Apigee → Segment API	Segment response status, latency, timeout, `429`, retries and request size	Enforce quotas, payload controls and predictable error handling
Kafka Connect source and sink	Polling duration, records per poll, connector lag, task restarts, retry backlog and DLQ age	Process incremental records within the agreed data-freshness SLA
Segment Source ingestion	Received, accepted, rejected, filtered and failed events where available	Source processing must remain within agreed success and freshness thresholds
Reverse ETL	Query duration, selected records, processing duration, rejected records, retry result and watermark lag	Complete scheduled synchronisation before the next approved processing window
Warehouse Destination / Profiles Sync	Ping/Load result, duration, rows synchronised, rejected rows and table freshness	Complete within the agreed target-data availability SLA
Redshift	CPU, concurrency, queue time, query duration, load performance, capacity, connection count and storage	Segment integrations and validation queries must not materially affect existing MDS services
Segment → S3 export	Delivery delay, object count, data volume, object-size distribution and latest-object time	Deliver raw-event data within the agreed export-freshness SLA
Monitoring pipeline	Poll duration, API rate limits, S3 delivery, queue backlog and Splunk ingestion latency	Monitoring must identify service degradation within the agreed detection window

End-to-end service indicators

The following SLIs will be defined and baselined during performance and operational-acceptance testing:

- event-ingestion success rate;
- API latency and error rate;
- source-to-Segment processing latency;
- Reverse ETL completion time;

- Warehouse Destination and Profiles Sync duration;
- end-to-end data freshness;
- source-to-target reconciliation variance;
- failed and pending-record rate;
- DLQ backlog and age;
- Redshift query and load duration;
- S3 export freshness;
- monitoring telemetry freshness;
- incident detection and recovery time.

Target values, warning thresholds and critical thresholds must be agreed with the relevant service owners and captured in the Splunk Integration and Dashboard Specification or the associated NFR catalogue.

Performance assurance

Performance testing must cover:

- normal, peak and burst traffic;
- concurrent client-side and server-side event ingestion;
- Apigee quotas and Segment API rate limits;
- Kafka Connect batch size, polling frequency and retry behaviour;
- Reverse ETL overlapping with data publication;
- Warehouse Destination and Profiles Sync execution;
- Redshift concurrent query and load activity;
- target-validation queries;
- S3 export and Glue processing where in scope;
- failure and recovery conditions;
- monitoring collection during peak processing.

Baseline and peak results must record throughput, latency, processing duration, resource consumption, failure rates, reconciliation outcomes and impact on existing workloads.

Data Backup Policy

The solution must be capable to back up and restore data with reasonable amount of timey. This capability must have tested and restore time recorded during the testing phase.

Business Continuity and Disaster/System Recovery

Describe the redundancy built into the systems (RTO/RPO) in case of system or site failure. Business Continuity, Remote Recovery, failover and fall-back scripts must be tested before changes are made to those scripts

Twilio maintains a security and continuity framework based on ISO 27001 and SOC 2 Type 2. The platform includes automated monitoring, traffic routing, capacity management, and infrastructure redundancy to support ongoing service continuity.

Twilio performs regular backups of customer data, retains backups redundantly across multiple availability zones, encrypts backups in transit and at rest, and maintains capabilities to restore service availability following infrastructure failures. Security incident response processes and customer notification obligations are also contractually defined.

Automation

Deployment Automation

CLMNBA-5099: [FR] CDP Implementation | Infrastructure as Code (IaC) for Configuration Deployment (TBC) Deprioritised/Delayed

CLMNBA-5913: [FR] CDP Implementation | Git-Based Configuration Storage & Version Contro | READY FOR CONFIRMATION

Segment SaaS configuration will be automated through GitHub Actions using an approved ephemeral runner, protected repository and environment controls, the Segment Terraform provider, environment-specific Public API credentials, and an organisation-controlled remote state backend. The runner will initiate outbound TLS connections to the Segment Public API, with no inbound connectivity into TPG required. Production deployment will be subject to pull-request review, environment approval, change control, least-privilege credentials, security scanning, audit retention and defined MDP operational ownership.

Setup of Github repositories for each environment is needed even if there is delay in setting up GitHub actions for CI/CD pipeline.

GitHub Actions Automation Setup for Segment SaaS

Automation area	Proposed setup	Key security and governance requirements

Repository	Use a private repository in the TPG Telecom GitHub organisation to store Segment Terraform configuration, workflow files and controlled Public API scripts. Repository onboarding requires a BMC Helix entry and workload ID when using the automated SSP process; manual creation is also available.	Apply role-based access, branch protection, mandatory pull-request review and prevent direct changes to the production branch.
Workflow	Define GitHub Actions workflows within the repository to validate and plan changes on a pull request and deploy approved changes to the applicable Segment workspace. The application team owns the workflow configuration and ongoing support. 9	Separate validation, plan and deployment jobs. Only approved deployments may access production credentials. Retain plans, approvals, commits and execution results as release evidence.
Runner	Use an approved ephemeral GitHub Actions runner. Confirm enterprise runner capacity, licensing and whether a standard or dedicated runner is required with the GitHub/SRE tools team.	A self-hosted or dedicated runner introduces additional ownership for network access, isolation, patching and operational support.

Terraform execution	Install a pinned Terraform version on the runner during workflow execution. Use the approved `segmentio/segment` provider version to calculate configuration differences and apply supported changes through the Segment Public API.	Pin Terraform, the Segment provider and reusable actions to approved versions. Use API scripts under change control where required configuration is not supported by the provider.
Environment promotion	Configure separate deployment environments aligned to Dev, SIT/UAT, Pre-Production and Production Segment workspaces.	Apply environment-specific credentials and required-reviewer approval gates for higher environments.
Connectivity	The GitHub Actions runner initiates outbound HTTPS connections to the Segment Public API. No inbound connectivity from Segment into TPG is required for configuration deployment.	Permit outbound TLS only to approved Segment and dependency endpoints. Review any marketplace action, provider or binary download as part of the pipeline supply-chain boundary.
API credentials	Create a dedicated Segment Public API token per managed environment and provide it to the Segment provider at runtime through `PUBLIC_API_TOKEN`.	Store tokens as protected GitHub Environment secrets or retrieve them from an approved enterprise secret manager. Do not store tokens in code, workflow YAML, variable files, artefacts or Terraform state.

Credential controls	Assign the minimum Segment role required, define token ownership, and establish scheduled rotation and emergency revocation procedures.	Production tokens must be accessible only to the protected production deployment job. GitHub recommends least-privilege workflow permissions and warns that automatic redaction may not protect every transformed secret value. [docs.github.com]
Terraform state	Store Terraform state in an approved, organisation-controlled remote backend with locking, encryption, backup and recovery. The final backend remains to be selected.	Restrict state access to the deployment workflow and authorised support personnel. Prevent state from being uploaded as a workflow artefact or exposed in logs. The internal architecture review identified state location and exfiltration protection as unresolved.
Pipeline security	Run formatting, validation, linting, configuration security scanning, dependency checking and secret scanning before deployment.	Set workflow permissions to read-only by default, prohibit unapproved third-party actions, pin approved actions to immutable references where required, and avoid verbose API or Terraform logging in production.
Change and operations	Link production deployment to the applicable change record and define support procedures for failed deployments, API errors, provider upgrades, token rotation, state recovery and configuration drift.	Restrict direct UI changes in higher Segment environments. Run scheduled Terraform plans to detect out-of-band configuration changes and retain deployment evidence for audit.

Operational Activities

Operational activities will enable authorised support teams to monitor, pause, resume, rerun, recover and reconcile Segment integrations without requiring uncontrolled changes to production configuration.

Detailed monitoring, dashboards, alerts and reconciliation controls are defined in the Service Dashboards and Alerting section. This section defines the operational actions and support processes that use those controls.

Scheduled operations

The following scheduled activities must be documented in the operating model and runbooks:

- Redshift source-data publication;
- Kafka Connect source polling;
- API-based identity and profile updates;
- Reverse ETL synchronisation;
- Warehouse Destination and Profiles Sync;
- Redshift target validation;
- Redshift-to-MDS publication;
- Segment Public API monitoring collection;
- raw-event S3 export validation;
- deletion and suppression processing;
- Terraform drift detection;
- retention and lifecycle processing.

For each scheduled activity, document:

- owner;
- schedule and timezone;
- upstream and downstream dependencies;
- expected duration;
- input watermark;
- success and failure criteria;
- retry policy;
- maximum retry count;
- pause and resume process;
- rerun or replay process;
- incident conditions;
- reconciliation method.

Pause, resume and rerun controls

Operational processes must support pausing or rescheduling an integration when:

- upstream data is incomplete;
- schema validation fails;
- destination availability is degraded;
- reconciliation variance exceeds the approved threshold;
- an unauthorised configuration change is detected;
- credential or connectivity failure occurs;
- processing could cause duplicate or inconsistent customer data.

The relevant runbook must state whether the process resumes from the last successful watermark, restarts the complete processing window, or requires a controlled replay.

Retry, DLQ and replay

Failed API requests and connector records will follow controlled retry and DLQ processes. Operational monitoring will record:

- original message or batch identifier;
- original processing time;
- failure reason;
- retry count;
- final DLQ status;
- replay approval;
- replay time and new correlation ID;
- replay outcome;
- target-validation result.

Replay must preserve traceability to the original record, prevent duplicate processing and update the appropriate reconciliation result. Only authorised support roles may initiate production replay.

Data reconciliation and recovery

When source, Segment and target counts do not reconcile:

1. Identify whether the variance represents an approved filter, validation rejection, retry, pending record or unexplained loss.
2. Confirm the affected processing window and watermark.
3. Determine whether the source dataset can be safely reprocessed.

4. Pause downstream publication where continued processing may propagate incomplete or inconsistent data.
5. Recover using the approved rerun, retry or replay process.
6. Validate the target after recovery.
7. Update the linked incident and retain the reconciliation evidence.

A successful rerun does not close the incident until target validation confirms completeness and the operational watermark has progressed.

Credential operations

Runbooks must cover:

- OAuth credential rotation and revocation;
- Segment Source Write Key replacement;
- Apigee KVM update and validation;
- Segment Public API token rotation;
- GitHub deployment-token rotation;
- emergency credential revocation;
- validation after credential change.

Authentication failures and deviations from the expected access schedule must remain visible during and after rotation.

Configuration operations

Segment configuration changes will be managed through the approved GitHub Actions and Terraform pipeline wherever supported. Operational procedures must cover:

- failed validation, plan and apply;
- approval rejection;
- provider or API failure;
- Public API rate limiting;
- Terraform state recovery;
- rollback or corrective deployment;
- configuration drift;
- direct UI changes outside the pipeline;
- features not covered by the Terraform provider.

Direct changes in higher environments must be restricted. Approved emergency changes must be linked to a change record and reconciled back into the authoritative repository.

Monitoring-pipeline operations

The monitoring solution must monitor itself. Runbooks must cover:

- Public API poller failure;
- bearer-token failure;
- Segment API rate limiting;
- S3 landing failure;
- KMS or IAM denial;
- queue backlog;
- Splunk indexing delay;
- missing telemetry;
- dashboard staleness;
- restoration and backfill of monitoring records.

A telemetry-silence alert must distinguish between a healthy process with no business records and failure of the monitoring feed.

Incident and support process

Operational alerts will be assigned according to the affected component:

Area	Primary support responsibility
Segment configuration and synchronisation	CDP/Segment application support
Kafka Connect and API integration	Core Integration support
Apigee	API Platform/Digital support
Redshift and data validation	MDP/Data Platform support
PrivateLink, NLB and AWS infrastructure	Cloud/MDP infrastructure support
S3 raw export	MDP/Data Platform support
Splunk collection and dashboards	OSS/Splunk support
Authentication anomalies and unauthorised access	Security Operations

Deletion and suppression failures	CDP Operations, Data Owner and Privacy

The final assignment groups, escalation contacts and L1 to L3 responsibilities must be confirmed in the Target Operating Model and BMC Helix service hierarchy.

Operational acceptance

Before production go-live:

- dashboards and alerts must be implemented and tested;
- support groups and incident routing must be validated;
- thresholds must be agreed;
- runbooks must be accepted by Operations;
- retry, DLQ and replay must be demonstrated;
- reconciliation and target validation must be tested;
- credential-rotation procedures must be exercised;
- monitoring-pipeline failure must be tested;
- Redshift access-anomaly alerts must be validated;
- open monitoring gaps must be recorded in the operational acceptance or risk process

Test Automation

Describe the automated test framework used to validate Deployment and Operational Activities including Heath Checks.

Data Migration

The thresholds are for data replication- acceptable latency and fall-back process must be defined in the Solution Architecture Design. This Solution Architecture Design must also advise what to do if data falls out of sync

There will be one off sync of identity information into segment of current active records. post that it would only delta changed or new records will be synched to Twilio segment.

Patching

Segment is a SaaS offering and rolls out updates to customer instances when applicable, including major/minor releases and security updates. If an update may impact a customer uniquely due to their configuration, their Customer Success Manager will notify them ahead of time and facilitate any necessary training.

Types of changes: New Integrations, Product Enhancements, SDK & Library Updates, Infrastructure Improvements, bugs fixes, Security upgrades etc.

Impacts from these changes: Functional Continuity and Stability, Performance Improvements, Expanded Capabilities, Infrastructure Scaling. Usually all changes/upgrades are backwards compatible, but the team provides reasonable window for any major impact changes/deprecate

Capacity

The integrity of the solution is dependent on capacity contributions from all elements spanning the breath of this solution. Hence the capacity description should address all the elements listed in section 2.4.1

System Name	Describe the impacts, if any, to the System because of the capacity requirements
Marketing Data Store	1. Additional nonproduction environments to integrate to non-production environments of Twilio segment. 2. New Redshift cluster component installed to manage current active and 13 months inactive status of customer accounts. 3. MDS reporting from data exported by Twilio segment to be made available online for 5-year period. 4. S3 raw events data to be stored for compliance purpose for 7 years. Use different storage types to minimize costs.

Table 26: Capacity Description

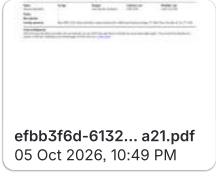
Include the baseline growth assumptions / projection / BVM version here if available.

Based on business projections, Project should size Application / Infrastructure to ensure that a minimum 18 months capacity is available post go live.

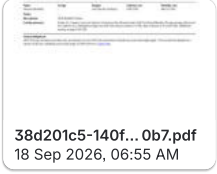
For every new component being added, it should be possible to report on its capacity utilization via capacity management tool.

Additional storage required for MDS workload to store 7 years of raw data from Twilio Segment in S3. 5 years of data should be available for consumption from MDS Redshift Serverless.

The estimated monthly charges to be around 3,550 AUD per month. rough sizing estimate attached. This is based on average volume of data across 4 years. The expectation is that initial data build up would be less in year 1 and will progressively increase year on year.



For CDP Redshift cluster, attached is the estimated based on average volume expected to be around $654 AUD.



Portability

Describe how the solution can be ported to a different environment. (E.g. from physical to cloud or from one cloud environment to another)

Audit

Describe how the logs can be audited.

Service Level Agreements (SLAs)

Describe the SLAs associated with the solution

SLA name	Description
Connection & Response time	*How long will be the connection and response time*
Switchover time	*In case of failover how long will it take to divert all traffic to the standby / redundant instance of the application*
Portability	*How long will it take for the solution to be ported from one environment to another*

Incident Management	*How soon a trouble ticket will be resolved by the vendor based on the priority level*
....	

Table 27: Service level agreements

Update of Operational Databases/Processes

If your solution has changes to the operational databases, please make sure that the respective systems and processes are updated. The changes could be new application, change to existing application, removal of application etc.

System Name	Description
SSP	*Self Service Portal updated so users can request access to the solution*
UAM Process	*User Access Management updated*
PUAM	*Solution onboarded into Azure Entra ID setup for privileged identity management.*
CMDB	*Solution logical and physical components and services offered (ideally updated dynamically).*
Active Directory	*Centralised Authentication and Authorisation used for all components*
IDM	*Solution onboarded to SailPoint Identity and Access Management for user access provisioning*
FIM	*Solution onboarded to File Integrity Monitoring system where required for PCI compliance*
PCI CDE	*No credit information is stored.*
SIEM	*Audit logs will be ingested into SIEM or equivalent solution.*

Table 28: Operational Databases

Environments

As part of the Onboarding SAD SAD - #901: Ignite - Customer Data Platform Onboarding/Standalone there are three workspaces provisioned from Segment for Vodafone brand. Two of them in Non Production zone i.e Dev and SIT/UAT and One Production zone.

S3 MDS: Role-based access

The core patten for IAM role access to S3 buckets for data transfer: The trust relationship uses the **Segment Workspace ID as the External ID**, and the role's permissions policy is scoped to a specific S3 prefix/bucket. This External-ID-per-workspace mechanic is exactly what you use to drive environment segregation — because if non-prod and prod are different Workspaces (or different Sources), they get **different Workspace IDs**, which means each needs its **own dedicated IAM role**, each with its own trust policy and its own scoped bucket/prefix.

CLMNBA-5116: [FR] CDP to MDS | S3 - Environment-Specific Buckets READY FOR CONFIRMATION

Redshift MDS: Role-Based Access, Segregated by Environment + Domain

CLMNBA-5128: [FR] CDP to MDS | Redshift - Environment-Specific Configuration READY FOR CONFIRMATION

Unlike S3 (bucket/prefix + IAM role per environment), Redshift MDS segregation works through a **naming-convention-driven RBAC model** rather than separate clusters. TDA team's approved pattern is:

Persona	Role Naming Convention	Example
Database Admin	`rs_<domainname>_<env>_dba_role`	`rs_consumer_dev_dba_role`
Operator (read/write, no admin)	`rs_<domainname>_<env>_operator_role`	`rs_consumer_dev_operator_role`
Data Domain Expert (governance)	`rs_<domainname>_<env>_<datagroup>_dataexpert_role`	`rs_consumer_dev_sales_dataexpert_role`
Data Consumer (Privileged/PII)	`rs_<domainname>_<env>_<datagroup>_priv_consumer_role`	`rs_consumer_dev_sales_priv_consumer_role`

Data Consumer (Non-PII)	`rs_<domainname>_<env>_<datagroup>_consumer_role`	`rs_consumer_dev_sales_consumer_role`

Environment (`dev` / `sit` / `prod`) is embedded directly in the role name, so a user's AD-group-to-Redshift-role mapping automatically scopes them to one environment only. This is already live in practice — e.g., Sidd Aggarwal's access request for you used the pattern `SGAZ-APP-AWS-Redshift-MDS-Prod-ReadOnlyNonPII` , confirming env + PII-tier is baked into the role/AD-group name itself. The same pattern extends to the Data Exploration RBAC rollout: `TPGT-Data-Exploration-prod-MDS-DataCons-NonPII` , `TPGT-Data-Exploration-prod-MDS-DataConsPII` , `TPGT-Data-Exploration-prod-MDS-Operator` , etc., each mapped to a dedicated IAM role and Azure AD group.

Key segregation mechanisms on top of the role naming:

- **Dynamic Data Masking (DDM)** — PII columns (e.g., email) are masked at query time based on the requesting role/persona, without altering stored data.
- **Prod → Non-Prod data flow is one-directional and masked**: sensitive production data in S3/Redshift is masked in the Production Zone first, then copied to non-prod via TPG-approved transfer mechanisms — never the other way round. Your iLLuminate SOP documents the actual mechanics: a Lambda-driven "Dynamic Redshift DataUnload Process" masks PII and unloads Prod → Non-Prod on a config-driven counter/trigger.
- **Cross-account data sharing** (CDS→MDS, or Central Governance→MDS) uses Lake Formation Tag-Based Access Control (TBAC) for the strategic model, or native Redshift RBAC/Dynamic Masking for the current tactical model — both enforce PII vs non-PII visibility per role regardless of which account the data physically lives in.
- **Network layer**: External authentication via IAM roles/Azure AD (not Redshift-internal passwords) is the default going forward, and the Segment↔Redshift path is fronted by an NLB so the cluster itself is never publicly accessible.

Redshift CDP cluster: Schema-Based Segregation (per the Segment CDP "Data Share" design)

For the specific CDP Redshift cluster instance used as the Segment reverse-ETL data share segregation is done via **dedicated schemas per data-flow direction and sensitivity tier**, sitting behind Multi-AZ, private-subnet-only deployment, KMS CMK encryption at rest, TLS in transit, and dedicated Redshift database IAM role per environment and integration, with schema-scoped privileges:

Schema	Contents	Direction	Access
`pii_identify_share`	MSISDN, Email, SHA-256 hashed identifiers (email) + change log	MDS → Segment (poller Lambda)	Read-only for Segment's Reverse ETL user
`nonpii_traits_share`	Non-PII trait model views — tenure, plan family, propensity, consent flags	MDS → Segment (Reverse ETL)	Read-only for Segment
`segment_export`	Landing schema for Segment Warehouse destination output (identifies/tracks/pages/screens/users/accounts)	Segment → MDS load-back	Write access limited to Segment's DB user only
`__segment_reverse_etl`	Segment-managed sync-state schema	Internal to Segment	Segment's DB user needs write access to this schema only

This mirrors the same least-privilege principle as Redshift: **Segment's Postgres user is scoped to specific schemas only**, never given superuser/admin, and PII vs non-PII is split into separate schemas rather than separate databases — similar in spirit to the Redshift dynamic-masking approach but implemented as schema-level segregation instead of column-level masking.

Environment segregation for CDP Redshift Cluster follows the same BRS mandate as S3/Redshift — BR039/FR037-038*"Environment-Specific Configuration for AWS S3"* and*"...for AWS Redshift"* — meaning each environment (Dev/SIT/Prod) should get its **own Redshift Cluster instance (or at minimum isolated schema set)**, its own dedicated Redshift database IAM role per environment and integration, with schema-scoped privileges, consistent with the full-segregation principle explicitly stated in your Ignite R3 BRS:*"full segregation across all AWS (MDS) infrastructure environments will be enforced for Segment CDP data... This includes dedicated IAM roles, isolated S3 buckets and folder structures, and separate Redshift schemas per environment to prevent cross-environment data contamination"*.

Consolidated View — All Three Layers Side by Side

Layer	Env Segregation Mechanism	PII Segregation Mechanism	Access Grant Mechanism
S3	Separate bucket/prefix + dedicated IAM role per env	Separate prefix/bucket for masked vs raw	IAM role trust scoped to Workspace ID (External ID)
Redshift Serverless MDS	Env embedded in role name (`_dev_` , `_sit_` , `_prod_`)	Dynamic Data Masking + separate consumer/priv_consumer roles	IAM role / Azure AD group → Redshift DB role mapping
CDP Redshift Cluster MDS	Dedicated instance/schema set + IAM DB role per env	Separate schemas (`pii_identify_share` vs `nonpii_traits_share`)	Dedicated Redshift database usernames and passwords per environment and integration, with schema-scoped privileges.

Production Environment & Pre-Production Environment

It is expected that all of the changes to the Elements described in Section 1.4 and Components described Section 2.4 will be introduced in to the Production environment. If there is any variation, then describe.

The Production workspace provisioned for Segment can be split into Production and Pre-Prod and the can be kept separate.

Single Workspace with environment-specific Sources (and Tags), e.g., `App_Preprod` and `App_Prod` , plus Tags for RBAC alignment and operational clarity.

How to prevent prepared activity from impacting prod?

- **Separate Write Keys per environment Source**
  Segment generates a **unique Write Key** per Source; Preprod applications must use the Preprod Write Key so test events remain isolated and do not pollute production.
- **Separate Destinations per environment**
  Configure Preprod Sources to route only to **sandbox/testing destinations**, and Prod Sources

to route only to **production destinations**.

- **Separate Engage Spaces (recommended)**
  Create environment aligned
- For Engage Spaces to isolate audience and activation behaviour between Preprod and Prod.
- **Tagging and RBAC alignment**
  Use **environment-specific Tags** (recommended by the document) to organise resources and align access controls.

The Pre-Prod application connections destination and sources connect to the other Preprod and sandbox environments only. The Production application in the workspace will access all the live integrations for source and target.

System Integration / User Acceptance Testing Environment

The above setup for production is replicated form SIT/UAT connections within the non-production workspace.

Development Environment

The above setup for production is replicated for DEV connections within the non-production workspace.

Possible Environment Mapping

The table below represents the proposed/draft environment mapping for Segment Workspaces/Sources, S3 buckets, Redshift schemas, and IAM roles. This mapping must be reviewed and finalised — including confirmation of naming conventions, bucket provisioning, and IAM trust/permission scoping — during the build phase and **prior to SIT commencing**, to ensure connectivity and access are correctly segregated before test execution begins.

Environment	Segment Workspace/Source	S3 Bucket	Redshift Schema	IAM Role
Dev	segment-dev-web, segment-dev-mobile	segment-data-dev	segment_dev_web	segment-dev-role
Non-Prod (SIT)	segment-uat-web, segment-uat-mobile	segment-data-uat	segment_uat_mobile	segment-uat-role

Production	segment-prod-web, segment-prod-mobile	segment-data-prod	segment_prod_web	segment-prod-role

Key items to confirm during build:

- Final naming convention for Workspaces/Sources, buckets, schemas and IAM roles (Dev / SIT / Prod), per brand if applicable.
- Each environment has its own **dedicated IAM role** with trust policy scoped to that environment's Segment Workspace ID (External ID) and permissions scoped to its own bucket/prefix — no shared roles across environments.
- Bucket-level controls applied consistently per environment: public access blocked, versioning enabled, encryption at rest (SSE-KMS) configured.
- Connectivity (network/security group, endpoint access) validated end-to-end for each environment ahead of SIT test data flow.

Testing Considerations

Testing should ensure and demonstrate, solution should meet Availability, Performance and Accuracy (Success ratio) requirement of new functionality being delivered as per TPG TELECOM benchmark. And same for existing processes/ services are not degraded, if not improved upon.

For database, demonstrate all database housekeeping activities required to maintain the integrity and performance of the database must be able to run at any time without impact to supported business process/ services.

If data replication/migration is involved, describe how

(a) The replication software and link would be tested,

(b) Monitors are active on the storage arrays and links and there is no performance drop for any of the business service supported while data migration activity is on

(c) Technology Operations staff and their Managed Service Providers are familiar with and have accepted the replication monitoring tool/s.

Business Continuity, Remote Recovery, failover and fall-back scripts must be tested before changes are made to those scripts

If project/ change is introducing new Application/Infrastructure components, recovery of OS, Application, DB from backup should be tested and demonstrated to meet within business approved outage for such scenarios

Monitoring, Alerting and Error handling scenarios introduced and modified should be validated and documented as part of testing.

Error handling: This standard requires, for Business Process/ Service added/ customised, Distinct Error Code and Error Messages should be generated for every failure scenario, across Application stack and meaningful message displayed on front end for end customer consumption. Each Error should have unique code and message to be stored in respective originating application for debugging and reprocessing

If project/ change is introducing new Application/Infrastructure components, recovery of OS, Application, DB from backup should be tested and demonstrated to meet within P1 SLA timeframe.

All feature and compliance against security requirements should be tested

Test area	Testing requirement
CloudFront reverse proxy and zero rating	Validate routing of approved Segment CDN and ingestion traffic through Vodafone CloudFront endpoints, with no unexpected direct Segment calls. Test WAF, TLS, path controls, environment isolation, caching and graceful failure. Confirm zero-rating behaviour for normal, barred and zero-balance scenarios without impacting customer journeys.
Inactive contact purge in Segment	Validate if purging of contact through API is reflected in Segment workspace.

Operation Design Standards Exceptions

Required for ARB presentation

Next

Back

Describe any of Operations Design standards that will not be met by said solution with reasoning. Current list of standards is available at here

Security Controls Compliance

Required for ARB presentation

Next

Back

Required for both SO and SA

If **ALL** of the three conditions below are met for this solution, please complete **Section A**. If **One or more** conditions fail, the project needs to complete **Section B**.

Condition 1: No new System being introduced by the initiative.

Condition 2: No new integration is being introduced by the initiative.

Condition 3: The initiative is NOT a life cycle upgrade.

Section A:

I acknowledge that the solution satisfies **ALL the three conditions** above and confirm that there will be **"NO CHANGE TO EXISTING MANDATORY SECURITY CONTROLS"** with this solution.

NOTE: As part of TRA, if the GRC team identifies a possible increase to the security risk from this initiative, the project MUST complete the mandatory security controls and provide all required information to complete the TRA.

Please acknowledge YES/NO by answering as "NO CHANGE to EXISTING" and confirming that the mandatory security controls will be completed during TRA, if requested by GRC.

Section B:

Please specify how this solution meets all required Mandatory Security Controls by populating the table below.

Detail descriptions on the requirements for these controls, along with links to the respective Enterprise Security Standard, can be found on the Mandatory Security Controls confluence page.

Control No	Name of the Control	Applicability of the Control	Compliant (Yes, No or N/A)	How? (Please provide the brief explanation on how the solution supports your answer)
1	NGFWL (Next Generation Firewall)	All traffic flows must pass through a NGFW between on-prem networks or when flowing between on-prem and public cloud. TPG's AWS Public cloud logs all traffic flows at the platform level. *Does this initiative deploy a new system into a new network on-prem?* *Are all firewalls that handle traffic for this initiative, NGFW's?*	Yes	TPG AWS traffic will use approved network controls, security groups and flow logging. Segment-to-Redshift connectivity will use private connectivity, with no public exposure of the Redshift cluster. Client traffic terminates at Vodafone-controlled CloudFront endpoints.
2	WAF (Web Access Firewall)	Any web application that exposes any interface externally must ONLY exist behind one of TPG's approved WAFs. *Is this initiative introducing any publicly exposed interfaces and, if so, are these interfaces protected by an approved WAF?*	Yes	Segment has WAF controls to protect internet-facing web services from common web threats and OWASP risks. Vodafone CloudFront endpoints will use

				AWS WAF. Only approved Segment origins, paths and methods will be allowed, with request limits and rate controls. The proxy will not operate as an unrestricted forward proxy.
3	DDoS Protection (Distributed Denial of Service)	All TPG-managed network subnets (on-premises, private cloud, and public cloud) must be protected by approved DDoS detection and mitigation controls. ***Is this** initiative **introducing** any publicly exposed interfaces and, if so, are these interfaces protected by an appropriate (D)DoS solution**?***	Yes	Twilio uses AWS Shield Advanced and associated AWS edge protections. Vodafone CloudFront endpoints will use approved AWS DDoS controls. Apigee applies SpikeArrest and per-Source quotas.
4	EDR (Endpoint Detection & Response)	*All in-scope TPG managed endpoints, including servers, workstations, and laptops, must have an approved EDR solution installed, active, and reporting to the central management platform.* *Are all in-scope endpoints covered by an approved EDR solution?* *The EDR Standard is available **here**.*	Yes	Twilio manages EDR within its SaaS environment. Applicable TPG-managed compute and administrative endpoints will use approved EDR controls and managed-device access.

5	NDR (Network Detection & Response)	*All core TPG networks, including data centre, cloud, and interconnect environments, must apply approved NDR controls to provide continuous monitoring and threat detection.* *If you are deploying a new network segment, additional steps may be required.* *You can request evidence of your compliance by emailing SAE-SecurityEngineering.* *Please include your IP(s) or IP ranges in your request.*	Yes	Twilio Segment applies network segmentation, isolation and hardened configurations to protect computing resources and customer environments from unauthorised access. Network security controls generate alerts used to inform ongoing hardening, supported by quarterly vulnerability scanning and annual penetration testing of segmentation controls. Remote access to production assets requires MFA over an encrypted VPN. AWS network telemetry, PrivateLink/NLB health and Redshift connection behaviour will be monitored. Alerts

				will cover unexpected access paths, abnormal connection frequency, authentication failures and unauthorised schema access.
6	SIEM (Security Incident & Event Management)	*All systems, applications, and platforms must forward relevant security logs and events to the approved SIEM platform for monitoring, alerting, and compliance reporting..* *Are all **OS, DB**, **and App** layers **of** this initiative, appropriately logging all system and security logs to the TPG **SIEM platform?***	Yes	Yes, Twilio offers integration capabilities with enterprise Security Information and Event Management (SIEM) tools, such as Splunk, for security event logging and monitoring. Twilio's Event Streams can be utilized to send consistent and structured data to various sinks, including Amazon Kinesis and Segment, which can then be integrated with SIEM tools like Splunk.

				This handled as part of on-boarding SAD. SAD - #901: Ignite - Customer Data Platform Onboarding/Standalone | Security Controls Compliance
7	Access Controls	*All systems, applications, and services must enforce RBAC and integrate with TPG's AD for identity and access management.* *Further information can be found in the **user and system access control standard (Sec 23-TS)**.*	Yes	Handled as part of on-boarding SAD SAD - #901: Ignite - Customer Data Platform Onboarding/Standalone | Security Controls Compliance
8	PUAM (Privileged User Access Management)	All privileged accounts across systems, applications, and infrastructure must comply with the PAM Federated Model. System owners must adopt one of the approved solutions to manage privileged access in line with TPG's control objectives. *Further information can be found in the **user and system access control standard (Sec 23-TS)**.*	Yes	Handled as part of on-boarding SAD SAD - #901: Ignite - Customer Data Platform Onboarding/Standalone | Security Controls Compliance Setup will be done using Entra ID.
9	VM (Vulnerability Management)	All TPG-managed systems, applications, and infrastructure must undergo regular vulnerability scanning, risk assessment, and remediation in line with enterprise security requirements.	Yes	Twilio Segment maintains a risk-based vulnerability management process covering corporate systems, cloud infrastructure,

				source code and custom-built solutions. Vulnerabilities identified through open-source and commercial scanning tools, third-party notifications and security reports are validated, risk-rated and assigned to the appropriate engineering teams for remediation through a managed workflow. Risk ratings consider exploit severity, affected scope and existing mitigating controls. TPG-managed integrations, pipelines, configurations and dependencies will use approved scanning, assessment and remediation processes
10	Data Security	*All data managed, processed, or stored by TPG must be handled in accordance with*	Yes	Twilio Segment uses Crowdstrike

		the Information Classification and Handling Policy. Systems and services must implement appropriate controls (e.g., encryption at rest and in transit) based on data classification. • *Information Classification and Handling Policy (Sec-03B)* • *Cryptographic Standard (STA_01)* *Does this initiative impact a system that stores, processes, or otherwise handles Highly Sensitive Information and, if so, is that information encrypted in transit and at rest?*		Falcon. Twilio Segment has a network-based data loss prevention (DLP) system in place. The system scans for sensitive data that may potentially be exposed publicly or improperly stored. i.e. SSN or credit card PAN and has alerting and quarantine capabilities. Data is protected using TLS in transit, KMS-backed encryption at rest, RBAC, masking, schema segregation, retention and deletion controls. PII classification for raw events and event data, data location, retention and deletion coverage must be reconciled.
11	API Security	*API Security ensures that all Application Programming Interfaces are protected against unauthorised access, misuse, and*	Yes	Data transmitted through the Twilio Segment platform

		data breaches. This control covers the full API lifecycle—design, deployment, maintenance, and retirement—focusing on authentication, authorisation, data protection, rate limiting, input validation, and monitoring. *Follow the TPG API Security Standard. A summary of its contents include:* • *Implement strong authentication and role-based access controls.* • *Protect sensitive data in transit and at rest, and prevent excessive data exposure.* • *Enforce rate limiting, input validation, and secure configurations.* • *Monitor and log all API activity to the central SIEM.* • *Maintain an up-to-date API inventory and documentation.* *Further information can be found in the API Mandatory Controls Standard (Sec-28TS)*		is encrypted in transit via TLS 1.2. CloudFront will enforce fixed Segment origins, approved paths and methods, request controls, WAF protection and monitoring. Client-side calls use Source-specific Write Keys with WAF, CSP, rate limits and schema controls. Server-side workloads use OAuth 2.0 to Apigee, with Source authorisation and Write Key injection from encrypted KVM. Public API tokens are segregated by environment and purpose.
12	Network Segmentation	*All TPG environments (on-prem, private cloud, and public cloud) must implement logical or physical segmentation aligned with security zoning principles. Critical systems must be placed in restricted zones with tightly controlled access.*	Yes	Twilio Segment will utilise separate Production and Non-Production workspaces, with source and

		Further information can be found in the [Network Security Standard (Sec-25TS)]		destination connections configured within the corresponding workspace to maintain environment segregation and prevent cross-environment data processing. For TPG AWS setup has separate account sets for Prod/NPE. Production and non-production use separate Segment Sources, credentials, IAM roles, S3 locations and database users. Redshift access is private and restricted by schema and dedicated integration accounts.
13	Application Security	*All applications built, deployed, or integrated within TPG's environment must follow secure development practices, undergo security testing, and address findings before production release. Development teams must align with TPG's Secure Development Lifecycle (SDLC) and*	Yes	Twilio Segment applies secure software development, vulnerability management, security testing

		integrate with approved security tools and processes. *Does this initiative:* • *Use a Static and Dynamic application Security Technology,* • *Appropriately handle Errors,* • *Log securely, ensuring sensitive information is not written to logs, and* • *Securely handle sessions,* *in accordance with the TPG Enterprise Application Security Policy?*		and penetration-testing practices to its vendor-managed SaaS platform. TPG-managed integrations, CloudFront proxy configuration and custom components will follow the TPG Secure Development Lifecycle, including approved CI/CD controls, security testing and remediation of identified findings before Production deployment. Secure logging, error handling and access controls will be implemented, with Segment platform access restricted through Entra ID SSO, MFA and role-based access controls. TPG-managed components will follow the Secure Development

				Lifecycle, including review, scanning, protected CI/CD, secure configuration and testing. CSP, WAF, secure logging and controlled error handling apply to Digital integrations.
14	Customer Security Control	*Does this initiative ensure customer-facing systems and access paths are designed and operated to protect customer access to TPG services and reduce the risk of unauthorised access, fraud, misuse, and customer harm, including secure customer authentication and authorisation, session security, monitoring of customer access activity, and protection of customer data in accordance with TPG's Data Security Standard and privacy obligations?* *Appropriate safeguards for vulnerable customers, where the risk of harm is higher, is especially important.*	N/A	Customers do not access the Segment tenant. Client analytics must fail gracefully and not affect customer journeys. Client-provided identity is not treated as verified identity without authoritative server-side corroboration.

Table 29: List of Mandatory Security Controls

For detail security design aspects, please check Security Design Non-Negotiable. Please call out below if any security design aspect is not achievable by this solution.

Legal and Regulatory Aspects of the Proposed Solution

Required for both SO and SA

Legal and Regulatory

Does the solution impact any area of TPG TELECOM's compliance obligations? If yes, please describe how the solution will ensure that we remain compliant to the obligation. The TPG TELECOM obligations register can be accessed here.

Compliance Area	Describe how the solution will comply to the respective compliance (if applicable)

Table 30: TPG TELECOM Compliance Obligations

PCI DSS Compliance

Required for ARB presentation Next
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Clearly specify how the solution meets the PCI Data Security Standards or if PCI compliance is Not Applicable.

No PIC data stored in Twilio Segment platform.

Legal Interception

If your solution includes and Core network element, you need to ensure that an LI assessment has been done and if LI is impacted its solution is reflected in the LI design document

Regulatory Codes and Acts

Required for ARB presentation Back

If the solution involves a change to any system / process that supports TPG TELECOM compliance to a Telecommunication Regulatory instrument as per TPG compliance obligations, pls describe how the solution would ensure TPG TELECOM compliance to the regulatory instruments. The list of regulatory obligations are available here.

Area	Relevant Apps	Describe how the solution will comply to the respective Act / Standard (if applicable)
Number Portability	MNP/Inp Port Admin (Paradigm one) Siebel; MNP Fusion; MNP	N/A

	UI; MNP OCSG; ONE MNP; PCMON	
Premium Services	AAA, Siebel	N/A
Emergency Services	MSC, GMLC (eMOLI), STP, MGW, IPIC/SGB*; NetScout	N/A
LI Services	LI systems	N/A
SOCI – End user Equipment/Management	ACS; Mountainview New CPE Mgmt System	N/A
SOCI – Network/Security Operations	Netcool, NI4CC, ITSM, SIEM New Monitoring/Control System	Audit trail from application passed to Splunk and Siem Application logs will be passed to Splunk
Legal Requirements	Siebel; Fusion; Symbio (VML & UC)	N/A
SOCI – Cloud Hosted Systems	Mission/Business Critical Apps that are hosted on the SaaS or a new Public Cloud environment	Business critical app
Customer Rights and Protection	Siebel, BRM, Tallyman, Contact Centre, ITSM; UIM(wholesale); Symbio; TCS; HOBS; ACS	N/A

Table 32: Legal and Regulatory Impacts

Non-Production Environment (Optional)

Required for SA

Describe the components that perform roles in the delivery of the end-to-end solution in the Pre-Production Environment.

Non-Production Environment Solution Overview

Provide a Summary description of the Model environment needed, to enable the solution to be implemented on the model

Changes to Non-Production Environment Architecture

Describe the Proposed architecture of the Model environment.

Model Building blocks (including testing tools)

Clearly describe what Building Blocks will be modified, consolidated, added or removed to enable the solution to be implemented

SYSTEM Name	New or Proposed Change	Contribution / Impact Description including impacted realised capability

Table 33: NPE Component Contribution / Impact

Deployment/Migration Strategy

ℹ Required for SA

Describe how deployment/ migration would be phased to ensure no/minimum outage to supported business process/ service

---

Future Considerations

ℹ Required for both SO and SA

Appendix

ℹ Optional for both SO and SA

Where necessary use this Appendix to provide additional information associated with this Solution Outline.

A. Auth0 to Segment direct connector

Auth0 is used for customer authentication signup registration, login checking for digital app and web. Auth0 has a connector that enables sending certain events to Twilio Segment.

Segment Integration with Auth0

- **Channel Applications** – Web, mobile, or API-driven channels initiating authentication flows through Auth0.
- **Auth0 Tenant** – Performs authentication and emits tenant logs representing authentication outcomes.
- **Auth0 Segment Connector (Log Stream)** – Native Auth0 integration configured with the Segment write key and optional filtering of event codes.
- **Twilio Segment Source** – Receives inbound events.

The integration uses the **Segment write key** configured in the Auth0 Segment stream. This should be managed as a controlled integration secret and handled in line with enterprise credential management practices. Auth0's marketplace setup for Segment requires the write key as part of stream configuration.

Only the minimum approved Auth0 event codes should be forwarded, reducing unnecessary identity and authentication telemetry exposure into downstream systems. Auth0 support guidance confirms that event type filtering is supported after the log stream is configured.

Events to enable

For your requirement, the minimum Auth0 event codes to stream are:

- `ss` = **Success Signup** [auth0.com]
- `s` = **Success Login** [auth0.com]
- `f` = **Failed Login** [auth0.com]

A practical optional addition is:

- `fs` = **Failed Signup** if you also want visibility of failed registrations. [auth0.com]

As part of build go review the Jira ticket and the data requirement sheet to finalise the events.

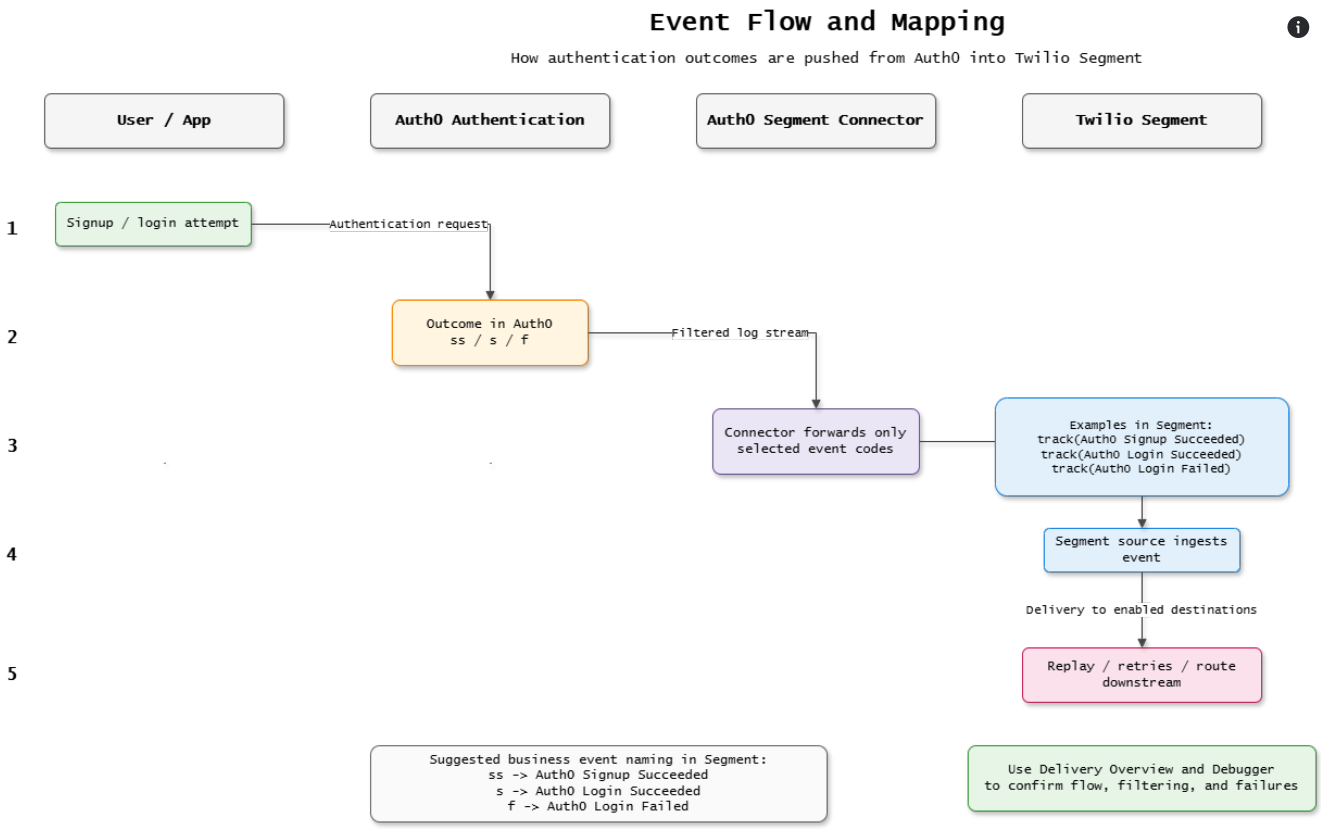


Figure A: Sequence diagram for Auth0 event collection

sample event from Auth0.

```
{
  "date": "2026-05-06T10:48:00.123Z",
  "type": "s",                                  // Auth0 Event Code (e.g., 's' = Successful
Login)
  "description": "Successful login",
  "connection": "sms",                          // or "email", "Username-Password-
Authentication"
  "connection_id": "con_TGfH8JhSZZVt",
  "client_id": "xv2nAqBz3NGFyH4jsK5d",
  "client_name": "OneApp",
  "client_ip": "103.197.74.228",
  "user_id": "auth0|667406c99c145b3b51dd4959",
  "user_name": "john.doe@example.com",
  "user_agent": "Mozilla/5.0 (iPhone; CPU iPhone OS 16_7 like Mac OS X)",
  "log_id": "90020260606104800000000876351923456789012345678 9",
  "isMobile": true,
  "details": {
    "auth_method": "pwdless-sms",
    "session_connection": "sms"
  },
  "hostname": "tpgtelecom.au.auth0.com",
  "audience": "https://tpgtelecom/api",
  "scope": "openid profile email",
  "strategy": "sms",
  "strategy_type": "passwordless",
  "tenant_name": "tpgtelecom"
}
```

B. Twilio Segment to Pega CDH

Twilio Segment CDP will integrate with Pega Customer Decision Hub (CDH) via a real-time event streaming pipeline, using Segment Destination Functions, Apigee as the API gateway/microservice layer, and AWS MSK Kafka as the message broker. This pattern aligns with TPG's enterprise integration standards and ensures decoupled, scalable, and secure event delivery for marketing decisioning.

Data Flow

1. **Segment CDP (Source)** — Collects first-party customer events (identify, track, page, screen) from digital sources (websites, OneApp, server-side). Segment's event pipeline processes, filters, and routes events to configured destinations.
2. **Segment Destination Function** — A custom JavaScript function hosted within Segment's serverless infrastructure acts as the outbound integration point. The function transforms the Segment event payload (JSON) into the agreed schema expected by Apigee and issues an HTTPS POST to the Apigee proxy endpoint. Destination Functions provide built-in retry logic, real-time monitoring, alerting, and enterprise-grade scalability (trusted by JPMC, handling over 200 billion calls/month). No external infrastructure is required from TPG for this layer.
3. **Apigee API Gateway (Microservice Layer)** — Apigee receives the HTTPS request from the Segment Function and acts as the secure integration and governance layer. It performs:
   - **Authentication & authorisation** (OAuth2/JWT validation)
   - **Schema validation** and payload transformation
   - **Rate limiting & throttling** policies
   - **Security policy enforcement** (DataMasking for PII, TLS/mTLS)
   - **Logging & observability** (Apigee analytics, integration with Splunk/SIEM)

   After validation, Apigee acts as a **Kafka Producer**, publishing the validated and transformed JSON message to the designated AWS MSK Kafka topic.
4. **AWS MSK Kafka (Message Broker)** — The Kafka topic serves as the durable, fault-tolerant message buffer between Apigee and Pega CDH. MSK ensures high availability, scalability, replay capability, and decoupling of producers (Apigee) from consumers (Pega). The topic is configured with appropriate partitioning, retention, and encryption (SSE-KMS). Kafka cluster configuration is jointly managed by Pega and TPG.
5. **Pega CDH (Consumer)** — Pega CDH subscribes to the Kafka topic using a **Kafka Data Set** and consumes messages via **Pega Stream Services** (or external Data Flow components). A real-time **Data Flow** processes each incoming event, transforms the JSON payload, and

feeds it into the CDH decisioning engine. This enables Pega to react to customer digital engagement events in near real-time for Next Best Action (NBA) decisioning, adaptive model updates, and journey orchestration.

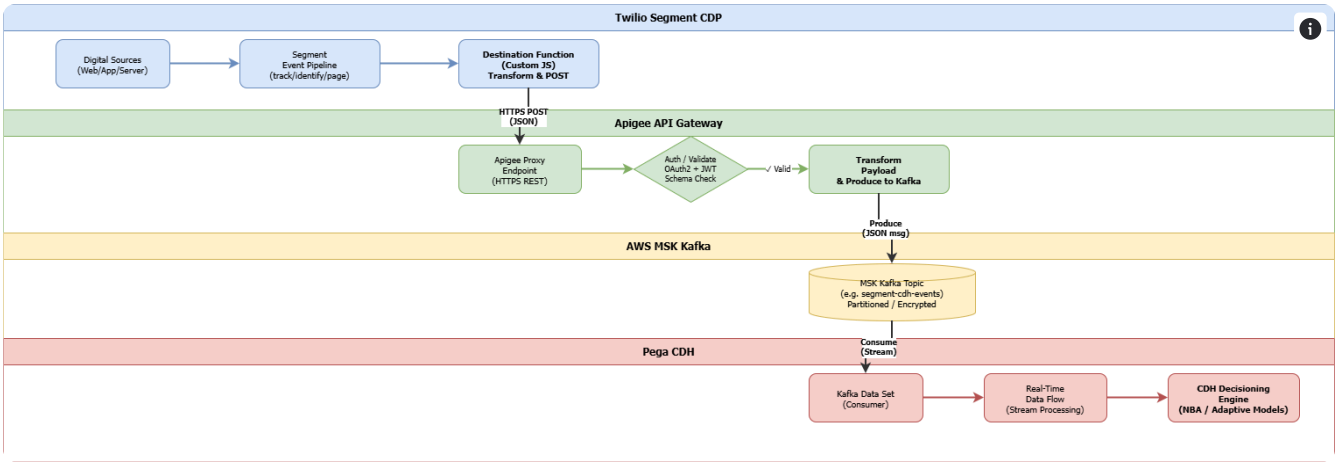


Figure: Segment to Pega CDH Data flow

Generic Microservice forwarder

The microservice should be a **dumb forwarder**, not a field-aware transformer. It should never need to know about individual traits or event properties — it just maps the Segment envelope structure to a Kafka message envelope structure.

Segment destinations can pass traits/events. Segment Engage can pass Audiences created in segment.

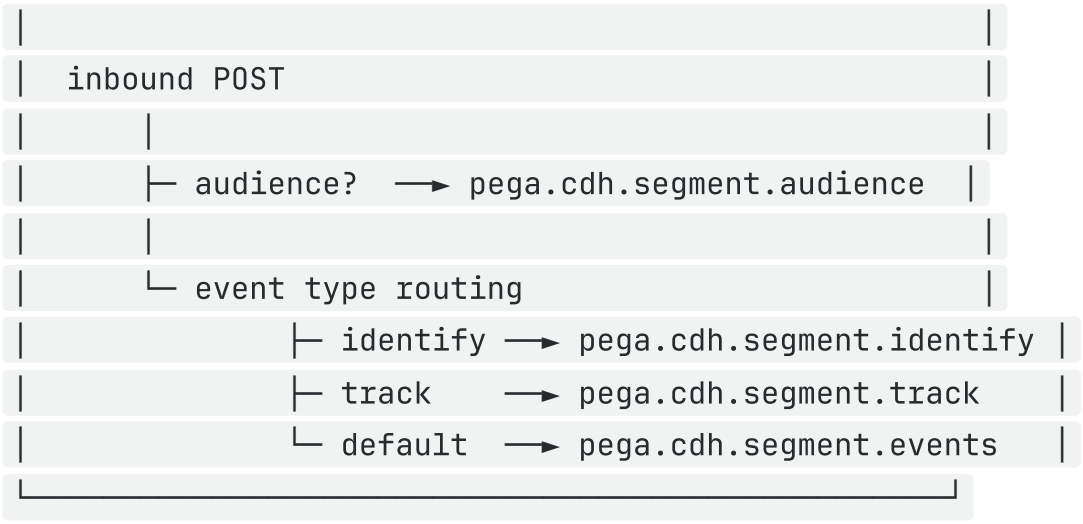
The micro service forwarder will send out to Pega with a wrapper for the segment payload

```
{
  "source": "segment",
  "segmentMessageId": "abc-456",
  "receivedAt": "2026-06-10T03:00:01Z",
  "segmentPayload": {
    /* entire Segment event, untouched */
  }
}
```

Pega consumes `segmentPayload` and maps fields using **Pega Data Transform rules** — keeping all field-mapping logic inside Pega where it belongs.

The micro service forwarder logic will be as below

```
┌──────────────────────────────────────────────┐
│            Generic Forwarder Microservice    │
```



One microservice. Routing config drives behaviour. New audience or new event type = config change only, no deployment.

Pega Side — Two Different Consumers

```
pega.cdh.segment.events  →  Pega Data Flow (real-time profile update)
pega.cdh.segment.track  →  Pega Data Flow (real-time profile update)
pega.cdh.segment.audience  →  Pega Audience Ingestion / Segment population rule
```

Pega needs to have different pattern for ingestion for audiences vs traits/events.

Possible sample examples of Kafka output for audience vs identify/traits.

Traits `pega.cdh.segment.identify`

```
{
  "source": "segment",
  "segmentMessageId": "msg-001",
  "receivedAt": "2026-06-10T03:00:01Z",
  "segmentPayload": {
    "type": "identify",
    "userId": "user-123",
    "traits": {
      "plan": "premium",
      "churnRisk": 0.72,
      "lastPurchaseDate": "2026-05-01"
    },
    "context": { ... },
    "timestamp": "2026-06-10T03:00:00Z",
    "messageId": "msg-001"
  }
}
```

Audiences `pega.cdh.segment.audience`

```
{
  "source": "segment",
  "segmentMessageId": "msg-002",
  "receivedAt": "2026-06-10T03:00:01Z",
  "segmentPayload": {
    "type": "identify",
    "userId": "user-123",
    "traits": {
      "high_value_prepaid": true
```

```
    },
    "context": {
      "personas": {
        "computation_class": "audience",
        "computation_key": "high_value_prepaid",
        "computation_id": "aud_abc123",
        "space_id": "spc_xyz"
      }
    },
    "timestamp": "2026-06-10T03:00:00Z",
    "messageId": "msg-002"
  }
}
```

Non-Functional Considerations

- **Latency**: End-to-end event delivery target of < 10 seconds (Segment Function → Apigee → Kafka → Pega)
- **Security**: TLS in transit, SSE-KMS at rest (Kafka), OAuth2/JWT at Apigee, PII masking enforced
- **Monitoring**: Segment built-in alerting, Apigee analytics, MSK CloudWatch metrics, Pega PDC stream monitoring, Splunk/SIEM integration
- **Error Handling**: Segment Function retries (built-in), Apigee error policies, Kafka consumer offset management, Pega Data Flow error handling with staging/audit tables
- **Segment Function timeout (30s)** — microservice must respond fast; Kafka produce should be async or use a fast broker path
- **Segment Function retry behaviour** — on 5xx it retries, so the `segmentMessageId` dedup key in Pega matters
- **Payload size** — Segment events with large trait sets can be a few KB; confirm Kafka topic max message size is configured to handle it (default 1MB is fine, but worth noting)

C. RIO pipeline with Terraform option

Program to setup a central Git-based repository for managing Infrastructure as Code (Terraform/OpenTofu), deployment configuration, and Segment platform artefacts. The repository acts as the authoritative source for configuration management and supports automated deployment, version control, change traceability, and promotion of approved changes across environments through TPG CI/CD pipelines.

RIO's Terraform flavour is built for AWS resources inside a TPG AWS account, so running the Segment provider through it needs enhancements in five areas — pipeline/provider support, credential handling, state management, security-check tooling, and change governance.

Where the gap sits

RIO's Terraform pattern (as used for Databricks and ReXI) assumes: a registered workload with an AWS account, CloudFormation-provisioned central pipeline resources, and Terraform confined to the SaaS control plane only. The ReXI SAD spells out the uplift checklist for the "Terraform flavour" — provider config must include RIO-injected default tags, IAM roles must carry the TPG permission boundary, resource identifiers must avoid conflicts, and required RIO variables must be included. None of these map cleanly to a provider that only talks to `api.segmentapis.com` over HTTPS with a bearer token.

Enhancements needed

#	Area	Enhancement required
1	**Workload & pipeline onboarding**	MDS AWS account registered as the RIO workload host, with a `.rio/config.yml` component whose trigger is `terraform/**` — same hybrid pattern proven for Databricks [RIO & Terr...rent State | TPG Confluence (Cloud)]
2	**Third-party provider allow-listing**	RIO CodeBuild must permit egress to `registry.terraform.io` and the Segment Public API endpoint, and allow the `segmentio/segment` provider (v1.6.x) to be initialised or mirrored into an internal registry [registry.t...rraform.io]
3	**Credential management**	Segment Public API token (Workspace Owner scope) stored in AWS Secrets Manager per workspace/environment and injected at runtime — RIO currently requires the Cloud team to load secrets on your behalf [twilio.com], [IC-RIO FAQ...924-004524 | PDF]
4	**State backend & environment segregation**	Separate remote state per Segment workspace (dev, SIT/UAT, pre-prod, prod), KMS-encrypted, matching the workspace model in the SAD [Draft+SAD+...1+(Drop+2) | Word]

5	**Security checks**	Checkov and the standard RIO security stages are AWS-resource aware; Segment resources will either be skipped or produce false negatives — needs a policy decision plus a Segment-specific ruleset [SAD - VH25...Migration | TPG Confluence (Cloud)]
6	**Plan/apply gating**	A manual approval stage before `terraform apply`, consistent with the change-control requirement that Terraform changes follow the existing CR and approval process [TPG CI/CD Framework | TPG Confluence (Cloud)]
7	**Framework registration**	The new pattern must be registered as a Registered Implementation and assessed against the applicable CI/CD Framework version to be deemed compliant [CI/CD Fram...quirements | TPG Confluence (Cloud)]

D. Segment HTTP API — Pseudo Code

The following pseudo code shows the exact payload structure Segment's HTTP Tracking API expects for **track**, **identify**, and **batch** calls.

Track Call

```
FUNCTION sendTrackEvent(userId, event, properties, context):

    // --- Payload Segment expects at api.segment.io/v1/track ---
    payload = {
        "userId":       userId,                          // REQUIRED (or anonymousId)
        "anonymousId":  context.anonymousId OR null,     // REQUIRED if no userId
        "event":        event,                           // REQUIRED — e.g. "Order Completed"
        "properties": {                                  // OPTIONAL — free-form dict
            "orderId":    properties.orderId,
            "revenue":    properties.revenue,
            "currency":   properties.currency,
            "brand":      properties.brand,
            ...any custom key-value pairs...
        },
        "context": {                                     // OPTIONAL — enrichment metadata
            "app":        { "name": config.appName, "version": config.appVersion },
            "channel":    "server",
            "ip":         context.clientIp,
            "userAgent":  context.userAgent,
```

```
            "library":    { "name": "http-api", "version": "1.0.0" },
            "locale":     "en-AU"
        },
        "timestamp":    ISO8601_NOW(),                  // OPTIONAL — ISO 8601 UTC
        "messageId":    UUID_V4(),                      // OPTIONAL — dedup key
        "integrations": {                               // OPTIONAL — destination toggles
            "All":   true,
            "Braze": true,
            "Google Analytics 4": true
        }
    }

    // --- HTTP Request to Apigee (which injects auth before forwarding) ---
    HTTP POST  {config.apigeeProxyUrl}/track
    Headers:
        Content-Type:       application/json
        Authorization:      Bearer {IAM_JWT_TOKEN}
        x-segment-source:   {config.segmentSource}       // e.g. "shop-server-side"
        x-correlation-id:   UUID_V4()
    Body:
        JSON.stringify(payload)

    // --- What Apigee forwards to Segment (auth injected) ---
    // POST https://api.segment.io/v1/track
    // Authorization: Basic base64({writeKey}:)
    // Body: payload (unchanged)

    RETURN response.status    // 200 = accepted, 400 = invalid, 401 = bad key
```

Identify Call

```
FUNCTION sendIdentifyEvent(userId, traits, context):

    // --- Payload Segment expects at api.segment.io/v1/identify ---
    payload = {
        "userId":       userId,                         // REQUIRED (or anonymousId)
        "anonymousId":  context.anonymousId OR null,    // REQUIRED if no userId
        "traits": {                                     // OPTIONAL — user profile traits
            "email":        traits.email,
            "firstName":    traits.firstName,
            "lastName":     traits.lastName,
            "phone":        traits.phone,
            "brand":        traits.brand,               // e.g. "vodafone", "tpg"
            "accountType":  traits.accountType,         // e.g. "postpaid", "prepaid"
            "loginMethod":  traits.loginMethod,         // e.g. "auth0", "social"
            "createdAt":    traits.createdAt,           // ISO 8601
            ...any custom key-value pairs...
        },
        "context": {
            "app":        { "name": config.appName, "version": config.appVersion },
            "channel":    "server",
            "ip":         context.clientIp,
            "library":    { "name": "http-api", "version": "1.0.0" }
        },
        "timestamp":    ISO8601_NOW(),
        "messageId":    UUID_V4(),
        "integrations": { "All": true }
    }

    // --- HTTP Request ---
    HTTP POST  {config.apigeeProxyUrl}/identify
    Headers:
        Content-Type:       application/json
        Authorization:      Bearer {IAM_JWT_TOKEN}
        x-segment-source:   {config.segmentSource}
        x-correlation-id:   UUID_V4()
```

```
    Body:
        JSON.stringify(payload)

    RETURN response.status
```

Batch Call

This can be used if there is high volume of calls.

```
FUNCTION sendBatchEvents(events[]):

    // --- Payload Segment expects at api.segment.io/v1/batch ---
    // Max 500KB per batch, max 32KB per individual message
    payload = {
        "batch": [
            {
                "type":       "track",                  // REQUIRED — "track" | "identify"
                "userId":     events[0].userId,
                "event":      events[0].event,
                "properties": events[0].properties,
                "context":    events[0].context,
                "timestamp":  events[0].timestamp,
                "messageId":  UUID_V4()
            },
            {
                "type":       "identify",
                "userId":     events[1].userId,
                "traits":     events[1].traits,
                "context":    events[1].context,
                "timestamp":  events[1].timestamp,
                "messageId":  UUID_V4()
            },
            ... up to N events ...
        ],
        "context": {                                    // OPTIONAL — shared context
            "library": { "name": "http-api", "version": "1.0.0" }
        }
    }

    // --- HTTP Request ---
    HTTP POST  {config.apigeeProxyUrl}/batch
    Headers:
        Content-Type:       application/json
        Authorization:      Bearer {IAM_JWT_TOKEN}
        x-segment-source:   {config.segmentSource}
        x-correlation-id:   UUID_V4()
    Body:
        JSON.stringify(payload)

    // --- Segment constraints ---
    // • Max batch size: 500 KB
    // • Max single message: 32 KB
    // • Recommended: flush every 20 events or every 10 seconds
    // • Response: 200 = all accepted, 207 = partial, 400 = validation error

    RETURN response.status
```